

Impact of stock market liberalization on returns volatility and economic growth: an evidence from Asian economies

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Abstract

The liberalization-privatization-globalization plays a role of catalyst and laid the pathways for economic development in emerging and developed economies. The financial market liberalization unlocked the new channels for foreign capital entry. Inflows led to appreciating real exchanges rate, to a rapid expansion of bank lending, and especially to increasing vulnerability to a reversal in capital inflows. The present study tries to quantify the dynamics persistence among and impact of, stock market liberalization on returns volatility in 14 Asian economies through modelling the GARCH and TGARCH model of volatility. The impact of stock market financial liberalization on returns volatility in Asian economies is witnessed subsequently by categorizing the study period into three phases, namely pre-, during, and post-liberalization periods. The dynamicity and behaviour in returns volatility further entangled through employing the daily, weekly and monthly frequency data set to witness the impact of financial stock market liberalization on volatility returns in Asian economies including global financial crunch of 2007-08. Successively, the second phase of the study examines the impact of stock market liberalization on industrial growth in the Indian economy. The literature insists and accounts for the direct relationship among financial stock market liberalization and economic growth. Thus, the third phase of the study tries to examine the impact of financial stock market liberalization on the economic growth in Asian economies owing to different course of liberalization, after controlling the macro-economic factors.

The findings confirms the positive impact of financial stock market liberalization on returns volatility in Asian economies. The evidences revealed that stock market liberalization significantly

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increased the returns volatility. In addition, the during- and post-liberalization periods experienced an incremental impact of financial stock market liberalization on returns volatility as compared to full period. Japan witnessed asymmetric (leverage) impact on returns volatility in post-liberalization and full periods, respectively and thus leads to increase in returns volatility. The industries indulging in financial and non-financial activities witnessed equivalent growth impact throughout the phases of stock market liberalization in Indian economy. Furthermore, the evidences revealed that, Hong Kong and Malaysia exerts positive and Republic of Korea exerts negative economic growth in during-liberalization period. Concurrently, China, Japan, and Republic of Korea felt significant positive and, India and Hong Kong has witnessed negative economic growth after stock market liberalization.

JEL classification: F37, G12, G15

Keywords: Asian stock market; economic crisis; economic growth; financial stock market liberalization; industrial growth; returns volatility.

1 Introduction

The financial market liberalization is the revolutionary phenomenon that persists in last two decades and simultaneously triggering the economic-growth of the nations. The equity market liberalization possess the favourable effects in various phases. However, the impacts of financial market liberalization for the developed and developing economies exerts manifolds. Financial market liberalization unwrapped new channels for foreign capital entry. Conversely, due to foreign investment inflows, led to gaining the positive momentum in real exchange rates, broadening the financial services with expansion of bank lending, and primarily increasing the vulnerability that prone to distortions due to reversal in capital inflows.

Though the empirical studies witnessed that financial market liberalization exerts positive impact on emerging countries mimicking declined cost of equity, improved private physical investment and increased returns (Bekaert et al. (2001), Bekaert et al. (2005), Henry (2000), Kim and Singal (2000), Klein and Olivei (2008), Stulz (1999)). Conversely, financial liberalization may perhaps prone a country skeptical to economic and political outbreak abroad resulting the domestic market more volatile. The impact of stock market liberalization on return volatility is primarily a significant concern that emerging economies necessarily has to take into consideration in advance prior to their decision to liberalize and perhaps even post-liberalization process. The returns volatility possess the characteristic feature that has adverse impact pertaining to decision making to the effective apportionment of resources, subsequently for investment (Bekaert and Harvey (1997), Kim and Singal (1993, 2000), Singh (1993)). Hence, the returns volatility and financial market liberalization is closely interlinked and consequently plausible justification is liberalization attracts the fresh group of investors, mostly the institutional investors. However, the institutional investors versed from developed markets and their decisions are pertaining to rational investment analysis. Thus, the possibility of reduction in returns volatility after liberalization. In contrast, the financial market liberalization may prone the liberating economy to uncertainties overseas which may witnessed domestic stock market volatility. Hence, likelihood of increased volatility in stock returns after liberalization. Moreover, financial market liberalization might not have significant impact on returns volatility due to offsetting the competing effects (Imegi (2014), Huana and Kracaw (1994), Rizwan and Khan (2007)). Bekaert and Harvey (1997) accounts that poorly developed stock market without financial liberalization perhaps expected to be categorized by greater stock market volatility.

The financial stock market liberalization and volatility literature suggests that financial market openness may leads to decrease (Beakert and Harvey (1997) De Santis and Imrohoroglu (1997), Kwan and Reyes (1997), Kim and Singal (1993), Koot and Padmanabhan (1993)) or increase (Beakert and Harvey (2001), Kim and Singal (1993), Levine and Zervos (1998)) volatility.

Thus, the present study first examines the impact of financial stock market liberalization on the returns volatility of 14 Asian economies in three different regime of the study period, namely, pre-, during and post-liberalization periods on daily, weekly and monthly data set. Consequently the second phase of the study examines the impact of stock market liberalization on industrial growth in Indian stock market through the different regimes of the study period on weekly and monthly frequency data set. Exceptionally, the foremost contribution of our work lies into the third phase that tries to investigate the impact of financial stock market liberalization on the economic growth of 14 Asian economies. The emerging economies literature on impact of stock market liberalization on economic growth proposes mixed response primarily the lack of evidences to justify the said relationship exerts triviality. Thus, in this conjecture, present study examines and tries to unwrap the dynamicity persistence in the relationship between financial stock market liberalization and economic growth among Asian economies.

2 Literature review

2.1 Financial liberalization and return volatility

The existing literature on the impact of financial stock market liberalization on volatility has been diverse. The literature embedded with the intensity of liberalization period dates to increase in volatility. Huang and Yang (2000) studied the effect of financial liberalization on the stock market volatility in ten emerging markets mimicking post-liberalization period. Findings revealed that volatility rises in Mexico, South Korea and Turkey, however, in turn volatility significantly decreases in Malaysia, Chile, Argentina and Philippines. Though, Huang and Yang (2000) evident no precise variation in volatility changes in Brazil, Thailand and Taiwan. Levine and Zervos (1998) revealed in their study of impact of capital control liberalization on volatility in 16 emerging economies, that persistence of higher volatility in the post-liberalization period. Uppal (1998) studied variations in volatility of stock returns in Karachi stock exchange in Pakistan and accounts for significant rise in volatility subsequently the market opening. Further the evidences implied that volatility normalized when the market opening at fresh and higher level. Grabel (1995) evident

substantial increment in volatility during financial liberalization in Venezuela, South Korea, Chile and Colombia, and in turn meagre significant volatility increment in Philippines and Argentina. Seminal study of Koot and Padmanbhan (1993) employed the GARCH models for Jamaican stock market and found higher volatility in post-liberalization period. However, the above studies evident considerably higher volatility in the post-liberalization period, conversely there exist significant evidence that liberalization cause lower volatility.

Kassimatis (2002) employed Exponential GARCH and revealed that volatility decreases (Umutlu et. al, 2010) after key liberalization guidelines were implemented in India, Argentina, the Philippines, Pakistan, South Korea, and Taiwan. Kim and Singal (2000) studied the impact of market opening pertaining to variations in the level and the volatility of stock returns in 14 emerging markets. It evident that the volatility levels through the first two years, post stock market opening were not significantly different from those before opening. Further, Kim and Singal (2000) found that a significant decline in stock return volatility in fourth and fifth years post-liberalization. Moreover, Bekaert and Harvey (1997) explored the impact of liberalization policies on stock returns volatility in seventeen emerging stock markets. Findings witnessed the significant decrease in volatility in Argentina, Brazil, Mexico, Taiwan, and Portugal and yet insignificant decline in volatility in rest of the stock markets (Li et. al, 2011). Kwan and Reyes (1997) examined the influence of stock market liberalization on returns volatility of Taiwan stock market through employing GARCH model and found that the post-liberalization period experienced decline in returns volatility. However, the stock market liberalization indicates variations in returns distribution to some extent. Further, Taiwan stock market became highly efficient in processing information compared to pre-liberalization period. However, contrasting with Kwan and Reyes (1997), De Santis and Imrohoroglu (1997), Bley and Saad (2011) unearth no significant relationship between liberalization and returns volatility in Asia, Europe/Middle-East and Latin America. Seminal study of Spyrou and Kassimatis (1999) revealed that no significant variation in volatility during post-liberalization period in Argentina, India, Chile, Mexico, Philippines, Pakistan, and Taiwan. Concurrently, Pagan and Schwert (1990), Engle and Ng (1993) suggests the bad news inclined to cause high volatility in returns rather than good news. However, the asymmetric information impact is ascribed to the usage of debt financing by firms is known as leverage effect (Pagan and Schwert (1990), Engle and Ng (1993)). Jaleel and Samarakoon (2009)

examined the influence of liberalization of Sri Lankan stock market to foreign stock investor on the return volatility and subsequently the autoregressive models of volatility indicates that market liberalization significantly drives the return volatility increment. Though mutually conditional and unconditional volatility measures are maximum in the liberalization period. Consequently, negative return shocks indicates minimum volatility signifying no leverage effect, and this resembles the very low levels of leverage employed by Sri Lankan companies. Jayasuriya (2005) evidenced that market openness to foreign investors frequently increased or decreased stock return volatility. Further the evidences revealed that countries witnessed minor post-liberalization volatility are commonly categorized through higher quality accounting standards and investor protection laws, and lower restrictions levied on the deportation of foreign income and capital. Such countries are comprised of improved quality institutions as lower level of corruption. Following evidences imparted spontaneous legacy that favourable market dynamics and quality institutions are significantly nurture more efficient markets where the implementation of financial liberalization policy is less prone to generate excessive volatility effects.

Bekaert et al. (2006) investigate the impact of equity market liberalization and capital account openness respectively on real consumption growth volatility. Bekaert et al. (2006) revealed that financial liberalization is specifically connected with lower consumption growth volatility. The findings are persistent even after controlling business cycle impacts, economic and financial development, and quality of institutions. Economies which have sound open capital accounts witnessed a greater decrease in consumption volatility, subsequently after equity market openings. Furthermore, the evidence suggests that the role financial liberalization involves with decrease in the ratio of consumption growth volatility to GDP growth volatility mimicking diversification in risk sharing. Seminal study of Kaminsky and Schmukler (2008) revealed that influential factor behind inadequate evidence for financial liberalization is the impacts are of time-varying in nature and that leads to enormous peak and troughs in short-horizon. However, in long-horizon the performance of institutions improve and stock market incline to stabilize. Das and Mohapatra (2003) studied inequality in income distribution and found a pattern resembling that income growth increased to the maximum for the top quintile of the income distribution at the cost of three middle quintiles of the income distribution. The influential findings followed by the lowest income share remained stagnant in the post-liberalization period. However, the patterns are persistent after controlling the global shocks, country-specific factors, policies implementations with respect to

privatization and stabilization. There exist close linkage between leverage and stock return volatility. The leverage claims that the negative stock return declines the stock price, in turn increasing the leverage of the firm, and subsequently increased leverage indicates rise in stock price volatility. Conversely, leverage causes inverse shocks that possess a greater effect on conditional volatility than positive shocks signifies, and leads to asymmetric variations in stock return volatility.

Since, the mixed view of literature about the relationship among stock market volatility and stock market liberalization in emerging and developed economies exerts wide array of differentiations owing to manifolds in distinct opinions. Therefore, to unearth the various dimensions and prospects underneath the financial liberalization and returns volatility, the present study tries to examine the impact of stock market liberalization on stock returns volatility in 14 economies through modelling heteroskedastic conditional volatility model in the first phase of the study. The study period comprised of three regimes, namely pre-, during and post-liberalization period to infer the magnitude of stock market liberalization impact on returns volatility. However, daily, weekly and monthly frequency data sets been employed to enhance the insights with a view point of analysis to compare and measure the behavioural aspect of dynamic relationship between stock market liberalization and returns volatility. In addition, we tries to investigate the impact of stock market financial liberalization on industrial growth on India economy mimicking three regimes of the study period on weekly and monthly frequency data-sets. The successive section compiles the literature on the impact of stock market financial liberalization on economic growth.

2.3 Financial liberalization and economic growth

The existing literature exhibit empirical indications that the growth impact of financial liberalization reveals contradictory conclusions. Conversely, meagre evidence supports financial liberalization positively affects economic growth, though the relationship is not robust. Concurrently there is no general consensus among economists that financial liberalization leads to economic growth. The path breaking study of Quinn (1997) reveals positive relationship between capital account liberalization and economic growth. Quinn (1997) empirical estimation includes cross-sectional data for 58 countries, over the period of 30 years. Subsequently, discovered that change in capital account liberalization possess a deep significant influence on real per capita gross

domestic product (GDP) growth. Seminal study of Arteta et al. (2001) used the identical indicator of liberalization as in Quinn (1997) on pooled data of 51 to 59 countries over the period of investigation 1973 to 1981, 1982 to 1987, and 1988 to 1992 respectively. The findings evident the first-order positive relationship among economic growth and capital account liberalization (Bekaert et al., 2001 & 2005). Stock market liberalization outgrows economic growth, and the link between these two variables is robust and prompt. Bekaert et al. (2001, 2005) reveals that financial liberalization indicates 1 percent increase in annual per capita GDP growth over the five-year periods and the impact is significantly persistent for the period 1980 to 1997. Further the robustness of this outcome being examined with different countries groupings 95, 75, 50 and 28 countries respectively, and consecutively with separate time horizons for estimating economic growth. Fuchs-Schundeln and Funke (2001) proposed a significant robust evidence supporting the fact of equity market liberalization boosts economic growth among 27 emerging economies, over the long-horizon period 1975 to 2002. Further, they accounts that financial market liberalization witnessed 4 percent rise in annual per capita GDP growth in post-liberalization period (Edison et al., 2002).

Through contrasting, Grilli and Milesi-Ferretti (1995) discover no relation exist between capital account liberalization and economic growth (Naceur et al., 2008) and the study period ranges through 1966 to 1989 including 61 countries. Consequently, Rodrik (1998) develops triviality on impact of liberalization account that used 100 developed and emerging economies sample simultaneously after controlling for other growth indicators for the period from 1975 to 1989. Rodrik (1998) revealed that capital account conversion ability is statistically insignificant with long-horizon economic performance. Kraay (1998) further supports the findings of Rodrik (1998) that examines the impact of capital account liberalization on investment, growth and inflation, and found insignificant relationship between capital account liberalization and economic growth. However, Kraay (1998) further reveals the regression that comprises of capital flow data emerges positive and statistically significant outcomes which includes the sample from 117 countries for the period 1985 to 1997.

The enormous and growing body of literature (Klein and Olivei, 2008; Edwards, 2001; Chanda, 2001; Edison et al., 2002; Loayza and Rancière, 2004; Klein, 2003; Eichengreen and Leblang, 2003; Fratzscher and Bussière, 2004; Calderon, Loayza and Schmidt-Hebbel, 2004; Bonfiglioli

and Mendicino, 2004) reveals that effects of financial liberalization are generally conditional. Klein and Olivei (2008) proposed that economies which ought to more open capital markets during the period 1976 to 1995 witnessed comparatively higher growth rates, however the result is mostly determined through developed economies in a different sample of 82 countries. The seminal study of Edwards (2001) supports conclusions drawn by Klein and Olivei (2008), and found liberalization drives economic growth and the impact is limited to comparatively developed countries mimicking a sample of 60 countries. Further, they reveals that the positive relationship among liberalization and per capita GDP infers more open capital account impact with the increase in country's initial development level. Successively, the negative coefficient of the openness index signifying that an open capital account entails a negative impact at low levels of development. The seminal study of Klein (2005) supports the findings of Klein and Olivei (2008). Klein (2005) employed a sample from 71 countries for the period through 1976 to 1995 and suggests that the insignificant interactions among capital account liberalization and economic growth depends on institution's quality. By contrasting, Klein (2003) found opposite result that used the identical indicators (Klein, 2005) of financial liberalization in a sample of 85 developed economies for the same period. Klein (2003) showed the existence of interactions among capital account liberalization and economic growth is due to the institutions quality. However, unable to identify a statistical significant impact of financial liberalization in economic growth. Though OLS tests in cross-sectional verifies liberalization exercises a positive impact on growth solely in medium income nations, in turn low and high income nations experienced insignificant impact of financial liberalization.

Calderon et al. (2004) validate the findings of Klein (2003) with generalized methods of moments for the period through 1970 to 2000 between 76 countries by confirming that there exist non-linear relationship among income and stock market. Further conclude that with the high level of economic development does not always leads to positive growth impact of openness. Edison et al. (2002) investigate the impact of financial market liberalization on the economic, financial and institutional level of development. The impact of financial market liberalization is examined through different econometric models with panel data of 57 nations with numerous indicators of liberalization for the period from 1980 to 2002. Further Edison et al. (2002) concludes that the

results are stronger to validate the assumption of positive relationship among liberalization and economic growth, after monitoring for financial, economic, institutional and political conditions. Arteta et al. (2001) examined the association of capital account liberalization with economic growth and simultaneously evident the weak positive association between capital account liberalization and growth. The impact is time-varying and the nature of capital account liberalization and its relationship with economic growth measurement owes how it is estimated. Artera et al. (2001) further revealed that the impacts of capital account liberalization is very weak, however, meagre evidence suggests that the positive growth effects of liberalization are stronger in the nations with strong organizations. Seminal study of Bekaert et al. (2005) confirms that equity market liberalization leads to increase in economic growth. Furthermore, financial market liberalization experienced statistically significant 1 percent rise in annual per capita GDP growth. The impact is persistent and robust with respect to alternative definitions of liberalization and successively no significant variation reflect in world business cycle. The impact remains stagnant after controlling the measure of growth opportunities in the regression. Bekaert et al. (2005) further revealed that capital account liberalization plays the part in future economic growth, however, it does not include the impact of capital market liberalization. The countries possess high quality institutions witnessed more economic growth.

Galindo et al. (2007) revealed that financial liberalization leads to an improvement in the efficiency with respect to which funds have been allocated for investment. Galindo et al. (2007) has developed an efficiency index with respect to pre- and post-liberalization regimes that suggest in majority of emerging countries instances financial reform experienced to surge in efficiency with which investment funds are allocated. However, Eichengreen and Leblang (2003) suggests that the inability of previous study to unearth the dynamicity in relationship between capital account liberalization and economic growth is due to their failure to measure and estimate the effect of crisis on economic growth and subsequently for capacity of controls to minimize the negative output impacts. Furthermore, the openness to crisis carries across the nations and with the structure and performance of international financial system, it is obvious that the impact of capital account liberalization on growth are conditional and definite. Diamandis (2008) observes the dynamic behaviour of stock market volatility for emerging stock markets and developed stock market. The study period witnessed a significant financial and economic growth in these emerging economies, which have experienced many instances of financial crisis. Furthermore, Diamandis (2008)

revealed that there exist strong market integration among Latin American stock markets and the developed market during the financial crisis of 1997-1998. From initial period distress the market efficiency bounced back to the normal levels. The study unearth the presence of multiple volatility regime and the significant increased volatility found during the crisis period, even though capital flows liberalization development spurs medium shifts in volatility. By contrasting, Rejeb and Boughrara (2014) and Ndako (2012) reveals that financial liberalization does not spur excessive volatility in emerging stock market rather the volatility declines gradually. Furthermore, emerging economies witnessed decline in the impact of financial liberalization on probability of crisis steadily.

Gupta and Yuan (2002) investigates the causal linkage between financial development and economic growth subsequently found no evidence of a constant shift in average industry growth across all segments in post-liberalization period. In turn the financial liberalization indicates comparatively higher growth in industries which relies further on external finance. Further, the findings shed the light towards liberalizations promotes economic growth through decreasing the incremental cost of external capital rather reducing the aggregate cost of capital in an economy (O'Connor, 2013). The findings obtains back up support from the evidence that growth emerges through formation of new establishments, which depends more on external financing, rather than through an expansion in size of an existing firms that relies more on internal financing. Seminal study of Gupta and Yuan (2009) found countries and industries with meagre entry constraints witnessed growth in new firm establishments following liberalization. Gamra (2009) studied the effect of financial liberalization on the economic growth in East Asian region and found the significant relationship between financial liberalization and economic growth in East Asian emerging countries. However, the financial liberalization on economic growth depends on the prominent factors namely, nature along with the intensity of financial sectors liberalization. Partial liberalization inclined positive association with growth effects in turn full liberalization results negative growth effect. Further, domestic financial liberalization exhibits economically significant positive growth effect. Henry (1999a) reveals that the stock market liberalizations leads to increase in growth rate of private physical investment.

Henry (2000) investigates the effect of stock market liberalization on emerging market stock prices and subsequently found aggregate equity price of a country on an average experienced excess

returns during eight-month window following until implementation of its initial stock market liberalization. Further, Henry (2000) proposed that stock market liberalization leads to decrease in liberalizing country's cost of equity capital through permitting the diversification of risk sharing among domestic and foreign investors. Quinn and Toyoda (2008) examined the relationship among economic growth and capital account liberalization through employing *de jure* measures of capital account and financial current account openness for 94 countries. They used pooled time-series data, cross-sectional OLS and GMM estimators to investigate the growth rates for 50 years and revealed that capital account liberalization possess a positive association with growth in developed and emerging market economies (Bumann et. al, 2013). Furthermore, Quinn and Toyoda (2008) ensures that equity market liberalization has an independent effect on economic growth. Prolonging, Levchenko et al. (2009) found strong empirical evidence that financial liberalization leads to increase in economic growth and volatility. However, the impact on the volatility is firmly lower and subsequently on growth is temporary.

Since the literature emphasizes diversified dimensional prospects about empirical relationship among stock market financial liberalization and economic growth. Hence, the third phase of the study attempts to examine the impact of stock market liberalization on economic growth in 14 Asian economies after controlling the economic factors. The present study shed the light on the various dimension persistence among financial liberalization and economic growth and subsequently the behavioral aspect in Asian economies.

3 Methodology

We employ autoregressive conditional heteroskedastic (ARCH) and generalized autoregressive conditional (GARCH) models for modelling the heteroskedastic conditional volatility to examine the objective of first phase, as impact of stock market financial liberalization on returns volatility in 14 Asian economies, namely, SHCOMP index, HSI index, Nifty index, SENSEX index, JCI index, NKY index, KOSPI index, FBMKLCI index, NZSE50FG index, KSE100 index, PCOMP index, FSSTI index, CSEALL index, TWSE index, and SET index. Engle (1982) coined ARCH models assuming that the variance of the error term in a given period depends on the squared error terms from previous periods. Hence, the volatility in past periods is captured by lags of the squared residuals. Bollerslev (1986) coined the GARCH models in addition with the ARCH model to allow for the variance of the error term to rely on its own lags in addition with lags of the squared errors.

Thus, the GARCH model can witness the volatility changes with restricted parameters than the ARCH models.

The autoregressive (AR) model for indexes returns specifies the current returns as a function of lagged returns,

$$R_t = \beta_0 + \sum_{i=1}^n \beta_i R_{t-i} + \varepsilon_t \quad (1)$$

Where R_t is the current indexes return, R_{t-1} is the lagged indexes return, and ε_t is an error term.

The residuals and squared residuals obtained from the equation 1 are tested for autocorrelation and heteroskedasticity simultaneously.

3.1 Generalized autoregressive conditional heteroskedasticity (GARCH) model of volatility

The univariate GARCH (p,q) method specification as follows,

$$R_t = \phi_0 + \phi_1 R_{t-1} + \varepsilon_t \quad (2)$$

$$\varepsilon_t / \psi_{t-1} \sim N(0, h_t)$$

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j} \quad (3)$$

Where R_t is the current indexes returns, R_{t-1} is the lagged return, ε_t is the error term, h_t is the current conditional volatility, h_{t-1} is the lagged conditional volatility. The coefficients $\alpha_0 > 0, \alpha_i \geq 0$ and $\beta_j \geq 0$ for all i and j , and $\left(\sum_{i=1}^q \alpha_i + \sum_{j=1}^p \beta_j \right) < 1$. The sum of the slope coefficients $\left(\sum_{i=1}^q \alpha_i + \sum_{j=1}^p \beta_j \right)$ measures the persistence of volatility.

The Maximum Likelihood (ML) method is employed to estimate the parameters of the conditional mean (Eq. 2), and variance equation (3). Multiple ARCH(p) and AR(s)-GARCH(p,q) models were estimated. The AR(1) process is adequate to overcome the serial correlation in all periods with an exception in post-liberalization, where an AR(2) process is necessary. Consequently, the model is specified as follows,

$$R_t = \phi_0 + \phi_1 R_{t-1} + \phi_2 R_{t-2} + \varepsilon_t \quad (4)$$

$$\varepsilon_t / \Psi_{t-1} \sim N(0, h_t)$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_j h_{t-j} + \lambda_1 \text{Prelib}_t \quad (5)$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_j h_{t-j} + \lambda_2 \text{Duringlib}_t \quad (6)$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_j h_{t-j} + \lambda_3 \text{Postlib}_t \quad (7)$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_j h_{t-j} + \lambda_4 \text{PreDuringPostlib}_t \quad (8)$$

To observe the impact of stock market liberalization on returns volatility in pre-, during, post-liberalization and full study period, the sub-set for respective phase has been included in the conditional variance equations (5), (6), (7) and (8) respectively. Hence, statistically significant $\lambda_1, \lambda_2, \lambda_3$, and λ_4 will indicate that the sub-sets of liberalization period possess a significant impact on returns volatility.

However, the GARCH model eliminates the asymmetric changes in stock returns volatility. The literature symbolizes the negative stock return shock cause additional volatility than the positive return shocks, as the reason been inferred due to leverage effect (Christie 1982; Black 1976). The leverage impact specifies that when stock prices goes down due to negative shocks, the leverage of the firm increases led to increase in stock price volatility. Thus, to comply with the asymmetric effect, we employ the threshold-GARCH (TGARCH) model coined by Golstan et al. (1993) and Zakoian (1994). The TGARCH model introduced a dummy variable to apprehend the effect of unexpected returns on the conditional variance of returns.

3.2 Threshold-GARCH (TGARCH) model and asymmetric shock

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j} + \gamma \varepsilon_{t-1}^2 d_{t-1} \quad (9)$$

Where $d_{t-1} = 1$ if $\varepsilon_{t-1} < 0$ and 0 otherwise. If $\gamma > 0$ and significant, then there exist a leverage effect. It means that adverse market conditions or negative shocks cause more volatility than positive shocks.

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j} + \gamma \varepsilon_{t-1}^2 d_{t-1} + \lambda_1 \text{Prelib}_t \quad (10)$$

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-1}^2 + \sum_{j=1}^p \beta_j h_{t-1} + \gamma \varepsilon_{t-1}^2 d_{t-1} + \lambda_2 \text{Duringlib}_t \quad (11)$$

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-1}^2 + \sum_{j=1}^p \beta_j h_{t-1} + \gamma \varepsilon_{t-1}^2 d_{t-1} + \lambda_1 \text{Postlib}_t \quad (12)$$

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-1}^2 + \sum_{j=1}^p \beta_j h_{t-1} + \gamma \varepsilon_{t-1}^2 d_{t-1} + \lambda_1 \text{PreDuringPostlib}_t \quad (13)$$

The second phase of the study examines the impact of stock market liberalization on industrial growth of Indian economy, and to capture the magnitude and intensity of impact the OLS has been employed. The 11 industrial sector comprised of both financial and non-financial, been taken into account to witness the impact of stock market liberalization in pre-, during, and post liberalization periods, respectively. The specifications to infer industrial growth are as follows.

$$C \& RE_t = \alpha_{0,t} + \beta \text{Nifty}_t + \varepsilon_t \quad (1)$$

Where $C \& RE_t$ is construction and real estate industry engaging in non-financial activities, Nifty_t is returns on CNX Nifty index at time t , ε_t captures the error-term.

$$ELE_t = \alpha_{0,t} + \beta \text{Nifty}_t + \varepsilon_t \quad (2)$$

Where ELE_t is electricity industry engaging in non-financial activities, Nifty_t is returns on CNX Nifty index at time t , ε_t captures the error-term.

$$\text{MANU}_t = \alpha_{0,t} + \beta \text{Nifty}_t + \varepsilon_t \quad (3)$$

Where MANU_t is the manufacturing industry engaging in non-financial activities, Nifty_t is returns on CNX Nifty index at time t , ε_t captures the error-term.

$$\text{MIN}_t = \alpha_{0,t} + \beta \text{Nifty}_t + \varepsilon_t \quad (4)$$

Where MIN_t is the mining industry engaging in non-financial activities, Nifty_t is returns on CNX Nifty index at time t , ε_t captures the error-term.

$$\text{SERV}_t = \alpha_{0,t} + \beta \text{Nifty}_t + \varepsilon_t \quad (5)$$

Where $SERV_t$ is the service industry engaging in non-financial activities, $Nifty_t$ is returns on CNX Nifty index at time t , ε_t captures the error-term.

$$AFS_t = \alpha_{0,t} + \beta Nifty_t + \varepsilon_t \quad (6)$$

Where AFS_t is asset financing services industry engaging in financial activities, $Nifty_t$ is returns on CNX Nifty index at time t , ε_t captures the error-term.

$$BSERV_t = \alpha_{0,t} + \beta Nifty_t + \varepsilon_t \quad (7)$$

Where $BSERV_t$ is the banking services industry engaging in financial activities, $Nifty_t$ is returns on CNX Nifty index at time t , ε_t captures the error-term.

$$FBFS_t = \alpha_{0,t} + \beta Nifty_t + \varepsilon_t \quad (8)$$

Where $FBFS_t$ is the fee based financial service industry engaging in financial activities, $Nifty_t$ is returns on CNX Nifty index at time t , ε_t captures the error-term.

$$INVS_t = \alpha_{0,t} + \beta Nifty_t + \varepsilon_t \quad (9)$$

Where $INVS_t$ is the investment service industry engaging in financial activities, $Nifty_t$ is returns on CNX Nifty index at time t , ε_t captures the error-term.

$$OTFS_t = \alpha_{0,t} + \beta Nifty_t + \varepsilon_t \quad (10)$$

Where $OTFS_t$ is the other financial services industry engaging in financial activities, $Nifty_t$ is returns on CNX Nifty index at time t , ε_t captures the error-term.

$$OFBFS_t = \alpha_{0,t} + \beta Nifty_t + \varepsilon_t \quad (11)$$

Where $OFBFS_t$ is the other fund based financial services industry engaging in financial activities, $Nifty_t$ is returns on CNX Nifty index at time t , ε_t captures the error-term.

The impact of stock market liberalization on industrial growth has witnessed in respective 11 industries engaging in financial and non-financial activities through employing OLS. The impact of financial liberalization further decomposed in pre-, during and post-liberalization period for each of OLS measurement to witness the magnitude and intensity of impact in variations.

Moreover, the third phase of the study examines the impact of stock market liberalization on economic growth in Asian economies, after controlling for the influence of macro-economic factors. To witness the impact of stock market liberalization on economic growth in Asian economies, the liberalization proxy and macro-economic proxies been regressed on the economic growth proxy through modelling the ordinary linear regression. The regression specification for each of 14 Asian economies been modelled and subsequently the specifications are as follows.

$$GDP_t = \alpha_{0,t} + \beta_1 SHCOMPindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (1)$$

Where GDP_t is the real GDP growth rate of China in month t , $SHCOMPindex_t$ is stock returns on Shanghai composite index in the month t , CPI_t is the consumer price index growth rate of China in the month t , $FOREX_t$ is foreign exchange rate for Chinese yuan of China relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 HSIindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (2)$$

Where GDP_t is the real GDP growth rate for Hong Kong in month t , $HSIindex_t$ is stock returns on Hang Seng index in the month t , CPI_t is the consumer price index growth rate for Hong Kong in the month t , $FOREX_t$ is foreign exchange rate for Hong Kong dollar of Hong Kong relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 NIFTYindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (3)$$

Where GDP_t is the real GDP growth rate of India in month t , $NIFTYindex_t$ is stock returns on NSE Nifty index in the month t , CPI_t is the consumer price index growth rate of India in the

month t , $FOREX_t$ is foreign exchange rate of Indian rupee for India relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 SENSEXindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (4)$$

Where GDP_t is the real GDP growth rate of India in month t , $SENSEXindex_t$ is stock returns on BSE SENSEX index in the month t , CPI_t is the consumer price index growth rate of India in the month t , $FOREX_t$ is foreign exchange rate of Indian rupee for India relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 JCIindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (5)$$

Where GDP_t is the real GDP growth rate of Japan in month t , $JCIindex_t$ is stock returns on Jakarta composite index in the month t , CPI_t is the consumer price index growth rate of Japan in the month t , $FOREX_t$ is foreign exchange rate of Japanese Yen for Japan relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 KOSPIindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (6)$$

Where GDP_t is the real GDP growth rate of Republic of Korea in month t , $KOSPIindex_t$ is stock returns on Korea composite stock price index in the month t , CPI_t is the consumer price index growth rate of Republic of Korea in the month t , $FOREX_t$ is foreign exchange rate of South Korean won for Republic of Korea relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 FBMKLCCIindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (7)$$

Where GDP_t is the real GDP growth rate of Malaysia in month t , $FBMKLCCIindex_t$ is stock returns on Kuala Lumpur composite index in the month t , CPI_t is the consumer price index growth rate

of Malaysia in the month t , $FOREX_t$ is foreign exchange rate of Malaysian ringgit for Malaysia relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 NZSE50FGindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (8)$$

Where GDP_t is the real GDP growth rate of New Zealand in month t , $NZSE50FGindex_t$ is stock returns on New Zealand exchange 50 gross index in the month t , CPI_t is the consumer price index growth rate of New Zealand in the month t , $FOREX_t$ is foreign exchange rate of New Zealand dollar for New Zealand relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 KSE100index_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (9)$$

Where GDP_t is the real GDP growth rate of Pakistan in month t , $KSE100index_t$ is stock returns on KARACHI 100 Index in the month t , CPI_t is the consumer price index growth rate of Pakistan in the month t , $FOREX_t$ is foreign exchange rate of Pakistani rupee for Pakistan relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 PCOMPindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (10)$$

Where GDP_t is the real GDP growth rate of or Philippines in month t , $PCOMPindex_t$ is stock returns on PSE Composite Index in the month t , CPI_t is the consumer price index growth rate of Philippines in the month t , $FOREX_t$ is foreign exchange rate of Philippine peso for Philippines relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 FSSTIindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (11)$$

Where GDP_t is the real GDP growth rate of Singapore in month t , $FSSTIindex_t$ is stock returns on Straits times index in the month t , CPI_t is the consumer price index growth rate of Singapore in the month t , $FOREX_t$ is foreign exchange rate of Singapore dollar for Singapore relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 CSEALLindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (12)$$

Where GDP_t is the real GDP growth rate of Sri Lanka in month t , $CSEALLindex_t$ is stock returns on Colombo all-share price index in the month t , CPI_t is the consumer price index growth rate of Sri Lanka in the month t , $FOREX_t$ is foreign exchange rate of Sri Lankan rupee for Sri Lanka relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 TWSEindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (13)$$

Where GDP_t is the real GDP growth rate of Taiwan in month t , $TWSEindex_t$ is stock returns on Taiwan weighted index in the month t , CPI_t is the consumer price index growth rate of Taiwan in the month t , $FOREX_t$ is foreign exchange rate of Taiwanese dollar for Taiwan relative to US dollar in the time t .

$$GDP_t = \alpha_{0,t} + \beta_1 SETindex_t + \beta_2 CPI_t + \beta_3 FOREX_t + \varepsilon_t \quad (14)$$

Where GDP_t is the real GDP growth rate of Thailand in month t , $SETindex_t$ is stock returns on stock exchange of Thailand index in the month t , CPI_t is the consumer price index growth rate of Thailand in the month t , $FOREX_t$ is foreign exchange rate of Thai baht for Thailand relative to US dollar in the time t .

Furthermore, for estimating the impact of stock market financial liberalization on economic growth in 14 Asian economies through pre-, during and post-liberalization periods, subsequently regression specification being modelled for respective indexes. Hence, in turn will witness the variations in magnitude and intensity among the phases.

4 Data and variables

The present study proceeds in the three phases-, firstly, through examining the impact of stock market liberalization on returns volatility in 14 Asian economies. Secondly, the impact of stock market liberalization on industrial growth in India, and finally the impact of stock market liberalization on economic growth in 14 Asian economies. The impact of stock market

liberalization on three objectives of the present study in terms of magnitude and intensity been further witnessed through categorizing the whole sample period pertaining to the respective Asian economies into three periods, namely pre-, during, and post-liberalization. Table 1 presents the detailed evidence with supportive literature for the bifurcation of the sample period owing to stock market liberalization for the 14 Asian economies.

Table 2 of panel A, B, and C represents the detailed specification of the variables used pertaining to the first, second and third objectives respectively for the present study. Panel A of table 2 comprised of the stock returns on value-weighted indexes for the 14 Asian economies, namely, Shanghai composite index, Hang Seng index, CNX Nifty index and S&P BSE SENSEX index, Jakarta composite index, Nikkei 225 index, Korea composite index, Kuala Lumpur composite index, New Zealand exchange 50 gross index, KARACHI 100 index, PSE composite index, Straits times index, Colombo all-share price index, Taiwan weighted index, and stock exchange of Thailand index. Subsequently, the date of bifurcation for various phases of stock market liberalization, data frequency and source of data collection been resembles in table 2 of panel A. Consequently table 2 of panel B summarizes the data and variables used to estimate the impact of stock market liberalization on industrial growth in India. The companies registered on national stock exchange (NSE) and simultaneously engaging in financial and non-financial activities, categorizing and comprising the industries namely, construction and real estate, electricity, manufacturing, mining, non-financing services, asset financing services, banking services, fee based financial services, investment services, other financial services, and other fund based financial services are accounted for to examine the impact of stock market liberalization. Hence, the 11 sector industrial returns pertaining to financial and non-financial activities are collected from the prowess database of Centre for monitoring Indian economy (CMIE). Panel B of table 2 indicates the variable specification and data source for the second objective of the study.

Table 2 of panel C shows the variable, data specification and source of data collection in relation with the third objective of the study. To examine the effect of stock market liberalization on economic growth in 14 Asian economies through the different time horizon (pre-, during and post-liberalization), GDP (real GDP growth rate) is consider as the proxy for the economic growth for the respective economies. However, to control and capture the effect of macro-economic influence, inflation is added in the right hand side of the economic growth regression model and is proxied by consumer price index (CPI) for the respective regression equations of the economies. In

addition the Foreign exchange rate (FOREX) of the respective economies is considered in the respective economic growth regression specification model as a control variable. Moreover, the macro-economic variables namely, GDP, CPI, FOREX represented by time-series data are transformed and simultaneously log of the macro-economic variables are taken into account. The GDP is available on quarterly frequency basis and hence to match the frequency with rest of the variables, quarterly GDP series is converted into monthly frequency by assuming constant economic growth in respective economies for the specific periods respectively. The time-series data pertaining to stock returns on value weighted indexes for the 14 Asian economies and the 11 industry sectorial returns engaging in financial and non-financial activities registered on NSE are obtained in INR (Indian rupee) currency unit. However, for the third phase of economic-growth regression analysis, the respective indexes returns of 14 Asian economies been transformed into log for normalization. The dynamicity in the form of big data analytics of the present study has the significance to proceed with diverse angle of elevation to foresee the impact of stock market liberalization on returns volatility, industrial growth and economic growth. The empirical tests using daily, weekly and monthly frequency data set will provide new and insightful mechanism for better understanding of the behaviour of financial liberalization in economies.

Table 1 Asian stock markets and stock market liberalization: Basic information

Serial No.	Country	Official liberalization/ Stock market liberalization	Literature supporting stock market liberalization	Reason for official liberalization dating
1	China	1991/1992 (January)	Lee and Wong (1992)	-
2	Hong Kong	1994 (August)	Kaminsky & Schmukler (2008)	-
3	India	1992 (May)	Bekaert, Harvey & Lundblad (2005, 2001), Henry (2000), Schündeln & Funke (2001), Gupta & Yuan (2003), Kim & Singal (2000), Bhattacharya & Daouk (2002)	Government announces that foreign portfolio investors will be able to invest directly in listed Indian securities (September).
4	Indonesia	1989 (April)	Bekaert, Harvey & Lundblad (2005, 2001), Henry (2000), Schündeln & Funke (2001), Gupta & Yuan (2002), Kim & Singal (2000), Bhattacharya & Daouk (2002)	Minister of finance allows foreigners to purchase up to 49 % of all companies listing shares on the domestic exchange excluding financial firms (September).
5	Japan	1983 (September)	Bekaert, Harvey & Lundblad (2005), Bhattacharya & Daouk (2002)	Finance Ministry announces easing restrictions on investments by stocks by foreigners (September).
6	Korea, Republic of	1992 (January)	Bekaert, Harvey & Lundblad (2005, 2001), Henry (2000), Schündeln & Funke (2001), Kim & Singal (2000), Bhattacharya Daouk (2002)	Partial opening of the stock market to foreigners. Foreigners can now own up to 10% of domestically listed firms. Five hundred sixty-five foreign investors registered with the Securities Supervisory Board (January).
7	Malaysia	1988 (December)	Bekaert, Harvey & Lundblad (2005), Henry (2000), Schündeln & Funke (2001), Gupta & Yuan (2002), Bhattacharya & Daouk (2002)	Budget calls for liberalization of foreign ownership policies to attract more foreign investors (October).
8	New Zealand	1987	Bekaert, Harvey Lundblad (2005)	Major reforms initiated in 1986.
9	Pakistan	1991 (February)	Schündeln and Funke (2001), Gupta & Yuan (2002), Kim & Singal (2000), Bhattacharya & Daouk (2002), Bae & Mao (2006), Cowitt, Edwards & Boyce (1996b), Jayasuriya (2005)	A new Foreign Investment Law passed in February 1991 allows foreigners to own 100% equity in any industrial or business venture in Pakistan

10	Philippines	1991 (June)	Henry (2000), Schündeln & Funke (2001), Gupta & Yuan (2002), Bhattacharya & Daouk (2002)	Country fund introduction: “The Thornton Philippines Redevelopment Fund Limited” (The Wilson Directory of Market Funds, p. 15).
11	Singapore	1983 ^a	Schündeln & Funke (2001)	-
12	Sri Lanka	1990	Schündeln & Funke (2001), Gupta & Yuan (2002)	-
13	Taiwan	1991 (January)	Henry (2000), Schündeln & Funke (2001), Kim & Singal (2000), Bhattacharya & Daouk (2002)	Opened stock market to foreign investment
14	Thailand	1988 (January)	Henry (2000), Kim & Singal (2000)	Liberalized capital and dividend repatriation

Note: ^a Fully liberalized

Table 2 Data description

Panel A: Data description for estimating impact of stock market liberalization on returns volatility of Asian economies								
Serial No.	Country	Stock market indexes (Indexes Specification)	Phases of the study				Data Frequency	Data Source
			<i>Pre-liberalization period</i>	<i>During-liberalization period</i>	<i>Post-liberalization period</i>	<i>Full period</i>		
1	China	Shanghai Composite Index (SHCOMP Index)	Jan 1991-Dec 1991	Jan 1992-Dec 2007	Jan 2008-Sept 2014	Jan 1991-Sept 2014	Daily, Weekly, Monthly	Bloomberg
2	Hong Kong	Hang Seng Index (HSI Index)	Jan 1975-July 1994	Aug 1994-Dec 2007	Jan 2008-Sept 2014	Jan 1975-Sept 2014	Daily, Weekly, Monthly	Bloomberg
3	India	CNX Nifty Index (NIFTY Index)	Jan 1991-Apr 1992	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1991-Sept 2014	Daily, Weekly, Monthly	Bloomberg
		S&P BSE SENSEX Index (SENSEX Index)	Jan 1980-Apr 1992	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1980-Sept 2014		
4	Indonesia	Jakarta Composite Index (JCI Index)	NA	Dec 1988-Jul 1997	Aug 1997-Sept 2014	Jan 1992-Sept 2014	Daily, Weekly, Monthly	Bloomberg
5	Japan	Nikkei 225 Index (NKY Index)	Jan 1974-Aug 1983	Sept 1983-Jan 1985	Feb 1985-Sept 2014	Jan 1974-Sept 2014	Daily, Weekly, Monthly	Bloomberg
6	Korea	Korea Composite Stock Price Index (KOSPI Index)	Jan 1982-Dec 1990	Jan 1991-Apr 1998	May 1998-Sept 2014	Jan 1982-Sept 2014	Daily, Weekly, Monthly	Bloomberg
7	Malaysia	Kuala Lumpur Composite Index (FBMKLCI Index)	-	July 1973-April 2001	May 2001-Sept 2014	Jan 1978-Sept 2014	Daily, Weekly, Monthly	Bloomberg
8	New Zealand	New Zealand Exchange 50 Gross Index (NZSE50FG Index)	-	-	Jan 2002-Sept 2014	Jan 2002-Sept 2014	Daily, Weekly, Monthly	Bloomberg
9	Pakistan	KARACHI 100 Index (KSE100 Index)	-	Feb 1991-Dec 1993	Jan 1994-Sept 2014	Jan 1992-Sept 2014	Daily, Weekly, Monthly	Bloomberg
10	Philippines	PSE Composite Index (PCOMP Index)	-	Mar 1986-Dec 1993	Jan 1994-Sept 2014	Jan 1992-Sept 2014	Daily, Weekly, Monthly	Bloomberg
11	Singapore	Straits Times Index (FSSTI Index)	-	-	Jan 2000-Sept 2014	Jan 2000-Sept 2014	Daily, Weekly, Monthly	Bloomberg

12	Sri Lanka	Colombo All-Share Price Index (CSEALL Index)	Jan 1986-Oct 1989	Nov 1989-Dec 1994	Jan 1995-Sept 2014	Jan 1986-Sept 2014	Daily, Weekly, Monthly	Bloomberg
13	Taiwan	Taiwan Weighted Index (TWSE Index)	Jan 1984-Dec 1986	Jan 1987-Mar 1998	Apr 1998-Sept 2014	Jan 1984-Sept 2014	Daily, Weekly, Monthly	Bloomberg
14	Thailand	Stock Exchange of Thailand Index (SET Index)	-	Jan 1988-Dec 1989	Jan 1990-Sept 2014	Jan 1988-Sept 2014	Daily, Weekly, Monthly	Bloomberg

Panel B: Data description for estimating impact of stock market liberalization on industrial growth in Indian economy

Serial No.	Industries	Industry specification	Phases of the study				Data Frequency	Data Source
			<i>Pre-liberalization period</i>	<i>During-liberalization period</i>	<i>Post-liberalization period</i>	<i>Full period</i>		
1	Non-financial	Construction and real estate - C&RE	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess
2	Non-financial	Electricity - ELE	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess
3	Non-financial	Manufacturing - MANU	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess
4	Non-financial	Mining - MIN	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess
5	Non-financial	Services (other than financial) - SERV	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess
6	Financial	Asset financing services - AFS	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess
7	Financial	Banking services - BSERV	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess
8	Financial	Fee based financial services - FBFS	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess
9	Financial	Investment services - INVS	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess
10	Financial	Other financial services - OTFS	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess

11	Financial	Other fund based financial services - OFBFS	NA	May 1992-Apr 2008	May 2008-Sept 2014	Jan 1995-Sept 2014	Weekly, Monthly	Prowess
Panel C: Data description for estimating impact of stock market liberalization on economic growth of Asian economies								
Serial No.	Countries	Stock market indexes / macro-economic variable specification	Phases of the study				Data frequency	Data source
			<i>Pre-liberalization period</i>	<i>During-liberalization period</i>	<i>Post-liberalization period</i>	<i>Full period</i>		
1	China	Shanghai Composite Index	Jan 1991-Dec 1991	Jan 1992-Dec 2007	Jan 2008-Jun 2014	Jan 1991-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	Jan 1991-Dec 1991	Jan 1992-Dec 2007	Jan 2008-Jun 2014	Jan 1991-Jun 2014	Quarterly	
		CPI (Consumer Price Index) growth rate	Jan 1991-Dec 1991	Jan 1992-Dec 2007	Jan 2008-Jun 2014	Jan 1991-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	Jan 1991-Dec 1991	Jan 1992-Dec 2007	Jan 2008-Jun 2014	Jan 1991-Jun 2014	Monthly	
2	Hong Kong	Hang Seng Index	Jan 1982-July 1994	Aug 1994-Dec 2007	Jan 2008-Jun 2014	Jan 1982-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	Jan 1982-July 1994	Aug 1994-Dec 2007	Jan 2008-Jun 2014	Jan 1982-Jun 2014	Quarterly	
		CPI (Consumer Price Index) growth rate	Jan 1982-July 1994	Aug 1994-Dec 2007	Jan 2008-Jun 2014	Jan 1982-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	Jan 1982-July 1994	Aug 1994-Dec 2007	Jan 2008-Jun 2014	Jan 1982-Jun 2014	Monthly	
3	India	CNX Nifty Index	Jan 1991-Apr 1992	May 1992-Apr 2008	May 2008-Jun 2014	Jan 1991-Jun 2014	Monthly	Bloomberg
		S&P BSE SENSEX	Jan 1991-Apr 1992	May 1992-Apr 2008	May 2008-Jun 2014	Jan 1980-Jun 2014	Monthly	
		Real GDP growth (%)	Jan 1991-Apr 1992	May 1992-Apr 2008	May 2008-Jun 2014	Jan 1980-Jun 2014	Quarterly	
		CPI (Consumer Price Index) growth rate	Jan 1991-Apr 1992	May 1992-Apr 2008	May 2008-Jun 2014	Jan 1980-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	Jan 1991-Apr 1992	May 1992-Apr 2008	May 2008-Jun 2014	Jan 1980-Jun 2014	Monthly	

4	Indonesia	Jakarta Composite Index	-	-	Jan 2001-Jun 2014	Jan 2001-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	-	-	Jan 2001-Jun 2014	Jan 2001-Jun 2014	Quarterly	
		CPI (Consumer Price Index) growth	-	-	Jan 2001-Jun 2014	Jan 2001-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	-	-	Jan 2001-Jun 2014	Jan 2001 - Jun 2014	Monthly	
5	Japan	Nikkei 225 Index	Jan 1980-Aug 1983	Sept 1983-Jan 1985	Feb 1985-Jun 2014	Jan 1980-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	Jan 1980-Aug 1983	Sept 1983-Jan 1985	Feb 1985- Jun 2014	Jan 1980-Jun 2014	Quarterly	
		CPI (Consumer Price Index) growth rate	Jan 1980-Aug 1983	Sept 1983-Jan 1985	Feb 1985- Jun 2014	Jan 1980-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	Jan 1980-Aug 1983	Sept 1983-Jan 1985	Feb 1985- Jun 2014	Jan 1980-Jun 2014	Monthly	
6	Korea	Korea Composite Stock Price Index or KOSPI	Jan 1982-Dec 1990	Jan 1991-Apr 1998	May 1998-Jun 2014	Jan 1982-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	Jan 1982-Dec 1990	Jan 1991-Apr 1998	May 1998- Jun 2014	Jan 1982-Jun 2014	Quarterly	
		CPI (Consumer Price Index) growth rate	Jan 1982-Dec 1990	Jan 1991-Apr 1998	May 1998- Jun 2014	Jan 1982-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	Jan 1982-Dec 1990	Jan 1991-Apr 1998	May 1998- Jun 2014	Jan 1982-Jun 2014	Monthly	
7	Malaysia	Kuala Lumpur Composite Index	-	Jan 1978-April 2001	May 2001-Jun 2014	Jan 1978-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	-	Jan 1978-April 2001	May 2001-Jun 2014	Jan 1978-Jun 2014	Yearly	
		CPI (Consumer Price Index) growth rate	-	Jan 1978-April 2001	May 2001-S Jun 2014	Jan 1978-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	-	Jan 1978-April 2001	May 2001-Jun 2014	Jan 1978-Jun 2014	Monthly	
8	New Zealand	New Zealand Exchange 50 Gross Index	-	-	Jan 2002-Jun 2014	Jan 2002-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	-	-	Jan 2002-Jun 2014	Jan 2002-Jun 2014	Quarterly	

		CPI (Consumer Price Index) growth rate	-	-	Jan 2002-Jun 2014	Jan 2002-Sept 2014	Quarterly	
		FOREX (Foreign Exchange Rate)	-	-	Jan 2002-Jun 2014	Jan 2002-Jun 2014	Monthly	
9	Pakistan	KSE100 Index	-	-	Jan 1999-Jun 2014	Jan 1999-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	-	-	Jan 1999-Jun 2014	Jan 1999-Jun 2014	Yearly	
		CPI (Consumer Price Index) growth rate	-	-	Jan 1999-Jun 2014	Jan 1999-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	-	-	Jan 1999-Jun 2014	Jan 1999-Jun 2014	Monthly	
10	Philippines	PSE Composite Index	-	-	Jan 1998-Jun 2014	Jan 1998-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	-	-	Jan 1998-Jun 2014	Jan 1998-Jun 2014	Quarterly	
		CPI (Consumer Price Index) growth rate	-	-	Jan 1998-Jun 2014	Jan 1998-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	-	-	Jan 1998-Jun 2014	Jan 1998-Jun 2014	Monthly	
11	Singapore	Straits Times Index	-	-	Jan 2000-Jun 2014	Jan 2000-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	-	-	Jan 2000-Jun 2014	Jan 2000-Jun 2014	Quarterly	
		CPI (Consumer Price Index) growth rate	-	-	Jan 2000-Jun 2014	Jan 2000-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	-	-	Jan 2000-Jun 2014	Jan 2000-Jun 2014	Monthly	
12	Sri Lanka	All Share Price Index	Jan 1986-Oct 1989	Nov 1989-Dec 1994	Jan 1995-Jun 2014	Jan 1986-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	Jan 1986-Oct 1989	Nov 1989-Dec 1994	Jan 1995-Jun 2014	Jan 1986-Jun 2014	Yearly	
		CPI (Consumer Price Index) growth rate	Jan 1986-Oct 1989	Nov 1989-Dec 1994	Jan 1995-Jun 2014	Jan 1986-Jun 2014	Monthly	

		FOREX (Foreign Exchange Rate)	Jan 1986-Oct 1989	Nov 1989-Dec 1994	Jan 1995- Jun 2014	Jan 1986-Jun 2014	Monthly	
13	Taiwan	Taiwan Weighted Index	Jan 1984-Dec 1986	Jan 1987-Mar 1998	Apr 1998-Jun 2014	Jan 1984-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	Jan 1984-Dec 1986	Jan 1987-Mar 1998	Apr 1998-Jun 2014	Jan 1984-Jun t 2014	Quarterly	
		CPI (Consumer Price Index) growth rate	Jan 1984-Dec 1986	Jan 1987-Mar 1998	Apr 1998-Jun 2014	Jan 1984-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	Jan 1984-Dec 1986	Jan 1987-Mar 1998	Apr 1998-Jun 2014	Jan 1984-Jun 2014	Monthly	
14	Thailand	Stock Exchange of Thailand Index	-	-	Jan 1998-Jun 2014	Jan 1998-Jun 2014	Monthly	Bloomberg
		Real GDP growth (%)	-	-	Jan 1998-Jun 2014	Jan 1998-Jun 2014	Quarterly	
		CPI (Consumer Price Index) growth rate	-	-	Jan 1998-Jun 2014	Jan 1998-Jun 2014	Monthly	
		FOREX (Foreign Exchange Rate)	-	-	Jan 1998-Jun 2014	Jan 1998-Jun 2014	Monthly	

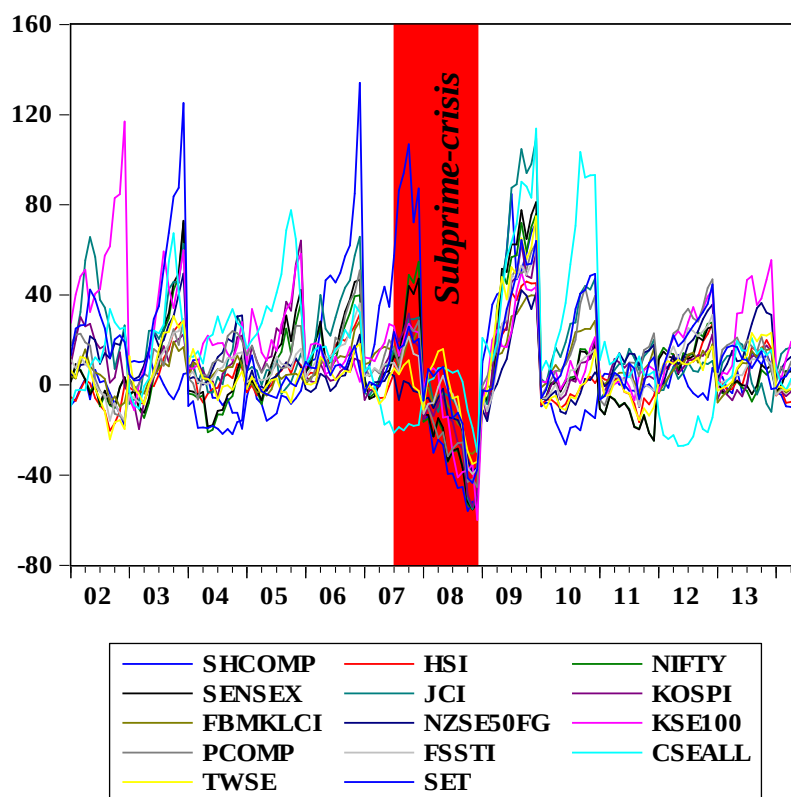
5 Descriptive statistics

The impact of financial stock market liberalization in China has witnessed decrease in stock returns through the period of during- and followed by post-liberalization phase. Table 3 shows the dynamicity persistence in descriptive statistics estimated through daily, weekly and monthly frequency data set. China experienced decrease in stock returns with mean average 63.75, 24.06 and 22.67 in pre-, during and post-liberalization period for the daily, weekly and monthly frequency data set, respectively. However, the mean average stock return for the full period is 19.11 with standard deviation 51.44 present in table 3 of panel A. The stock market liberalization in China sustains decrease in stock returns volatility owing standard deviation of 70.58 in pre-liberalization followed by 55.06 in during- and successively 22.67 in post-liberalization period. Hong Kong subsequently follows the identical trend as China experienced with decrease in stock returns and volatility after stock market liberalization. The average mean for pre-, during, post-liberalization and full periods (23.49, 4.63, -2.31, and 12.16) with standard deviation (31.21, 16.79, 16.23 and 25.42) successively expressed proportionately decrease in returns volatility shown in table 3 of panel A. Simultaneously India follows the same pattern as China and Hong Kong being witnessed with respect to impact of stock market liberalization on returns volatility from dual value weighted indexes average mean returns with standard deviation. The average mean (42.90, 7.96, 4.88 and 8.83) and standard deviation (32.98, 21.16, 25.95 and 24.65) for Nifty index resembles decrease in mean returns through the phases of stock market liberalization owing to decrease in the returns volatility. Proportionate evidence resembles in the value weighted SENSEX index with mean average (17.24, 8.45, 5.48 and 10.87) and standard deviation (27.57, 23.00, 26.54 and 25.26) for pre-, during-, post- liberalization and full period, respectively as shown in table 3 of panel A. In India the mean stock returns and volatility simultaneously high in pre-liberalization period followed by decline in mean returns and volatility after liberalization of financial stock market as witnessed in table 3 of panel A. Conversely, Indonesia experienced an average mean stock returns (13.80, 8.80, and 10.03) with high standard deviation (27.24, 32.56 and 31.39) successively for the phases, during, post-liberalization, and full period, respectively. Thus, the financial stock market liberalization in Indonesia causes increase in stock returns volatility. Table 3 of panel A presents mean average (6.56, 15.30, 16.71, and 7.61) and standard deviation (13.96, 9.20, 26.16 and 19.67) of stock returns in Japan for the pre-, during, post-liberalization and full periods, respectively, witnessed stock market liberalization leads to increase in mean average stock returns

and volatility. Republic of Korea resembles meagre decline in returns volatility in during-liberalization phase, however, rise in post-liberalization period. The average mean stock returns (17.12, 4.57, 8.97, and 10.26) and standard deviation (34.01, 17.41, 23.33 and 26.11) in pre-, during, post-liberalization and full periods, respectively denotes Republic of Korea has experienced mixed impact of stock market liberalization on returns volatility. It witnessed decline in volatility in during-liberalization phase in turn meagre increase in volatility returns in post-liberalization period. Malaysia has witnessed decrease in returns volatility in post-liberalization period with a mean average stock returns (10.54, 5.97, and 11.23) and standard deviation (26.79, 11.66, and 17.05) for during, post-liberalization and full periods, respectively present in table 3 of panel A. New Zealand experienced the average mean stock returns of 7.83 with standard deviation 14.31 for post-liberalization and full periods, respectively denoting persistence of high volatility in stock returns. Pakistan liberalized the stock market and successively perceived increase in average mean stock returns and volatility. The mean average stock returns (-3.33, 10.74, and 9.58) with standard deviation (16.70, 24.27, and 24.05) of Pakistan for during, post-liberalization and full periods, respectively evidenced capital inflow reiterates increase in returns volatility with high pay-offs. Philippines possess mean average returns (38.36, 4.75, and 7.71) with standard deviation (26.83, 21.13, and 23.68) for during, post-liberalization and full periods, respectively which in turn resembles increase in returns volatility after stock market liberalization.

Singapore resembles meager mean returns 3.78 with high degree of variation in returns volatility (16.38) subsequently for the post-liberalization and full periods, respectively. Sri Lanka possess average mean returns (4.00, 38.74, 7.39, 12.59) with standard deviation (19.01, 55.24, 26.60, 35.05) for the pre-, during, post-liberalization phases and full period, respectively as shown in table 3 of panel A. Thus, Sri Lanka experienced great surge in returns volatility with high mean returns in during phase of the liberalization followed by post-liberalization periods (Jaleed and Samarkoon, 2009). Taiwan in line with Sri Lanka follows the identical trend in returns volatility after stock market liberalization. Mean average returns (11.36, 40.02, 4.90 and 18.80) with standard deviation (14.83, 73.48, 16.69, and 49.78) for the pre-, during, post-liberalization phases and full period indicates increase in returns volatility in during-liberalization phase followed by the full period. However, Thailand follows the declining trend in mean returns and returns volatility. The mean average returns (59.07, 7.69, and 11.79) with standard deviation (32.93, 26.09,

and 29.84) for during, post-liberalization phases and full period denotes decline in mean returns and returns volatility as present in table 3 of panel A .



**Fig. 1 Graphical representation of stock returns in 14 Asian economies for the period Jan 2000
- Jun 2014**

Fig. 1 plots the stock returns of 14 Asian stock market returns uniformly for the period January 2000 to June 2014. The period (January 2007 to August 2008) comprises of sub-prime crisis (Rahul and Santhakumar, 2014) instance which is the true intervention in the Asian stock market to unlock the dynamicity in terms of response through mean average and returns volatility. Though Fig. 1 evidently resembles the declining trend in average stock market mean returns approximately in all 14 Asian economies through the period of sub-prime crisis from the end of 2007 sustains till last of 2008.

Table 3 of panel B presents the descriptive statistics pertaining to industrial returns of Indian stock market for the period, post, during-liberalization and full-period employing weekly and monthly

frequency data set. The industry engaging in non-financial activities namely, construction and real estate grossing average mean returns (0.06, 0.20, and 0.11) with standard deviation (2.18, 1.90, and 2.09) in during, post-liberalization and full period. Further, it witnessed increase in mean returns and subsequently decrease in returns volatility in during-liberalization phase.

The rest of the non-financial industries comprising respective mean average returns and standard deviation, namely, electricity (0.065, 0.23, 0.12) and (2.17, 1.65, 2.02), manufacturing (0.04, 0.25, 0.23) and (1.44, 1.36, 1.43), mining (-0.09, 0.28, 0.04) and (2.62, 1.62, 2.36), services (0.05, 0.26, -0.12) and (2.26, 1.58, 2.08), respectively. The five industries engaging in non-financial activities has witnessed increase in mean returns and subsequently decrease in returns volatility comparing relative to during and post-liberalization period with a sizeable standard deviation. However, the industries engaging in financial activities with respective mean returns and standard deviation namely, asset financing services (-0.23, 0.41, 0.10) and (1.89, 1.96, 1.98), banking services (0.02, 0.33, -0.37) and (2.12, 1.37, 2.81), fee based financial services (-1.17, 0.12, -0.25) and (4.05, 1.98, 2.76), investment services (1.20, 0.12, 0.84) and (17.84, 0.95, 29.4), other financial services (1.59, 0.005, -0.32) and (2.20, 1.70, 2.72), other fund based financial services (2.38, 0.07, 0.55) and (17.84, 1.12, 2.28), respectively. Hence, the industries engaging in financial activities witnessed increase in average mean returns and decrease in returns volatility with an exception of investment services, other financial services, and other fund based financial services, in turn experienced decline in average mean returns and returns volatility as shown in table 3 of panel B.

Table 3 Descriptive statistics: stock returns in Asian economies and industries of Indian stock market, macro-economic proxy

Panel A: Descriptive statistics of stock returns in 14 Asian countries																			
Sl. No.	Asian economies		Data frequency	Phases of the study period															
				Pre-liberalization period				During-liberalization period				Post-liberalization period				Full period			
				M/N	Med	S/D	J-B	M/N	Med	S/D	J-B	M/N	Med	S/D	J-B	M/N	Med	S/D	J-B
1	China	SHCOMP Index	D	61.16	32.15	67.50	31.59	23.78	10.60	53.83	45955.47	0.62	-1.33	26.97	321.71	18.93	5.33	50.50	65541.28
			W	62.17	34.76	68.26	6.47	23.76	10.35	54.93	10297.69	-12.71	-12.17	14.24	0.52***	18.96	5.17	50.98	15716.99
			M	67.93	39.24	75.98	1.43***	24.65	10.80	56.42	2031.00	-0.05	-2.62	26.82	16.98	19.45	5.05	52.85	2985.19
2	Hong Kong	HSI Index	D	20.92	16.01	28.64	880.12	3.49	2.40	16.23	266.21	2.35	-0.27	16.77	225.66	11.72	6.59	24.95	4889.52
			W	21.60	16.49	29.30	211.47	6.47	7.30	16.86	13.74	-11.97	-12.06	15.02	0.86***	12.25	6.88	25.32	1220.50
			M	27.97	18.61	35.70	41.98	3.94	2.37	17.29	17.54***	2.67	0.77	16.92	7.91***	12.53	7.36	26.00	246.52
3	India	NIFTY Index	D	39.49	27.85	30.44	23.69	7.08	4.20	20.71	439.57	3.93	1.18	25.94	146.12	7.87	3.98	23.83	1154.12
			W	41.13	31.32	32.96	5.24	8.18	5.29	21.17	129.37	5.37	1.96	25.74	68.81	9.27	5.05	24.54	392.71
			M	48.09	54.96	35.54	0.71***	7.74	4.73	21.62	17.16	5.36	2.14	26.19	5.51	9.36	6.22	25.59	72.22
		SENSEX Index	D	15.89	7.00	26.42	1127.68	7.83	3.74	21.38	402.52	5.13	2.54	25.73	285.86	9.80	4.50	24.30	2396.34
			W	17.24	7.282	27.57	261.03	9.11	2.92	25.37	113.88	5.65	2.04	26.75	100.46	11.30	5.44	25.36	631.58
			M	18.61	7.99	28.74	52.05	8.41	4.82	22.27	15.79	5.67	1.71	27.15	10.28	11.53	5.89	26.13	115.02
4	Indonesia	JCI Index	D	-	-	-	-	13.50	10.14	26.48	1182.67	8.26	6.60	32.39	386.92	9.55	7.51	31.12	894.88
			W	-	-	-	-	13.82	9.84	27.28	273.03	9.36	7.21	32.08	129.32	10.46	7.89	31.02	273.47
			M	-	-	-	-	14.09	10.42	27.97	81.14	8.80	7.83	33.23	13.74	10.10	8.32	32.05	42.65
5	Japan	NKY Index	D	6.30	6.44	13.79	138.72	14.65	14.56	8.76	21.59	20.62	15.98	29.00	36.14	7.35	3.95	19.50	4152.15
			W	6.29	6.27	13.79	27.21	14.92	15.09	8.94	4.18	7.55	3.11	21.03	563.99	7.51	4.22	19.29	888.71
			M	7.09	7.66	14.31	5.89	16.34	16.12	9.92	1.01	21.96	17.17	28.46	4.23***	7.97	4.72	20.24	179.61
6	Republic of Korea	KOSPI Index	D	16.26	6.77	33.24	583.38	4.75	3.33	16.72	470.21	8.83	6.42	23.04	1489.02	10.06	5.45	25.71	4845.35
			W	16.85	7.83	33.38	93.84	4.32	2.82	16.75	90.53	9.01	6.19	22.99	366.30	10.10	5.40	25.55	941.33
			M	18.27	8.65	35.41	22.15	4.66	3.02	18.78	16.43	9.09	6.40	23.98	82.85	10.62	6.08	27.08	211.90
7	Malaysia	FBMKLCI Index	D	-	-	-	-	10.12	6.86	26.54	67.10	5.68	4.81	11.46	419.42	10.10	7.24	22.13	775.91
			W	-	-	-	-	10.98	7.17	26.31	14.91	6.14	5.27	11.78	117.68	10.73	7.47	22.01	157.27
			M	-	-	-	-	10.52	7.94	27.53	4.80	6.10	5.36	11.75	12.93	12.87	13.24	7.02	26.88
8	New Zealand	NZSE50FG Index	D	-	-	-	-	-	-	-	-	7.23	5.83	13.94	233.57	7.23	5.83	13.94	233.57
			W	-	-	-	-	-	-	-	-	8.12	6.43	14.45	82.92	8.12	6.43	14.45	82.92
			M	-	-	-	-	-	-	-	-	8.15	7.16	14.56	6.81	8.15	7.16	14.56	6.81
9	Pakistan	KSE100 Index	D	-	-	-	-	-3.71	-3.62	16.47	286.65	11.10	10.50	23.88	206.73	10.09	8.98	23.75	191.02
			W	-	-	-	-	-2.29	-2.61	14.80	148.31	10.26	9.99	23.94	68.47	9.15	7.85	23.55	82.72
			M	-	-	-	-	-3.99	-4.70	18.85	31.23	10.88	10.76	25.01	20.23	9.5	8.55	24.87	20.62
10	Philippines	PCOMP Index	D	-	-	-	-	37.56	33.56	24.22	613.53	4.42	5.84	20.75	189.87	7.34	7.58	23.08	1185.38
			W	-	-	-	-	36.21	32.08	25.18	133.70	4.96	6.35	20.98	41.08	7.73	7.99	23.15	275.18
			M	-	-	-	-	41.31	35.18	31.11	52.61	4.88	6.45	21.66	9.59***	8.08	9.21	24.82	240.61
11			D	-	-	-	-	-	-	-	-	3.40	3.41	16.01	430.48	3.40	3.41	16.01	430.48

	Singapore	FSSTI Index	W	-	-	-	-	-	-	-	-	4.16	3.83	16.49	153.78	4.16	3.83	16.49	153.78
			M	-	-	-	-	-	-	-	-	3.78	3.93	16.65	18.84	3.78	3.93	16.65	18.84
12	Sri Lanka	CSEALL Index	D	3.96	6.98	19.45	3234.38	37.84	13.21	54.76	249.78	7.06	3.21	26.08	2186.23	12.22	4.95	34.71	11576.67
			W	3.49	6.96	18.54	0.24***	38.34	12.47	53.67	57.54	7.45	3.37	26.42	602.05	12.61	5.47	34.62	2439.46
			M	4.56	7.21	19.06	0.09***	40.04	14.35	57.30	11.97	7.66	4.06	27.30	113.28	12.95	6.23	35.92	599.26
13	Taiwan	TWSE Index	D	12.22	16.81	19.87	58.60	39.35	18.60	72.76	4343.67	4.51	2.30	16.43	1764.11	18.93	7.51	50.17	8210.19
			W	12.30	16.67	19.85	10.04	39.67	20.17	71.91	910.83	5.59	3.53	16.58	830.03	18.77	8.04	48.38	19645.17
			M	9.57	8.15	4.78	2.56***	41.04	20.79	75.79	221.30	4.62	2.51	17.08	84.36	18.72	7.47	50.81	4821.90
14	Thailand	SET Index	D	-	-	-	-	58.33	62.16	32.21	7.80	7.59	5.62	25.56	465.93	11.38	7.28	29.33	1143.69
			W	-	-	-	-	57.46	59.49	32.74	2.43	8.30	6.15	25.63	107.53	12.00	7.73	29.25	267.80
			M	-	-	-	-	61.44	63.61	33.86	1.28	8.01	6.38	27.08	44.03	12.00	7.71	30.96	79.82

Panel B: Descriptive statistics of stock returns in industrial sector of Indian stock market

Sl. No.	Industry	Industry specification	Data frequency	Pre-liberalization period				During-liberalization period				Post-liberalization period				Full period			
				M/N	Med	S/D	J-B	M/N	Med	S/D	J-B	M/N	Med	S/D	J-B	M/N	Med	S/D	J-B
1	Non-financial	C&RE	W	NA	NA	NA	NA	0.007	0.13	2.11	2.9.85	0.025	0.05	1.99	2708.39	0.01	0.09	2.07	1450.76
			M	NA	NA	NA	NA	0.13	0.23	2.25	26.13	0.38	0.33	1.82	10.71	0.21	0.25	2.12	45.22
2	Non-financial	ELE	W	NA	NA	NA	NA	0.12	0.01	2.15	514.99	0.15	0.15	1.79	370.44	0.13	0.03	2.04	892.70
			M	NA	NA	NA	NA	0.01	0.12	2.20	47.35	0.32	0.41	1.51	12.04	0.11	0.21	2.01	95.31
3	Non-financial	MANU	W	NA	NA	NA	NA	-0.01	0.040	1.35	468.11	0.02	0.10	1.57	323.65	-0.003	0.05	1.43	866.39
			M	NA	NA	NA	NA	0.09	0.25	1.54	28.04	0.49	0.49	1.16	94.67	0.22	0.32	1.44	74.40
4	Non-financial	MIN	W	NA	NA	NA	NA	-0.06	-0.06	2.53	573.01	-0.0005	0.02	1.67	1754.71	-0.04	-0.03	2.28	1466.90
			M	NA	NA	NA	NA	-0.12	0.06	2.72	19.20	0.66	0.58	1.58	1.70***	0.13	0.32	2.44	54.98
5	Non-financial	SERV	W	NA	NA	NA	NA	-0.09	0.03	2.14	326.50	0.01	0.06	1.86	95.38	-0.54	0.03	2.06	468.24
			M	NA	NA	NA	NA	0.19	0.24	2.38	35.14	0.51	0.43	1.31	21.64	0.30	0.36	2.10	100.58
6	Financial	AFS	W	NA	NA	NA	NA	-0.14	-0.20	1.81	283.67	0.12	0.11	2.05	251.52	0.05	-0.01	2.00	555.66
			M	NA	NA	NA	NA	-0.09	-0.11	1.97	39.13	0.70	0.58	1.87	131.82	0.16	0.11	1.97	125.75
7	Financial	BSERV	W	NA	NA	NA	NA	0.02	-0.06	1.97	784.48	-0.03	0.03	0.93	152.95	-0.98	-0.74	3.47	855.88
			M	NA	NA	NA	NA	0.02	0.02	2.28	58.51	0.70	0.55	1.81	2.31***	0.24	0.31	2.16	77.89
8	Financial	FBFS	W	NA	NA	NA	NA	-1.38	-1.23	3.92	2016.52	0.05	0.13	1.99	525.09	-0.07	-0.11	1.87	4478.10
			M	NA	NA	NA	NA	-0.96	-1.00	4.19	243.56	0.19	0.18	1.97	5.75	-0.58	-0.25	3.66	629.50
9	Financial	INVS	W	NA	NA	NA	NA	1.76	0.03	32.65	133380329	-0.05	-0.01	0.84	62.88	1.16	0.01	26.83	43592954
			M	NA	NA	NA	NA	0.64	0.19	3.03	11.11	0.30	0.37	1.07	300.19	0.53	0.23	2.57	66.20

10	Financial	OTFS	W	NA	NA	NA	NA	0.95	0.95	2.22	259.93	-0.15	-0.20	2.05	426.10	-0.98	-0.74	3.47	505.13
			M	NA	NA	NA	NA	0.90	1.01	2.18	40.07	0.16	0.11	1.36	13.31	0.66	0.60	1.98	83.83
11	Financial	OFBFS	W	NA	NA	NA	NA	1.75	0.03	32.65	13381448	-0.05	0.03	1.16	335.31	0.62	0.44	1.99	43608835
			M	NA	NA	NA	NA	0.63	0.15	3.03	11.20	0.20	0.10	1.08	551.67	0.49	0.12	2.57	69.18
Panel C: Descriptive statistics of macro-economic variables of 14 economies																			
Sl. No.	Country/Variable specification		Data frequency	Pre-liberalization period				During-liberalization period				Post-liberalization period				Full period			
				M/N	Med	S/D	J-B	M/N	Med	S/D	J-B	M/N	Med	S/D	J-B	M/N	Med	S/D	J-B
1	China	GDP	M	9.20	9.20	-	-	9.90	9.75	2.16	8.31	8.67	8.40	1.37	3.11***	9.53	9.50	1.99	18.54
		CPI	M	3.55	4.40	1.40	1.73***	5.28	2.00	7.32	83.28	3.13	2.85	2.42	0.04	4.61	2.70	6.25	290.49
		FOREX	M	5.33	5.36	0.07	1.38***	7.91	8.27	0.88	187.78	6.54	6.54	0.31	5.24***	7.42	8.09	1.06	33.88
2	Hong Kong	GDP	M	6.46	6.30	4.59	4.14***	3.96	4.80	4.18	23.37	2.61	2.90	3.55	13.87	4.65	5.20	4.49	15.43
		CPI	M	8.39	9.00	2.72	11.59	1.32	0.90	4.33	9.83	3.51	3.70	1.88	3.64***	4.49	4.55	4.64	16.44
		FOREX	M	7.60	7.79	0.50	387.18	7.77	7.77	0.02	15.31	7.76	7.75	0.01	12.66	7.70	7.77	0.32	9452.57
3	India	GDP	M	0.16	-0.93	1.95	3.85***	6.18	5.63	2.29	7.08	6.64	5.90	2.11	5.12	5.96	5.64	2.64	6.00
		CPI	M	13.8	13.60	1.16	0.88***	6.67	5.65	3.36	37.58	9.98	9.73	2.13	5.74	7.94	8.22	3.60	6.71
		FOREX	M	23.86	25.79	3.54	1.58***	40.42	42.81	6.23	18.21	50.77	48.77	6.02	6.97	42.39	44.07	8.93	2.04***
4	Indonesia	GDP	M	-	-	-	-	-	-	-	-	-	-	-	-	5.44	5.76	1.04	56.68
		CPI	M	-	-	-	-	-	-	-	-	-	-	-	-	83.92	83.81	22.65	11.07
		FOREX	M	-	-	-	-	-	-	-	-	-	-	-	-	9529.60	9180.50	915.11	50.45
5	Japan	GDP	M	3.74	3.00	3.22	0.93***	4.488	4.70	2.39	0.26***	1.77	1.70	4.63	28.89	2.09	2.05	4.50	41.27
		CPI	M	0.31	0.20	0.63	1.32***	0.25	0.30	0.65	0.36***	0.04	0.00	0.39	317.49	0.08	0.05	0.44	202.51
		FOREX	M	233.12	234.04	16.47	0.65***	238.23	235.03	8.19	0.53***	118.99	115.86	29.56	1002.36	136.02	119.26	49.58	115.32
6	Republic of Korea	GDP	M	10.06	9.85	2.83	4.09***	6.97	7.30	3.14	234.52	4.38	4.50	3.63	49.20	6.54	6.60	4.09	45.73
		CPI	M	0.37	0.30	0.53	9.29***	0.51	0.50	0.54	51.37	0.22	0.20	0.38	0.93***	0.33	0.30	0.48	123.43
		FOREX	M	779.84	783.53	71.33	6.92	843.28	797.43	181.67	693.60	1136.76	1136.44	119.86	1.50***	971.70	928.23	208.44	21.98
7	Malaysia	GDP	M	-	-	-	-	3.88	4.89	3.96	357.77	4.92	5.50	2.87	165.64	4.26	5.15	3.63	702.51
		CPI	M	-	-	-	-	58.51	55.46	12.66	12.61	94.48	93.81	8.87	12.35	71.49	70.83	20.72	26.39
		FOREX	M	-	-	-	-	2.68	2.54	0.51	135.58	3.48	3.51	0.29	16.72	2.97	2.71	0.58	43.09
8	New Zealand	GDP	M	-	-	-	-	-	-	-	-	-	-	-	-	2.53	2.90	1.96	14.21
		CPI	M	-	-	-	-	-	-	-	-	-	-	-	-	0.60	0.55	0.51	17.23
		FOREX	M	-	-	-	-	-	-	-	-	-	-	-	-	0.69	0.70	0.10	11.05

9	Pakista n	GDP	M	-	-	-	-	-	-	-	-	-	-	-	-	113.73	116.27	20.96	13.80
		CPI	M	-	-	-	-	-	-	-	-	-	-	-	-	74.40	61.11	32.13	22.57
		FOREX	M	-	-	-	-	-	-	-	-	-	-	-	-	70.67	60.72	16.33	21.94
10	Philippi nes	GDP	M	-	-	-	-	-	-	-	-	-	-	-	-	4.57	4.85	2.31	19.58
		CPI	M	-	-	-	-	-	-	-	-	-	-	-	-	4.93	4.45	2.22	21.96
		FOREX	M	-	-	-	-	-	-	-	-	-	-	-	-	47.13	46.26	5.37	13.04
11	Singapo re	GDP	M	-	-	-	-	-	-	-	-	-	-	-	-	5.65	5.80	5.14	4.29***
		CPI	M	-	-	-	-	-	-	-	-	-	-	-	-	0.17	0.15	0.55	9.73
		FOREX	M	-	-	-	-	-	-	-	-	-	-	-	-	1.52	1.53	0.20	15.99
12	Sri Lanka	GDP	M	1.19	1.08	0.86	4.35***	4.18	4.15	1.13	2.69	4.79	5.02	2.03	0.98***	4.19	4.15	2.15	1.59***
		CPI	M	10.05	9.83	1.19	3.65***	18.03	17.95	2.86	3.71	64.69	53.02	32.76	23.12	48.88	35.48	35.82	41.55
		FOREX	M	2.06	2.04	0.09	3.66***	1.67	1.64	0.11	2.54***	1.52	1.52	0.18	17.65	1.62	1.63	0.24	9.36
13	Taiwan	GDP	M	0.01	0.00	1.30	68.82	7.22	6.73	1.98	12.41	3.91	4.69	4.14	28.82	5.54	6.02	3.84	120.34
		CPI	M	0.10	-0.01	0.70	0.25***	0.008	-0.02	1.34	0.58***	1.04	1.02	1.44	8.42	1.68	1.44	1.86	11.76
		FOREX	M	-0.12	-0.15	0.24	0.31***	27.53	27.16	2.02	123.06	-0.01	-0.01	0.39	16.3	-0.02	-0.026	0.39	233.54
14	Thailan d	GDP	M	-	-	-	-	-	-	-	-	-	-	-	-	3.18	4.55	5.36	96.86
		CPI	M	-	-	-	-	-	-	-	-	-	-	-	-	2.80	2.52	2.39	42.35
		FOREX	M	-	-	-	-	-	-	-	-	-	-	-	-	36.89	37.10	4.79	7.80

Note: ^a M/N represents Mean

^b Med represents Median

^c S/D represents Standard Deviation

^d J-B represents Jarque-Bera

^e *, **, and *** indicates significance at 10, 5 and 1 % level, respectively

Table 3 of panel C presents descriptive statistics of macro-economic variables namely, gross domestic product (GDP), consumer price index (CPI) and foreign exchange rate (FOREX) as a measure of economic growth for 14 Asian economies. The impact of financial stock market liberalization has witnessed a decline in economic growth of China. The GDP (9.20, 9.90, 8.67, and 9.53) of China for pre-, during, post-liberalization and full period has experienced no impact of stock market liberalization as the mean (9.90) of GDP in during-liberalization has decline to 8.67 for the post-liberalization period. The CPI of China follows the similar trend as the GDP with mean 3.55 in pre-liberalization period followed by decline to 3.13 in post-liberalization period. However, the relative foreign exchange rate of China has assured increment from pre-liberalization (5.33) period to post-liberalization (6.54) period as per table 3 of panel C.

The GDP (6.46, 3.96, 2.61 and 4.65) with standard deviation (4.59, 4.18, 3.55, and 4.49) of Hong Kong has shown decreasing tendency for pre-, during, post-liberalization period respectively followed by full period. However, CPI with average mean (8.39, 1.32, 3.51, and 4.49) with standard deviation (2.72, 4.33, 1.88, and 4.64) depicts decrease in price index with meager deviation. The FOREX of Hong Kong possess mean (7.60, 7.77, 7.76 and 7.70) with inadequate standard deviation as shown in figure 3 of panel C. India experienced a higher impact of financial stock market liberalization on economic growth justifying with the mean (0.16, 6.18, 6.64, and 5.96) and meager standard deviation (1.95, 2.29, 2.11, and 2.64). The consumer price index of India depicts mean (13.8, 6.67, 9.98, and 7.94) surge in post-liberalization period with meager standard deviation (1.16, 3.36, 2.13, and 3.60). Subsequently the relative FOREX mean (23.86, 40.42, 50.77, and 42.39) of India for pre, during, post-liberalization and full period has increased with standard deviation (3.54, 6.23, 6.02, and 8.93).

The GDP of Indonesia depicts mean 5.44 with standard deviation 1.04 as shown in table 3 of panel C. The high mean 83.92 of CPI with high standard deviation 22.65 followed by the FOREX mean 9529.60 for the full period. The GDP mean (3.74, 4.48, 1.77, and 2.09) growth rate of Japan for the pre, during, post-liberalization and full period witnessed decline tendency with least standard deviation (3.22, 2.39, 4.63, and 4.50). The mean (0.31, 0.25, 0.04, and 0.08) CPI of Japan indicates meagre increment in price of consumer goods with least standard deviation (0.63, 0.65, 0.39, and 0.44) for the multi-liberalization phase along with full period. Subsequently the relative FOREX mean (233.12, 238.23, 118.99 and 136.02) with standard deviation (16.47, 8.19, 29.56 and 49.58)

witnessed declining trend in post-liberalization period as compared to during-liberalization period. The GDP mean (10.06, 6.97, 4.38, and 6.54) growth rate increased after stock market liberalization with a meagre standard deviation (2.83, 3.14, 3.63, and 4.09) for the periods of pre-, during, and post-liberalization and full period. CPI shown least increment in all the phase of liberalization periods with less deviation. However, the relative FOREX mean (233.12, 238.23, 118.99, and 136.04) witnessed decline in post-liberalization period successively with increase in standard deviation (16.47, 8.19, 29.56, and 49.58). The GDP (10.06, 6.97, 4.38, and 6.54) of Republic of Korea for the pre-, during, post-liberalization and full period surges after stock market liberalization with standard deviation (2.83, 3.14, 3.63 and 4.09). However, The CPI in line with GDP follows declining tendency for the phases of liberalization period with mean (0.37, 0.51, 0.22, and 0.33) and standard deviation (0.53, 0.54, 0.38, and 0.48). The FOREX relatively increased for pre-, during, and post-liberalization along with full period, respectively with mean (779.84, 843.28, 1136.76, and 971.70) and standard deviation (71.33, 181.67, 119.86, and 208.44).

The mean GDP (3.88, 4.92, and 4.26) and standard deviation (3.96, 2.87, and 3.63) of Malaysia for the pre-, during, post-liberalization and full period witnessed increment after stock market liberalization. However, CPI with mean (58.51, 94.48, 71.49) and standard deviation (12.66, 8.87, 20.72) experienced increment in consumer prices index with decrease in standard deviation. The relative FOREX mean (2.68, 3.48, and 2.97) and standard deviation (0.51, 0.29, and 0.58) supports that after stock market liberalization foreign exchange rate has increased with decline in standard deviation. The GDP mean (2.53) of New Zealand with standard deviation (1.96) for the full period indicates nominal economic growth. The CPI and relative FOREX mean (0.60 and 0.69) with standard deviation (0.51 and 0.10) witnessed increase in variation in CPI with meagre deviation in FOREX. Pakistan average mean GDP (113.73) with standard deviation (20.96) experienced stable economic growth with minimum variations. CPI and FOREX mean (74.40 and 70.67) with standard deviation (32.13 and 16.33) indicates high variations in CPI of Pakistan with minimum variations in relative FOREX. Philippines mean GDP (4.57) with standard deviation (2.31) witnessed nominal economic growth in full period. However, the CPI and relative FOREX mean (4.93 and 47.13) with standard deviation (2.22 and 5.37) subsequently indicates substantial increment in mean average with minimal variations. The mean (5.65) with standard deviation (5.14) signifies high variations in economic growth in Singapore. However, the mean of CPI and

FOREX (0.17 and 1.52) with standard deviation (0.55 and 0.20) indicates high variations in CPI successively minimum in relative FOREX for the full period.

The mean (1.19, 4.18, 4.79, and 4.19) GDP with standard deviation (0.86, 1.13, 2.03, and 2.15) of Sri Lanka for the pre-, during, post-liberalization and full period witnessed a continuous growth subsequently increase in variation. The mean CPI (10.05, 18.03, 64.69, and 48.88) with standard deviation (1.19, 2.86, 32.76, and 35.82) indicates increment and subsequently larger size of variations. However, relative FOREX mean (2.06, 1.67, 1.52, and 1.62) and standard deviation (0.09, 0.11, 0.18, and 0.24) signifies decline in relative foreign exchange rate with increase in variations. The mean GDP (0.01, 7.22, 3.91, and 5.54) and standard deviation (1.30, 1.98, 4.14, and 3.84) of Taiwan signifies increment in economic growth after stock market liberalization yet a meagre decline in post-liberalization period with increment in variations of GDP. The mean (0.10, 0.008, 1.04, 1.68) CPI of Taiwan witnessed surge with increasing tendency in variations (0.70, 1.34, 1.44, and 1.86). The relative FOREX mean (-0.12, 27.53, -0.01, and -0.02) and variations (0.24, 2.02, 0.39, and 0.39) expressed stable increment in foreign exchange rate leads to decline in variations. Thailand mean 3.18 GDP and standard deviation 5.36 for the full period represent meager growth with high variations. The mean 2.80 CPI with standard deviation 2.39 reflects consumer price sustainability and considerable increment in variations. The relative mean 36.89 and standard deviation 4.79 of FOREX for the full period reveals meager increment with large variations in foreign exchange rate.

Table 4 Covariance-correlation matrix of 14 Asian stock returns for the period Jan 2002-May 2014

Covariance Correlation	SHCOMP	HSI	NIFTY	SENSEX	JCI	KOSPI	FBMKLCI	NZSE50FG	KSE100	PCOMP	FSSTI	CSEALL	TWSE	SET
SHCOMP	-	334.28	471.81	484.58	439.16	274.73	233.63	128.16	154.84	332.40	287.11	63.41	264.81	211.28
HSI	0.72	-	304.98	316.82	264.66	199.54	140.92	144.53	150.05	220.21	206.67	144.99	203.84	205.12
NIFTY	0.66	0.88	-	544.56	457.35	319.87	208.97	194.92	270.05	342.23	301.79	276.43	28,6.15	381.90
SENSEX	0.65	0.89	0.99	-	479.28	334.48	213.98	206.46	280.16	352.09	314.69	314.95	302.03	385.16
JCI	0.55	0.69	0.77	0.78	-	319.49	230.11	217.82	268.54	350.92	298.16	430.00	293.84	431.00
KOSPI	0.50	0.75	0.78	0.79	0.70	-	150.82	168.25	267.56	240.82	212.02	244.87	202.35	229.29
FBMKLCI	0.64	0.80	0.77	0.76	0.76	0.72	-	101.90	111.40	188.80	152.02	158.44	138.35	167.82
NZSE50FG	0.28	0.66	0.57	0.59	0.57	0.64	0.59	-	237.33	178.23	149.74	122.30	148.68	207.90
KSE100	0.21	0.43	0.50	0.50	0.44	0.64	0.40	0.69	-	185.22	145.95	171.79	140.90	290.85
PCOMP	0.57	0.79	0.79	0.79	0.73	0.73	0.86	0.65	0.42	-	243.87	216.61	204.15	259.26
FSSTI	0.61	0.92	0.87	0.88	0.77	0.79	0.86	0.68	0.41	0.87	-	185.00	212.87	221.21
CSEALL	0.07	0.34	0.42	0.46	0.58	0.48	0.47	0.29	0.25	0.40	0.43	-	200.36	252.51
TWSE	0.51	0.82	0.74	0.76	0.68	0.68	0.71	0.61	0.36	0.66	0.85	0.42	-	233.644
SET	0.29	0.59	0.71	0.69	0.72	0.55	0.61	0.61	0.53	0.60	0.63	0.38	0.60	-

Note: ^a Covariance are in **bold** face

Table 4 presents the covariance-correlation matrix of Asian stock market for the period January 2002 to May 2014. The SHCOMP stock index consists of high correlation coefficients (0.72, 0.66, 0.65, 0.64, 0.61, and 0.57) and covariances (334.28, 471.81, 484.58, 233.63, 287.11, and 332.40) with HSI, Nifty, SENSEX, FBMKLCI, FSSTI, and PCOMP. The HSI index shows high correlation (0.88, 0.89, 0.69, 0.75, 0.80, 0.79, 0.92, 0.82) and covariance coefficients (304.98, 316.82, 264.66, 199.54, 140.92, 220.21, 206.67, 203.84) with Nifty, SENSEX, JCI, KOSPI, FBMKLCI, PCOMP, FSSTI, and TWSE stock indexes as shown in Table 4. The Nifty index has shown high correlation (0.77, 0.78, 0.77, 0.79, 0.87, 0.74, 0.71) and covariances (457.35, 319.87, 208.97, 342.23, 301.79, 286.15, 381.90) with JCI, KOSPI, FBMKLCI, PCOMP, FSSTI, TWSE, and SET indexes and subsequently emphasizes that the Indian stock market is highly integrated with major stock indices in the Asian region. However, in line with the above result of Nifty index, the SENSEX stock index also unveils the identical result of stock market integration as of correlation coefficients (0.78, 0.79, 0.76, 0.79, 0.88, 0.76, and 0.69) and covariances (479.28, 334.48, 213.98, 352.09, 314.69, 302.03, and 385.16) with JCI, KOSPI, FBMKLCI, PCOMP, FSSTI, TWSE, and SET indexes. Hence, there exists a strong indication that the Indian stock market is highly integrated with the rest of the major Asian stock markets.

The JCI stock index possesses high correlation (0.69, 0.77, 0.78, 0.76, 0.73, 0.77, and 0.72) and covariances (264.66, 457.35, 479.28, 230.11, 350.92, 298.16, and 431.00) with HSI, Nifty, SENSEX, FBMKLCI, PCOMP, FSSTI, SET indexes; however, the rest of the indexes follows a very meagre association with the JCI index. The KOSPI index accounts for correlation (0.75, 0.78, 0.79, 0.72, 0.73, 0.79) and covariances (199.54, 319.87, 334.48, 150.82, 240.82, 212.02) with HSI, Nifty, SENSEX, FBMKLCI, PCOMP, and FSSTI, subsequently signifying higher stock market integration. However, the rest of the seven stock indexes possess scanty association with the KOSPI index. FBMKLCI index has correlation (0.80, 0.77, 0.76, 0.76, 0.72, 0.86, 0.86) and covariances (140.92, 208.97, 213.98, 230.11, 150.82, 188.80, 150.02) with HSI, Nifty, SENSEX, JCI, KOSPI, PCOMP, and FSSTI indexes, which reflects high market integration among the Asian stock markets. NZSE50FG index accounts for an average correlation coefficient (0.64) and covariance (176.38) with HSI, KOSPI, KSE100, PCOMP, FSSTI, TWSE, and SET indexes. Successively, the rest of the six Asian stock indices signifies meagre association with NZSE50FG.

The KSE100 index implies an average correlation (0.66) and covariance (252.44) with KOSPI and NZSE50FG indexes; however, the rest of the Asian stock indices are meagerly correlated. PCOMP

stock index indicates an average correlation (0.79) and covariance (257.92) with HSI, Nifty, SENSEX, JCI, KOSPI, FBMKLCI, NZSE50FG, FSSTI, and TWSE as shown in the table 4. Though, the rest of Asian stock indexes signifies low associationship with PCOMP. FSSTI stock index average correlation coefficient (0.83) and covariance (232.42) with HSI, Nifty, SENSEX, JCI, KOSPI, FBMKLCI, NZSE50FG, FSSTI, and TWSE signifies a strong associationship. CSEALL stock index denotes a significant correlation (0.58) and covariance (430.00) with JCI stock index which implies stock market integration. However, rest of the Asian stock indexes possess meager associationship with CSEALL stock index. TWSE stock index indicates an average correlation coefficient (0.74) and covariance (234.20) with HSI, Nifty, SENSEX, FBMKLCI, FSSTI, JCI, and KOSPI as in table 4. The SET stock index denotes an average correlation coefficient (0.70) and covariance (399.35) with Nifty and SENSEX stock indexes. It reveals that there exist a high market integration among SET index and Indian stock indexes.

Table 5 Covariance-correlation matrix of 11 sectorial industrial returns for the period Jan 1995-Sept 2014

Covariance Correlation	C&RE	ELE	MANU	MIN	SERV	AFS	BSERV	FBFS	INVS	OTFSERV	OFBFSERV
C&RE	-	2.87	2.47	2.61	3.21	3.06	3.39	3.55	0.58	2.15	0.52
ELE	0.67	-	2.06	2.59	2.28	2.43	2.96	2.45	0.80	1.25	0.73
MANU	0.81	0.71	-	2.08	2.13	2.21	2.48	2.40	0.61	1.40	0.55
MIN	0.50	0.52	0.59	-	2.37	2.23	2.72	1.75	0.67	1.32	0.58
SERV	0.72	0.54	0.70	0.46	-	2.52	2.75	3.97	0.37	2.33	0.33
AFS	0.73	0.61	0.78	0.46	0.61	-	3.35	2.95	0.62	1.35	0.50
BSERV	0.74	0.68	0.79	0.51	0.60	0.78	-	3.51	1.26	1.71	1.19
FBFS	0.45	0.33	0.45	0.19	0.51	0.41	0.44	-	0.80	2.92	0.73
INVS	0.10	0.15	0.16	0.10	0.07	0.12	0.22	0.08	-	0.37	6.55
OTFSERV	0.51	0.31	0.49	0.27	0.56	0.34	0.40	0.40	0.07	-	0.39
OFBFSERV	0.09	0.14	0.14	0.09	0.06	0.10	0.21	0.07	0.99	0.07	-

Note: ^a Covariance are in **bold** face

Table 5 presents the covariance-correlation matrix of sectorial industrial returns for the period from January 1995 through September 2014. The C&RE sectorial index indicates higher correlation coefficient (0.67, 0.81, 0.72, 0.73, and 0.74) and covariance (2.87, 2.47, 3.21, 3.06, and 3.39) with ELE, MANU, SERV, AFS, and BSERV indexes, respectively. ELE sectorial index denotes higher correlation coefficient (0.67, 0.71, and 0.68) and covariance (2.87, 2.06, and 2.96) with ELE, MANU and BSERV sectorial indexes. MANU sectorial index signifies correlation (0.81, 0.71, 0.70, 0.78, 0.79) and covariance (2.47, 2.06, 2.13, 2.21, and 2.48) with C&RE, ELE, SERV, AFS, and BSERV sectorial indexes. MIN sectorial index denotes high correlation (0.59) and covariance (2.08) with MANU sectorial index as shown in table 5, however, rest of the sectorial indexes possess meager associationship with MIN index. SERV sectorial index specifies correlation coefficient (0.72, 0.70, and 0.61) and covariance (3.21, 2.13, and 2.52) with C&RE, MANU, and AFS sectorial indexes. AFS sectorial index possess high correlation coefficient (0.73, 0.78, 0.61 and 0.78) and covariance (3.06, 2.21, 2.52, and 3.35) with C&RE, MANU, SERV, and BSERV indexes.

Table 5 shows BSERV sectorial stock index indicates correlation coefficient (0.74, 0.68, 0.79, and 0.78) and covariance (3.39, 2.96, 2.48, and 3.35) with C&RE, ELE, MANU, and AFS sectorial stock indexes. FBFS sectorial stock index signifies low correlation and covariance with all the financial and non-financial sectorial stock indexes. However, INVS sectorial stock index denotes significant high correlation coefficient (0.99) and covariance (6.55) with OFBFSERV sectorial stock index. OTFSERV sectorial stock index indicates low correlation coefficient and covariance with rest of the sectorial stock indexes as shown in table 5.

Table 6 Variance-covariance matrix of macro-economic variables for the period Jan 1991-Sept 2014

Covariance Correlation	NIFTY	SENSEX	GDP	CPI	FOREX
NIFTY	-	683.78	-8.98	10.96	-57.93
SENSEX	0.99	-	-9.72	14.40	-63.66
GDP	-0.13	-0.13	-	-2.20	9.17
CPI	0.11	0.14	-0.23	-	-8.88
FOREX	-0.25	-0.27	0.39	-0.28	-

Note: ^a Covariance are in **bold** face

Table 6 represents the covariance-variance matrix of macro-economic variables for the period January 1991 to September 2014. The Nifty and SENSEX indexes indicates meager negative correlation (-0.13) with GDP. The Nifty and SENSEX indexes further shows meagre negative correlation (-0.25 and -0.27) with FOREX. However, CPI shows low positive correlation (0.11 and 0.14) with Nifty and SENSEX indexes. GDP and CPI are negatively correlated (-0.23) and subsequently GDP and FOREX are positively correlated (0.39). CPI and FOREX are negatively correlated (-0.28).

The Nifty and SENSEX indexes possess negative covariance (-8.98 and -9.72) with GDP, followed by negative covariance (-57.93 and -63.66) with FOREX as shown in table 6. The Nifty and SENSEX indexes denotes positive covariance (10.96 and 14.40) with CPI. The GDP signifies positive covariance (9.17) with FOREX and negative covariance (-2.20) with CPI. However, CPI shows negative covariance (-8.88) with FOREX.

6 Test of stationarity

Table 7 presents the result of unit-root test for stationarity of stock returns in Asian economies, industries of Indian stock market, and macro-economic variables for the pre-, during, post-liberalization, and full period employing daily, weekly, and monthly frequency data set, respectively.

Panel A of table 7 shows the result of test for stationarity of 14 Asian stock index returns for pre-liberalization period. The daily and weekly SHCOMP stock index returns appears to be stationary at first difference, however, the monthly stock returns eyes stationary at second difference in case of both ADF and PP test of stationarity. The daily and weekly HSI stock index returns acts as stationary at first difference and are significant at all conventional levels. However, the monthly HSI stock returns appears stationary at second difference and are significant at all conventional levels as in both ADF and PP test of stationarity. The daily, weekly, and monthly Nifty returns

follows stationary at first difference and are statistically significant at all conventional levels in ADF and PP test of stationarity. However, the daily and monthly SENSEX index returns follows stationary at levels and are significant at 5 percent level with an exception of weekly frequency that appears stationary at first difference and is significant at all conventional levels in ADF test of stationarity. Moreover, PP test of stationarity signifies the weekly SENSEX index returns follows stationary process at level and are significant at 10 percent level. The daily, weekly and monthly frequencies of NKY stock index follows stationary process at level and are statistically significant at 5 percent level, respectively as in both ADF and PP test of stationarity. However, the daily and weekly frequencies of KOSPI stock index returns appears stationary at level and are statistically significant at 5 percent level in both ADF and PP test of stationarity with an exception of monthly frequency, appears stationary process at level and are statistically significant at 10 percent level for both ADF and PP test of stationarity . Furthermore, daily, weekly and monthly frequencies of CSEALL and TWSE stock index returns appears to be stationary at first difference and are found significant at all conventional levels both in case of ADF and PP test of stationarity. However, an exception of daily CSEALL stock returns, follows stationarity at level and is statistically significant at all conventional levels in case of PP test of stationarity.

Panel A of table 7 shows the result of test for stationarity of 14 Asian stock index returns for during-liberalization period. The daily, weekly and monthly SHCOMP stock index returns appears to be stationary at levels and are statistically significant at all conventional levels both in case of ADF and PP test for stationarity. Daily and monthly HSI stock index returns follows stationary process at levels and are statistically significant at all conventional levels. However, weekly HSI stock returns follows stationarity process at first difference and is significant at all conventional levels both in ADF and PP test for stationarity. The daily, weekly, and monthly Nifty and SENSEX stock indexes returns appears to be stationary at levels and are statistically significant at all conventional levels both in case of ADF and PP test for stationarity as shown in table 7 of panel A for during-liberalization period. Though the daily, weekly, and monthly JCI stock index returns appears stationary at levels and are statistically significant at 10 percent level for both ADF and PP test for stationarity. Moreover, daily and weekly NKY stock index follows stationary process at levels and are statistically significant at 10 percent level, however, the monthly stock returns appears to be stationary at first difference and statistically significant at all conventional levels

both in ADF and PP test of stationarity. The daily, weekly, and monthly KOSPI and FBMKLCI stock indexes returns follows stationary process at level and are statistically significant at all conventional levels both in the case of ADF and PP test for stationarity as shown in table 7 of panel A.

Table 7 Test of stationarity: stock returns in Asian economies and industries of Indian stock market, macro-economic proxy

Panel A: Unit root test of stock returns in 14 Asian countries										
Sl. No.	Asian economies		Data frequency	Pre-liberalization period						
				ADF t statistics			PP t statistics			
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	3 rd Diff
1	China	SHCOMP Index	Daily	2.76	-6.42***	-	3.68	-14.64***	-	-
			Weekly	3.62	-4.52***	-	2.96	-4.54***	-	-
			Monthly	2.34	-1.34	-3.95***	2.17	-1.54	-7.21***	-
2	Hong Kong	HSI Index	Daily	-5.23***	-	-	-5.30***	-	-	-
			Weekly	-5.21***	-	-	-5.16***	-	-	-
			Monthly	-2.10	-15.83***	-	-1.95	-16.02***	-	-
3	India	NIFTY Index	Daily	-1.29	-14.52***	-	-1.37	-14.46***	-	-
			Weekly	-0.40	-7.16***	-	-0.49	-7.16***	-	-
			Monthly	-1.06	-4.91***	-	-1.26	-6.52***	-	-
		SENSEX Index	Daily	-2.84**	-	-	-3.05**	-	-	-
			Weekly	-2.22	-25.15***	-	-2.71*	-	-	-
			Monthly	-2.93**	-	-	-2.98**	-	-	-
4	Indonesia	JCI Index	Daily	NA	NA	NA	NA	NA	NA	NA
			Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
5	Japan	NKY Index	Daily	-3.12**	-	-	-3.00**	-	-	-
			Weekly	-3.10**	-	-	-3.14**	-	-	-
			Monthly	-3.01**	-	-	-3.28***	-	-	-
6	Republic of Korea	KOSPI Index	Daily	-3.05**	-	-	-3.00**	-	-	-
			Weekly	-2.96**	-	-	-2.81*	-	-	-
			Monthly	-2.85*	-	-	-2.86*	-	-	-
7	Malaysia	FBMKLCI Index	Daily	NA	NA	NA	NA	NA	NA	NA
			Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
8	New Zealand	NZSE50FG Index	Daily	NA	NA	NA	NA	NA	NA	NA
			Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
9	Pakistan	KSE100 Index	Daily	NA	NA	NA	NA	NA	NA	NA
			Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA

10	Philippines	PCOMP Index	Daily	NA	NA	NA	NA	NA	NA	NA	
			Weekly	NA	NA	NA	NA	NA	NA	NA	
			Monthly	NA	NA	NA	NA	NA	NA	NA	
11	Singapore	FSSTI Index	Daily	NA	NA	NA	NA	NA	NA	NA	
			Weekly	NA	NA	NA	NA	NA	NA	NA	
			Monthly	NA	NA	NA	NA	NA	NA	NA	
12	Sri Lanka	CSEALL Index	Daily	-2.25	-22.77***	-	-6.20***	-	-	-	
			Weekly	-1.89	-11.84***	-	-1.99	-11.97***	-	-	
			Monthly	-1.80	-5.40***	-	-1.98	-5.44***	-	-	
13	Taiwan	TWSE Index	Daily	-0.52	-29.90***	-	-0.53	-29.90***	-	-	
			Weekly	-0.55	-12.35***		-0.67	-12.37***	-	-	
			Monthly	-0.12	-5.67***	-	0.14	-6.0***	-	-	
14	Thailand	SET Index	Daily	NA	NA	NA	NA	NA	NA	-	
			Weekly	NA	NA	NA	NA	NA	NA	NA	-
			Monthly	NA	NA	NA	NA	NA	NA	NA	-
Sl. No.	Asian economies		Data frequency	During-liberalization period							
				ADF t statistics			PP t statistics				
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	3 rd Diff	
1	China	SHCOMP Index	Daily	-4.19***	-	-	-4.05***	-	-	-	
			Weekly	-3.58***	-	-	-4.05***	-	-	-	
			Monthly	-3.98***	-	-	-4.01***	-	-	-	
2	Hong Kong	HSI Index	Daily	-4.33***	-	-	-4.36***	-	-	-	
			Weekly	-2.30	-26.30***	-	-2.22	-26.38***	-	-	
			Monthly	-4.33***	-	-	-4.35***	-	-	-	
3	India	NIFTY Index	Daily	-4.99***	-	-	-5.04***	-	-	-	
			Weekly	-5.29***	-	-	-5.35***	-	-	-	
			Monthly	-4.66***	-	-	-4.84***	-	-	-	
		SENSEX Index	Daily	-4.93***	-	-	-4.97***	-	-	-	
			Weekly	-3.31***	-	-	-3.07**	-	-	-	
			Monthly	-4.60***	-	-	-4.78***	-	-	-	
4	Indonesia	JCI Index	Daily	-2.72*	-	-	-2.66*	-	-	-	
			Weekly	-2.70*	-	-	-2.69*	-	-	-	
			Monthly	-2.73*	-	-	-2.77*	-	-	-	
5	Japan	NKY Index	Daily	-2.68*	-	-	-2.69*	-	-	-	
			Weekly	-2.64*	-	-	-2.66*	-	-	-	

			Monthly	-2.58	-3.84***	-	-2.41	-4.40***	-	-
6	Republic of Korea	KOSPI Index	Daily	-4.30***	-	-	-3.71***	-	-	-
			Weekly	-4.45***	-	-	-3.95***	-	-	-
			Monthly	-4.29***	-	-	-4.30***		-	-
			Daily	-4.25***	-	-	-4.23***	-	-	-
7	Malaysia	FBMKLCI Index	Weekly	-4.48***	-	-	-4.51***	-	-	-
			Monthly	-4.53***	-	-	-4.57***	-	-	-
			Daily	NA	NA	-	NA	NA	NA	NA
8	New Zealand	NZSE50FG Index	Weekly	NA	NA	-	NA	NA	NA	NA
			Monthly	NA	NA	-	NA	NA	NA	NA
			Daily	2.70	-16.18***	-	2.03	-16.44***	-	-
9	Pakistan	KSE100 Index	Weekly	2.26	-7.94***	-	1.18	-8.48***	-	-
			Monthly	-1.47	-3.85***	-	-1.47	-3.82***	-	-
			Daily	2.69	-22.77***	-	2.85	-22.78***	-	-
10	Philippines	PCOMP Index	Weekly	2.46	-8.85***	-	2.20	-8.84***	-	-
			Monthly	0.17	0.13***	-	1.21	-3.74***	-	-
			Daily	NA	NA	-	NA	NA	NA	NA
11	Singapore	FSSTI Index	Weekly	NA	NA	-	NA	NA	NA	NA
			Monthly	NA	NA	-	NA	NA	NA	NA
			Daily	-2.32	-34.57***	-	-2.35	-34.57***	-	-
12	Sri Lanka	CSEALL Index	Weekly	-2.39	-16.76***	-	-2.40	-16.76***	-	-
			Monthly	-2.44	-8.14***	-	-2.58*	-	-	-
			Daily	-2.65*	-	-	-3.03**	-	-	-
13	Taiwan	TWSE Index	Weekly	-3.17**	-	-	-3.16**	-	-	-
			Monthly	-3.06**	-	-	-3.14**	-	-	-
			Daily	0.09	-22.58***	-	-0.02	-22.59***	-	-
14	Thailand	SET Index	Weekly	0.13	-10.40***	-	-0.14	-10.47***	-	-
			Monthly	0.11	-3.21**	-	0.41	-3.26**	-	-
			Sl. No.	Asian economies		Data frequency	Post-liberalization period			
ADF t statistics							PP t statistics			
Level	1 st Diff	2 nd Diff					Level	1 st Diff	2 nd Diff	3 rd Diff
1	China	SHCOMP Index	Daily	-2.13	-39.91***	-	-2.12	-39.92***	-	-
			Weekly	-1.61	-18.25***	-	-1.44	-18.67***	-	-
			Monthly	-2.47	-9.09***	-	-2.60*	-	-	-
2	Hong Kong	HSI Index	Daily	-2.75*	-	-	-2.54	-42.19***	-	-

			Weekly	-1.21	-18.50***	-	-0.95	-18.94***	-	-
			Monthly	-2.65*	-	-	-	-2.99**	-	-
3	India	NIFTY Index	Daily	2.32	-37.25***	-	-2.29	-37.26***	-	-
			Weekly	-2.37	-19.30***	-	-2.49	-19.27***	-	-
			Monthly	-2.51*	-	-	-2.61*	-	-	-
		SENSEX Index	Daily	-2.34	-39.40***	-	-2.31	-39.41***	-	-
			Weekly	-2.35	-19.32***	-	-2.47	-19.28***	-	-
			Monthly	-2.48	-8.76***	-	-2.60*	-	-	-
4	Indonesia	JCI Index	Daily	-3.97***	-	-	-4.05***	-	-	-
			Weekly	-4.14***	-	-	-4.43***	-	-	-
			Monthly	-4.22***	-	-	-4.27***	-	-	-
5	Japan	NKY Index	Daily	-2.65*	-	-	-2.62*	-	-	-
			Weekly	-5.30***	-	-	-5.21***	-	-	-
			Monthly	-2.68*	-	-	-2.69*	-	-	-
6	Republic of Korea	KOSPI Index	Daily	-4.99***	-	-	-5.01***	-	-	-
			Weekly	-5.18***	-	-	-5.18***	-	-	-
			Monthly	-5.43***	-	-	-5.46***	-	-	-
7	Malaysia	FBMKLCI Index	Daily	-4.15***	-	-	-4.17***	-	-	-
			Weekly	-4.14***	-	-	-4.28***	-	-	-
			Monthly	-4.40***	-	-	-4.50***	-	-	-
8	New Zealand	NZSE50FG Index	Daily	-4.21***	-	-	-4.10***	-	-	-
			Weekly	-4.23***	-	-	-4.31***	-	-	-
			Monthly	-4.09***	-	-	-4.22***	-	-	-
9	Pakistan	KSE100 Index	Daily	-4.94***	-	-	-4.98***	-	-	-
			Weekly	-4.80***	-	-	-4.85***	-	-	-
			Monthly	-5.06***	-	-	-5.16***	-	-	-
10	Philippines	PCOMP Index	Daily	-4.50***	-	-	-4.49***	-	-	-
			Weekly	-4.61***	-	-	-5.02***	-	-	-
			Monthly	-4.66***	-	-	-4.97***	-	-	-
11	Singapore	FSSTI Index	Daily	-3.86***	-	-	-3.90***	-	-	-
			Weekly	-3.84***	-	-	-3.97***	-	-	-
			Monthly	-3.97***	-	-	-4.14***	-	-	-
12	Sri Lanka	CSEALL Index	Daily	-3.82***	-	-	-3.89***	-	-	-
			Weekly	-3.77***	-	-	-3.99***	-	-	-
			Monthly	-3.89***	-	-	-3.89***	-	-	-

13	Taiwan	TWSE Index	Daily	-4.59***	-	-	-4.64***	-	-	-
			Weekly	-4.66***	-	-	-4.81***	-	-	-
			Monthly	-4.95***	-	-	-5.13***	-	-	-
14	Thailand	SET Index	Daily	-5.36***	-	-	-5.47***	-	-	-
			Weekly	-5.57***	-	-	-6.01***	-	-	-
			Monthly	-6.01***	-	-	-5.98***	-	-	-
Sl. No.	Asian economies		Data frequency	Full period						
				ADF t statistics			PP t statistics			
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	3 rd Diff
1	China	SHCOMP Index	Daily	-5.22***	-	-	-5.16***	-	-	-
			Weekly	-5.22***	-	-	-5.24***	-	-	-
			Monthly	-3.41***	-	-	-5.20***	-	-	-
2	Hong Kong	HSI Index	Daily	-6.98***	-	-	-6.97***	-	-	-
			Weekly	-7.00***	-	-	-7.07***	-	-	-
			Monthly	-7.06***	-	-	-6.95***	-	-	-
3	India	NIFTY Index	Daily	-4.93***	-	-	-5.00***	-	-	-
			Weekly	-4.86***	-	-	-5.19***	-	-	-
			Monthly	-5.10***	-	-	-5.32***	-	-	-
		SENSEX Index	Daily	-5.91***	-	-	-6.03***	-	-	-
			Weekly	-5.80***	-	-	-6.21***	-	-	-
			Monthly	-6.01***	-	-	-6.28***	-	-	-
4	Indonesia	JCI Index	Daily	-4.74***	-	-	-4.72***	-	-	-
			Weekly	-4.90***	-	-	-5.22***	-	-	-
			Monthly	-4.93***	-	-	-5.11***	-	-	-
5	Japan	NKY Index	Daily	-6.41***	-	-	-6.18***	-	-	-
			Weekly	-6.33***	-	-	-6.27***	-	-	-
			Monthly	-6.01***	-	-	-6.26***	-	-	-
6	Republic of Korea	KOSPI Index	Daily	-6.48***	-	-	-6.66***	-	-	-
			Weekly	-6.64***	-	-	-6.67***	-	-	-
			Monthly	-6.85***	-	-	-6.79***	-	-	-
7	Malaysia	FBMKLCI Index	Daily	-6.08***	-	-	-6.16***	-	-	-
			Weekly	-6.34***	-	-	-6.63***	-	-	-
			Monthly	-2.12	-22.90***	-	-1.95	-23.06***	-	-
8	New Zealand	NZSE50FG Index	Daily	-4.21***	-	-	-4.10***	-	-	-
			Weekly	-4.23***	-	-	-4.31***	-	-	-

			Monthly	-4.09***	-	-	-4.22***	-	-	-
9	Pakistan	KSE100 Index	Daily	-5.18***	-	-	-5.20***	-	-	-
			Weekly	-5.08***	-	-	-5.12***	-	-	-
			Monthly	-5.41***	-	-	-5.46***	-	-	-
10	Philippines	PCOMP Index	Daily	-5.45***	-	-	-5.29***	-	-	-
			Weekly	-5.49***	-	-	-5.50***	-	-	-
			Monthly	-5.76***	-	-	-5.86***	-	-	-
11	Singapore	FSSTI Index	Daily	-3.86***	-	-	-3.90***	-	-	-
			Weekly	-3.84***	-	-	-3.97***	-	-	-
			Monthly	-3.97***	-	-	-4.14***	-	-	-
12	Sri Lanka	CSEALL Index	Daily	-5.02***	-	-	-5.08***	-	-	-
			Weekly	-4.75***	-	-	-5.09***	-	-	-
			Monthly	-4.99***	-	-	-5.34***	-	-	-
13	Taiwan	TWSE Index	Daily	-4.16***	-	-	-4.73***	-	-	-
			Weekly	-4.95***	-	-	-4.99***	-	-	-
			Monthly	-4.90***	-	-	-4.73***	-	-	-
14	Thailand	SET Index	Daily	-5.44***	-	-	-5.48***	-	-	-
			Weekly	-5.60***	-	-	-6.02***	-	-	-
			Monthly	-5.96***	-	-	-5.93***	-	-	-

Panel B: Unit root test of stock returns in industrial sector of Indian stock market

Sl. No.	Industry	Industry specification	Data frequency	Pre-liberalization period						
				ADF t statistics			PP t statistics			
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	
1	Non-financial	Construction and real estate - C&RE	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
2	Non-financial	Electricity-ELE	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
3	Non-financial	Manufacturing - MANU	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
4	Non-financial	Mining - MIN	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
5	Non-financial	Services (other than	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA

		financial)- SERV								
6	Financial	Asset financing services-AFS	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
7	Financial	Banking services- BSERV	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
8	Financial	Fee based financial services-FBFS	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
9	Financial	Investment services-INV5	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
10	Financial	Other financial services-TFS	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
11	Financial	Other fund based financial services-FBFS	Weekly	NA	NA	NA	NA	NA	NA	NA
			Monthly	NA	NA	NA	NA	NA	NA	NA
Sl. No.	Industry	Industry specification	Data frequency	During-liberalization period						
				ADF t statistics			PP t statistics			
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	3 rd Diff
1	Non-financial	Construction and real estate - C&RE	Weekly	-16.34***	-	-	-25.41***	-	-	-
			Monthly	-13.33***	-	-	-13.31***	-	-	-
2	Non-financial	Electricity - ELE	Weekly	-24.65***	-	-	-24.77***	-	-	-
			Monthly	-12.82***	-	-	-13.06***	-	-	-
3	Non-financial	Manufacturing - MANU	Weekly	-15.75***	-	-	-25.26***	-	-	-
			Monthly	-13.15***	-	-	-13.20***	-	-	-
4	Non-financial	Mining - MIN	Weekly	-25.62***	-	-	-25.73***	-	-	-
			Monthly	-13.16***	-	-	-13.17***	-	-	-
5	Non-financial	Services (other than	Weekly	-15.52***	-	-	-25.27***	-	-	-
			Monthly	-15.13***	-	-	-15.02***	-	-	-

		financial) - SERV								
6	Financial	Asset financing services -AFS	Weekly	-26.07***	-	-	-26.07***	-	-	-
			Monthly	-15.13***	-	-	-15.02***	-	-	-
7	Financial	Banking services- BSERV	Weekly	-16.48***	-	-	-25.62***	-	-	-
			Monthly	-14.47***	-	-	-14.42***	-	-	-
8	Financial	Fee based financial services-FBFS	Weekly	-12.72***	-	-	-23.05***	-	-	-
			Monthly	-14.21***	-	-	-14.18***	-	-	-
9	Financial	Investment services-INV5	Weekly	-26.49***	-	-	-26.49***	-	-	-
			Monthly	-4.86***	-	-	-7.10***	-	-	-
10	Financial	Other financial services- OTFS	Weekly	-3.85***	-	-	-27.03***	-	-	-
			Monthly	-4.67***	-	-	-13.26***	-	-	-
11	Financial	Other fund based financial services- OFBFS	Weekly	-26.49***	-	-	-26.49***	-	-	-
			Monthly	-4.89***	-	-	-7.15***	-	-	-
Sl. No.	Industry	Industry specification	Data frequency	Post-liberalization period						
				ADF t statistics			PP t statistics			
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	3 rd Diff
1	Non-financial	Construction and real estate - C&RE	Weekly	-17.30***	-	-	-17.60***	-	-	-
			Monthly	-9.79***	-	-	-10.11***	-	-	-
2	Non-financial	Electricity - ELE	Weekly	-16.75***	-	-	-16.77***	-	-	-
			Monthly	-9.90***	-	-	-10.22***	-	-	-
3	Non-financial	Manufacturing - MANU	Weekly	-17.04***	-	-	-17.06***	-	-	-
				-10.14***	-	-	-10.14***	-	-	-
4	Non-financial	Mining - MIN	Monthly	-17.51***	-	-	-17.53***	-	-	-
			Weekly	-8.83***	-	-	-8.83***	-	-	-

5	Non-financial	Services (other than financial) - SERV	Weekly	-16.87***	-	-	-16.92***	-	-	-
			Monthly	-7.73***	-	-	-7.75***	-	-	-
6	Financial	Asset financing services - AFS	Weekly	-20.15***	-	-	-20.17***	-	-	-
			Monthly	-8.42***	-	-	-8.433***	-	-	-
7	Financial	Banking services - BSERV	Weekly	-18.62***	-	-	-18.62**	-	-	-
			Monthly	-8.42***	-	-	-8.43***	-	-	-
8	Financial	Fee based financial services-FBFS	Weekly	-10.98***	-	-	-8.36***	-	-	-
			Monthly	-8.37***	-	-	-8.36***	-	-	-
9	Financial	Investment services-INVS	Weekly	-19.29***	-	-	-19.30	-	-	-
			Monthly	-8.37***	-	-	-8.36***	-	-	-
10	Financial	Other financial services - OTFS	Weekly	-16.44***	-	-	-16.43***	-	-	-
			Monthly	-8.87***	-	-	-8.87***	-	-	-
11	Financial	Other fund based financial services - OFBFS	Weekly	-19.08***	-	-	-19.08***	-	-	-
			Monthly	-8.87***	-	-	-8.87***	-	-	-
Sl. No.	Industry	Industry specification	Data frequency	Full period						
				ADF t statistics			PP t statistics			
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	3 rd Diff
1	Non-financial	Construction and real estate - C&RE	Weekly	-19.70***	-	-	-31.25***	-	-	-
			Monthly	-16.49***	-	-	-16.46***	-	-	-
2	Non-financial	Electricity - ELE	Weekly	-29.87***	-	-	-29.84***	-	-	-
			Monthly	-15.83***	-	-	-16.09***	-	-	-
3	Non-financial	Manufacturing - MANU	Weekly	-20.05***	-	-	-30.40***	-	-	-
			Monthly	-16.15***	-	-	-16.20***	-	-	-
4	Non-financial	Mining - MIN	Weekly	-31.11***	-	-	-31.29***	-	-	-
			Monthly	-15.62***	-	-	-15.62***	-	-	-

5	Non-financial	Services (other than financial) - SERV	Weekly	-19.66***	-	-	-30.51***	-	-	-
			Monthly	-17.67***	-	-	-17.59***	-	-	-
6	Financial	Asset financing services - AFS	Weekly	-33.02***	-	-	-33.01***	-	-	-
			Monthly	-15.10***	-	-	-15.13***	-	-	-
7	Financial	Banking services - BSERV	Weekly	-31.66***	-	-	-31.65***	-	-	-
			Monthly	-16.71***	-	-	-16.67***	-	-	-
8	Financial	Fee based financial services - FBFS	Weekly	-12.56***	-	-	-29.54***	-	-	-
			Monthly	-16.68***	-	-	-16.62***	-	-	-
9	Financial	Investment services - INVS	Weekly	-32.24***	-	-	-32.24***	-	-	-
			Monthly	-6.12***	-	-	-9.51***	-	-	-
10	Financial	Other financial services - OTFS	Weekly	-4.51***	-	-	-33.92***	-	-	-
			Monthly	-5.94***	-	-	-16.00***	-	-	-
11	Financial	Other fund based financial services - OFBFS	Weekly	-32.23***	-	-	-32.24***	-	-	-
			Monthly	-6.12***	-	-	-9.55***	-	-	-

Panel C: Unit root test of macro-economic variables of 14 Asian economies

Sl. No.	Country/Variable specification		Data frequency	Pre-liberalization period						
				ADF t statistics			PP t statistics			
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	3 rd Diff
1	China	GDP	Monthly	1.00***	-	-	1.00***	-	-	-
		CPI	Monthly	-1.01	-2.64	-4.14**	-1.01	-2.61	-4.54***	-
		FOREX	Monthly	-2.72	-2.94*	-	-0.72	-1.99	-2.16	-3.77**
2	Hong Kong	GDP	Monthly	-2.42	-12.06***	-	-2.70*	-	-	-
		CPI	Monthly	-1.44	-3.81***	-	-2.33	-13.71***	-	-
		FOREX	Monthly	-6.93***	-	-	-4.32***	-	-	-

3	India	GDP	Monthly	1.00***	-	-	1.00***	-	-	-
		CPI	Monthly	-4.29***	-	-	-2.50	-2.12	-4.31***	-
		FOREX	Monthly	-0.84	-3.62**	-	-0.81	-3.62**	-	-
4	Indonesia	GDP	Monthly	NA	NA	NA	NA	NA	NA	NA
		CPI	Monthly	NA	NA	NA	NA	NA	NA	NA
		FOREX	Monthly	NA	NA	NA	NA	NA	NA	NA
5	Japan	GDP	Monthly	-3.23**	-	-	-2.73*	-	-	-
		CPI	Monthly	-4.59***	-	-	-5.65***	-	-	-
		FOREX	Monthly	-0.24	-4.67***	-	-1.65	-4.49***	-	-
6	Republic of Korea	GDP	Monthly	-2.37	-9.31***	-	-3.20**	-	-	-
		CPI	Monthly	-9.11***	-	-	-9.08***	-	-	-
		FOREX	Monthly	-1.22	-3.59***	-	-1.13	-3.41***	-	-
7	Malaysia	GDP	Monthly	NA	NA	NA	NA	NA	NA	NA
		CPI	Monthly	NA	NA	NA	NA	NA	NA	NA
		FOREX	Monthly	NA	NA	NA	NA	NA	NA	NA
8	New Zealand	GDP	Monthly	NA	NA	NA	NA	NA	NA	NA
		CPI	Monthly	NA	NA	NA	NA	NA	NA	NA
		FOREX	Monthly	NA	NA	NA	NA	NA	NA	NA
9	Pakistan	GDP	Monthly	NA	NA	NA	NA	NA	NA	NA
		CPI	Monthly	NA	NA	NA	NA	NA	NA	NA
		FOREX	Monthly	NA	NA	NA	NA	NA	NA	NA
10	Philippines	GDP	Monthly	NA	NA	NA	NA	NA	NA	NA
		CPI	Monthly	NA	NA	NA	NA	NA	NA	NA
		FOREX	Monthly	NA	NA	NA	NA	NA	NA	NA
11	Singapore	GDP	Monthly	NA	NA	NA	NA	NA	NA	NA
		CPI	Monthly	NA	NA	NA	NA	NA	NA	NA
		FOREX	Monthly	NA	NA	NA	NA	NA	NA	NA
12	Sri Lanka	GDP	Monthly	-1.86	-6.54***	-	-1.87	-6.54***	-	-
		CPI	Monthly	1.72	-5.48***	-	2.86	-5.38***	-	-
		FOREX	Monthly	-0.87	-4.22***	-	-0.53	-3.91***	-	-
13	Taiwan	GDP	Monthly	-1.05	-5.65***	-	-1.17	-5.67***	-	-
		CPI	Monthly	-1.37	-6.39***	-	-1.38	-6.40***	-	-
		FOREX	Monthly	-0.87	-3.15**	-	1.61	-3.19**	-	-
14	Thailand	GDP	Monthly	NA	NA	NA	NA	NA	NA	NA
		CPI	Monthly	NA	NA	NA	NA	NA	NA	NA

		FOREX	Monthly	NA	NA	NA	NA	NA	NA	NA
Sl. No.	Country/Variable specification		During-liberalization period							
			Data frequency	ADF t statistics			PP t statistics			
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	3 rd Diff
1	China	GDP	Monthly	-2.18	-7.91***	-	-2.11	-15.67***	-	-
		CPI	Monthly	-1.68	-4.12***	-	-1.26	-8.88***	-	-
		FOREX	Monthly	-2.80	-13.58***	-	-2.80	13.58***	-	-
2	Hong Kong	GDP	Monthly	-1.98	-12.53***	-	-2.40	-12.66***	-	-
		CPI	Monthly	-1.89	-11.77***	-	-1.89	-11.85***	-	-
		FOREX	Monthly	-1.95	-9.93***	-	-2.02	-9.83***	-	-
3	India	GDP	Monthly	-2.28	-13.65***	-	-2.31	-13.65***	-	-
		CPI	Monthly	-3.22	-10.78***	-	-3.36	-10.74***	-	-
		FOREX	Monthly	-1.99	-9.76***	-	-2.00	-9.76***	-	-
4	Indonesia	GDP	Monthly	NA	NA	-	NA	NA	NA	-
		CPI	Monthly	NA	NA	-	NA	NA	NA	-
		FOREX	Monthly	NA	NA	-	NA	NA	NA	-
5	Japan	GDP	Monthly	-3.78***	-	-	-2.36	-1.69	-3.54**	-
		CPI	Monthly	-4.58***	-	-	-9.97***	-	-	-
		FOREX	Monthly	-0.38	-3.87***	-	-0.48	-3.87***	-	-
6	Republic of Korea	GDP	Monthly	-0.19	-3.24**	-	1.17	-9.10***	-	-
		CPI	Monthly	-6.77***	-	-	-6.46***	-	-	-
		FOREX	Monthly	2.54	-7.43***	-	-0.56	-3.99***	-	-
7	Malaysia	GDP	Monthly	-2.98**	-	-	-3.10**	-	-	-
		CPI	Monthly	0.30	-13.80***	-	0.27	-13.79***	-	-
		FOREX	Monthly	-0.29	-14.49***	-	-0.44	-14.45***	-	-
8	New Zealand	GDP	Monthly	NA	NA	-	NA	NA	NA	-
		CPI	Monthly	NA	NA	-	NA	NA	NA	-
		FOREX	Monthly	NA	NA	-	NA	NA	NA	-
9	Pakistan	GDP	Monthly	NA	NA	-	NA	NA	NA	-
		CPI	Monthly	NA	NA	-	NA	NA	NA	-
		FOREX	Monthly	NA	NA	-	NA	NA	NA	-
10	Philippines	GDP	Monthly	NA	NA	-	NA	NA	NA	-
		CPI	Monthly	NA	NA	-	NA	NA	NA	-
		FOREX	Monthly	NA	NA	-	NA	NA	NA	-
11	Singapore	GDP	Monthly	NA	NA	-	NA	NA	NA	-

		CPI	Monthly	NA	NA	-	NA	NA	NA	-
		FOREX	Monthly	NA	NA	-	NA	NA	NA	-
12	Sri Lanka	GDP	Monthly	-3.53***	-	-	-3.70***	-	-	-
		CPI	Monthly	-1.11	-5.64***	-	-1.18	-5.68***	-	-
		FOREX	Monthly	-1.98	-5.65***	-	-1.84	-5.65***	-	-
13	Taiwan	GDP	Monthly	-2.91**	-	-	-3.04**	-	-	-
		CPI	Monthly	-2.33	-4.21***	-	-4.71***	-	-	-
		FOREX	Monthly	-3.06**	-	-	-2.96**	-	-	-
14	Thailand	GDP	Monthly	NA	NA	-	NA	NA	NA	-
		CPI	Monthly	NA	NA	-	NA	NA	NA	-
		FOREX	Monthly	NA	NA	-	NA	NA	NA	-
Sl. No.	Country/Variable specification		Post-liberalization period							
			Data frequency	ADF t statistics			PP t statistics			
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	3 rd Diff
1	China	GDP	Monthly	-2.80	-8.65***	-	-2.19	-8.82***	-	-
		CPI	Monthly	-3.82***	-	-	-2.11	-9.08***	-	-
		FOREX	Monthly	-1.59	-4.93***	-	-2.27	-4.78***	-	-
2	Hong Kong	GDP	Monthly	-1.95	-8.62***	-	-2.24	-8.71***	-	-
		CPI	Monthly	-1.93	-3.55***	-	-2.38	-13.22***	-	-
		FOREX	Monthly	-3.09	-6.89***	-	-2.90	-6.65***	-	-
3	India	GDP	Monthly	-2.19	-8.39***	-	-2.19	-8.39***	-	-
		CPI	Monthly	-2.48	-6.87***	-	-1.89	-6.74***	-	-
		FOREX	Monthly	-1.07	-6.08***	-	-0.95	-6.03***	-	-
4	Indonesia	GDP	Monthly	NA	NA	-	NA	NA	-	-
		CPI	Monthly	NA	NA	-	NA	NA	-	-
		FOREX	Monthly	NA	NA	-	NA	NA	-	-
5	Japan	GDP	Monthly	-4.49***	-	-	-7.11***	-	-	-
		CPI	Monthly	-2.47	-14.29***	-	-14.77***	-	-	-
		FOREX	Monthly	-5.51***	-	-	-5.79***	-	-	-
6	Republic of Korea	GDP	Monthly	-5.03***	-	-	-3.91***	-	-	-
		CPI	Monthly	-10.72***	-	-	-11.02***	-	-	-
		FOREX	Monthly	-2.65	-7.02***	-	-2.67	-9.31***	-	-
7	Malaysia	GDP	Monthly	-4.62***	-	-	-3.32	-12.66***	-	-
		CPI	Monthly	-0.16	-8.68***	-	-0.40	-8.68***	-	-
		FOREX	Monthly	-1.06	-8.95***	-	-0.95	-8.60***	-	-

8	New Zealand	GDP	Monthly	NA	NA	-	NA	NA	-	-
		CPI	Monthly	NA	NA	-	NA	NA	-	-
		FOREX	Monthly	NA	NA	-	NA	NA	-	-
9	Pakistan	GDP	Monthly	NA	NA	-	NA	NA	-	-
		CPI	Monthly	NA	NA	-	NA	NA	-	-
		FOREX	Monthly	NA	NA	-	NA	NA	-	-
10	Philippines	GDP	Monthly	NA	NA	-	NA	NA	-	-
		CPI	Monthly	NA	NA	-	NA	NA	-	-
		FOREX	Monthly	NA	NA	-	NA	NA	-	-
11	Singapore	GDP	Monthly	NA	NA	-	NA	NA	-	-
		CPI	Monthly	NA	NA	-	NA	NA	-	-
		FOREX	Monthly	NA	NA	-	NA	NA	-	-
12	Sri Lanka	GDP	Monthly	-2.52	-7.28***	-	-2.62	-15.16***	-	-
		CPI	Monthly	-2.25	-5.57***	-	-2.66	-10.15***	-	-
		FOREX	Monthly	-0.71	-11.65***	-	-0.47	-11.37***	-	-
13	Taiwan	GDP	Monthly	-3.09	-6.92***	-	-3.59***	-	-	-
		CPI	Monthly	-3.58***	-	-	-4.95***	-	-	-
		FOREX	Monthly	-2.31	-9.27***	-	-1.69	-9.00***	-	-
14	Thailand	GDP	Monthly	NA	NA	-	NA	NA	-	-
		CPI	Monthly	NA	NA	-	NA	NA	-	-
		FOREX	Monthly	NA	NA	-	NA	NA	-	-
Sl. No.	Country/Variable specification		Full period							
			Data frequency	ADF t statistics			PP t statistics			
				Level	1 st Diff	2 nd Diff	Level	1 st Diff	2 nd Diff	3 rd Diff
1	China	GDP	Monthly	-2.58*	-	-	-2.64*	-	-	-
		CPI	Monthly	-1.66	-5.60***	-	-1.86	-12.95***	-	-
		FOREX	Monthly	-1.93	-16.36***	-	-1.96	-16.36***	-	-
2	Hong Kong	GDP	Monthly	-4.29***	-	-	-4.10***	-	-	-
		CPI	Monthly	-1.57	-6.86***	-	-2.25	-23.01***	-	-
		FOREX	Monthly	-11.48***	-	-	-7.38***	-	-	-
3	India	GDP	Monthly	-3.35***	-	-	-3.38***	-	-	-
		CPI	Monthly	-3.56***	-	-	-3.39***	-	-	-
		FOREX	Monthly	-1.71	-12.38***	-	-1.74	-12.33***	-	-
4	Indonesia	GDP	Monthly	-3.18**	-	-	-3.18**	-	-	-
		CPI	Monthly	0.04	-9.89***	-	0.14	-9.64***	-	-

		FOREX	Monthly	-2.23	-11.29***	-	-2.04	-11.23***	-	-
5	Japan	GDP	Monthly	-4.50***	-	-	-7.74***	-	-	-
		CPI	Monthly	-3.37***	-	-	-16.95***	-	-	-
		FOREX	Monthly	-2.17	-14.70***	-	-1.95	-14.70***	-	-
6	Republic of Korea	GDP	Monthly	-3.42***	-	-	-3.72***	-	-	-
		CPI	Monthly	-3.83***	-	-	-13.94***	-	-	-
		FOREX	Monthly	-2.08	-15.14***	-	-2.22	-11.06***	-	-
7	Malaysia	GDP	Monthly	-3.85***	-	-	-4.34***	-	-	-
		CPI	Monthly	0.65	-15.50***	-	0.70	-15.16***	-	-
		FOREX	Monthly	-1.51	-17.62***	-	-1.43	-17.64***	-	-
8	New Zealand	GDP	Monthly	-2.54	-6.61***	-	-2.82*	-	-	-
		CPI	Monthly	-4.12***	-	-	-4.73***	-	-	-
		FOREX	Monthly	-2.41	-8.65***	-	-2.18	-8.66***	-	-
9	Pakistan	GDP	Monthly	-0.82	-2.08	-21.36***	-0.03	-16.95***	-	-
		CPI	Monthly	5.47	-4.73***		4.57	-11.39***	-	-
		FOREX	Monthly	0.13	-10.33***	-	0.07	-10.66***	-	-
10	Philippines	GDP	Monthly	-2.73*	-	-	-2.92***	-	-	-
		CPI	Monthly	-3.822***	-	-	-2.79*	-	-	-
		FOREX	Monthly	-1.33	-13.56***	-	-1.39	-13.57***	-	-
11	Singapore	GDP	Monthly	-2.55	-6.25***	-	-3.15***	-	-	-
		CPI	Monthly	-5.73***	-	-	-15.26***	-	-	-
		FOREX	Monthly	-0.45	-10.04***	-	-0.17	-9.90***	-	-
12	Sri Lanka	GDP	Monthly	-2.52	-7.28***	-	-2.62*	-	-	-
		CPI	Monthly	2.25	-5.57***	-	2.66	-10.15***	-	-
		FOREX	Monthly	-0.71	-11.65***	-	-0.47	-11.37***	-	-
13	Taiwan	GDP	Monthly	-3.09***	-	-	-3.59***	-	-	-
		CPI	Monthly	-3.58***	-	-	-4.95***	-	-	-
		FOREX	Monthly	-2.31	-9.27***	-	-1.69	-9.00***	-	-
14	Thailand	GDP	Monthly	-3.53***	-	-	-3.59***	-	-	-
		CPI	Monthly	-4.52***	-	-	-3.52***	-	-	-
		FOREX	Monthly	-3.25***	-	-	-3.26***	-	-	-

Note: ^a ADF denotes Augmented-Dickey-Fuller test

^b PP denotes Phillips-Perron test

^c *, **, and *** denotes significance at 10, 5 and 1 % level, respectively

^d **D, W, M** denotes Daily, Weekly, Monthly respectively

Table 7 of panel A presents daily, weekly, and monthly KSE100 and PCOMP stock index returns for the during-liberalization period owing stationary process at first difference and are statistically significant at all conventional levels in both the case of ADF and PP test for stationarity. Daily, weekly and monthly CSEALL stock index returns appears to be stationary at level and are statistically significant at all conventional levels in both the instance of ADF and PP test for stationarity. However, with an exception of monthly CSEALL stock returns follows stationarity at level and is statistically significant at 10 percent level of PP test for stationarity. The daily, weekly, and monthly TWSE stock index returns follows stationary process at level and are statistically significant at 10, 5, and 5 percent levels, respectively both in ADF and PP test for stationarity. However, the daily, weekly and monthly SET stock index returns follows stationarity process at first difference and are statistically significant at 1, 1, 5 percent levels, respectively in both the ADF and PP test for stationarity as shown in panel A of table 7.

Panel A of table 7 presents the result of unit root for stationarity of 14 Asian economies stock return for the post-liberalization period. The daily, weekly, and monthly SHCOMP stock index returns follows stationarity process at first difference for the post-liberalization period and are statistically significant at all conventional levels in both the case of ADF and PP test for stationarity. However, with an exception of monthly SHCOMP stock index returns appears to be stationary at level and is statistically significant at 10 percent level in case of PP test for stationarity. The daily HSI stock index returns follows the stationarity process at level and first difference and are statistically significant at 10 and 1 percent levels respectively, of the ADF and PP test for stationarity. Though weekly HSI stock index returns follows the stationarity process at first difference levels and are statistically significant at all conventional levels, of ADF and PP test for stationarity. Monthly HSI stock index returns follows the stationarity process at level and first difference subsequently are statistically significant at 10 and 5 percent levels respectively, both in case of ADF and PP test for stationarity as present in table 7 of panel A. The daily and weekly Nifty index returns follows stationarity process at first difference level for the post-liberalization

period, and are statistically significant at all conventional levels for both ADF and PP test for stationarity. However, monthly Nifty index returns appears to be stationary at level and are statistically significant 10 percent level for post-liberalization period both in ADF and PP test for stationarity. The daily, weekly, and monthly SENSEX index returns follows stationarity process at first difference and are statistically significant at all conventional levels both in ADF and PP test for stationarity, except monthly SENSEX index returns appears to be stationary at level and statistically significant at 10 percent level as in case of PP test for stationarity. The daily, weekly, and monthly stock returns of JCI, KOSPI, FBMKLCI, NZSE50FG, KSE100, PCOMP, FSSTI, CSEALL, TWSE, and SET indexes follows stationarity process at level for post-liberalization period and are found statistically significant at all conventional levels in both ADF and PP test for stationarity. However, daily and monthly NKY stock index returns follows stationarity process at levels and are statistically significant at 10 percent level both in ADF and PP test for stationarity as shown in table 7 of panel A. However, weekly returns of NKY appears to be stationarity at level and statistically significant at all conventional levels for both ADF and PP test for stationarity.

Table 7 of panel A depicts the result of unit root test for stationarity of 14 Asian stock markets for the full period. The daily, weekly, and monthly stock returns of SHCOMP, HSI, Nifty, SENSEX, JCI, NKY, KOSPI, NZSE50FG, KSE100, PCOMP, FSSTI, CSEALL, TWSE, and SET indexes follows stationarity process at level and are statistically significant at all conventional levels both in ADF and PP test for stationarity. However, daily and weekly FBMKLCI stock index returns appears to be stationary at level and are statistically significant at all conventional levels. The monthly FBMKLCI stock index returns follows the stationarity process at first difference and is statistically significant at all conventional levels both in case of ADF and PP test for stationarity as shown in the table 7 of panel A.

Table 7 of panel B presents the result of unit root test of stock returns in industrial sector of Indian stock market for during-, post-liberalization and full periods. The weekly and monthly stock returns of C&RE, ELE, MANU, MIN, SERV, AFS, BSERV, FBFS, INVS, OTFS, and OFBFS follows stationarity process at level and are statistically significant at all conventional levels both in ADF and PP test of stationarity for the during, post-liberalization and full period respectively.

Table 7 of panel C presents the result of unit root test of macro-economic variables of 14 merging economies for the pre-liberalization period. The GDP of China appears to be stationary at level and statistically significant at all conventional levels. The CPI of China follows stationary at second difference and is statistically significant at 5 and 1 percent levels for the ADF and PP test for stationarity. However, the Forex of China signifies stationary at first and third difference level for ADF and PP test for stationarity, respectively and are statistically significant at 10 and 5 percent levels. The monthly GDP of Hong Kong follows stationary process at first difference and level and are statistically significant at 1 and 10 percent levels, respectively for ADF and PP test for stationarity. The CPI of Hong Kong follows stationarity at first difference and are statistically significant at all conventional levels for the ADF and PP test for stationarity. The monthly FOREX reports stationary at level and are statistically significant at all conventional levels for the ADF and PP test for stationarity. The monthly GDP of India follows stationarity process at level and are statistically significant at all conventional levels for the ADF and PP test for stationarity. The CPI of India reports stationary at level and second difference for the ADF and PP test for stationarity and are statistically significant at all conventional levels. The monthly relative FOREX of India appears to be stationary at first difference and are statistically significant at 5 percent level for the ADF and PP test for stationarity, as shown in table 7 of panel C for the pre-liberalization period. The GDP and CPI of Japan follows stationary process at levels and are statistically significant at 5 and 1 percent levels, respectively of ADF and PP test for stationarity. Subsequently, relative FOREX of Japan follows stationary process at first difference and is statistically significant 1 percent level of ADF and PP test for stationarity. The GDP of Republic of Korea reports stationary at first difference and level for ADF and PP test for stationarity and are statistically significant at 1 and 5 percent levels. The CPI appears to be stationary at level and is statistically significant at all conventional level for the ADF and PP test for stationarity. The FOREX of Republic of Korea follows stationary at first difference and is statistically significant at all conventional levels of the ADF and PP test for stationarity. The GDP, CPI, and relative FOREX of Sri Lanka reports stationary at first difference and are statistically significant at all conventional levels of the ADF and PP test for stationarity as shown in table 7 of panel C. Moreover, The GDP, CPI, and relative FOREX of Taiwan follows stationary at first difference and are statistically significant at all conventional levels of the ADF and PP test for stationarity with an exception of relative FOREX which is significant at 5 percent level.

The GDP, CPI, and relative FOREX of China, Hong Kong, and India for the during-liberalization period follows stationarity process at first difference and are statistically significant at all conventional levels for the ADF and PP test of stationarity. However, the GDP of Japan signifies stationary at level for ADF test of stationarity, but PP test for stationarity reports at second difference and is statistically significant at 1 and 5 percent levels, respectively. The CPI of Japan reports stationary at level and is statistically significant at all conventional levels for ADF and PP test for stationarity. The relative FOREX follows stationarity process at first difference and are statistically significant at all conventional levels for ADF and PP test for stationarity. The GDP, CPI, and FOREX of Republic of Korea follows stationarity process at first difference, level, and first difference respectively and are statistically significant at all conventional levels for the ADF and PP test for stationarity as present in table 7 of panel C. The monthly GDP, CPI, and FOREX of Malaysia reports stationary at level and first differences and are statistically significant at 5, 1 and 1 percent level, respectively for the ADF and PP test for stationarity. The GDP, CPI, and FOREX of Sri Lanka reports stationary at level and first differences, respectively and are significant at all conventional levels for the ADF and PP test of stationarity. The GDP of Taiwan appears to be stationary at level and is statistically significant at 5 percent level for the ADF and PP test for stationarity. The CPI of Taiwan follows stationary at first difference and level for ADF and PP test for stationarity, respectively and is significant at all conventional levels. However, the FOREX of Taiwan reports stationary at level and is statistically significant at 5 percent level for the ADF and PP test for stationarity as shown in table 7 of panel C.

Table 7 of panel C reports the result of Unit root test for stationarity of macro-economic variables of 14 economies for post-liberalization period. The monthly GDP and FOREX of China reports stationary at first difference and are statistically significant at all conventional levels for the ADF and PP test for stationarity. However, the CPI of China follows stationarity process at level for ADF and first difference for PP test for stationarity test and is significant at all conventional levels. The monthly GDP, CPI and relative FOREX of Hong Kong and India follows stationarity process at first difference and are statistically significant at all conventional levels for the ADF and PP test of stationarity. The GDP, CPI, and FOREX of Japan reports stationary at level and are statistically significant at all conventional level for the ADF and PP test of stationarity. However, an exception

of monthly CPI follows stationarity at first difference for the ADF test of stationarity. The monthly GDP, CPI and relative FOREX of Republic of Korea reports stationary at levels and first difference and are statistically significant at all conventional levels for the ADF and PP test of stationarity. The GDP, CPI and relative FOREX of Malaysia and Sri Lanka reports stationary process at first difference and are statistically significant at all conventional levels for the test of ADF and PP test of stationarity, though an exception of monthly GDP of Malaysia appears stationary at level in ADF test of stationarity. The GDP of Taiwan follows stationary process for the ADF test of stationarity at first difference and for PP test of stationarity at level and both are statistically significant at all conventional levels. The CPI of Taiwan indicates stationary process at level and are statistically significant at all conventional levels of both the ADF and PP test for stationarity.

The table 7 of panel C presents the result of unit root test for stationarity of 14 Asian economies for the full period. The GDP, CPI, and relative FOREX of China reports stationary at level, and first differences and are statistically significant at 10, 1 and 1 percent level respectively for the ADF and PP test of stationarity. The GDP and relative FOREX of Hong Kong follows stationary process at level and are statistically significant at all conventional levels for the ADF and PP test of stationarity. However, the CPI of Hong Kong reports stationary at first difference and are statistically significant at all conventional for the ADF and PP test of stationarity. The GDP, CPI, and relative FOREX of India reports stationarity at levels and first difference and are statistically significant at all conventional levels for the ADF and PP test for stationarity. The GDP, CPI, and relative FOREX of Indonesia follows stationarity process at level and first difference, and are statistically significant at 5, 1 and 1 percent levels, respectively for ADF and PP test of stationarity. The GDP, CPI, and relative FOREX of Japan and Republic of Korea reports stationary at levels and first difference and are statistically significant at all conventional levels for the ADF and PP test of stationarity. The GDP, CPI, and relative FOREX series of Malaysia follows stationary process at level and first differences and are statistically significant at all conventional levels for the ADF and PP test for stationarity. The GDP of New Zealand reports stationary at first difference and level for ADF and PP test for stationarity and is statistically significant at 1 and 10 percent levels, respectively. Though the CPI of New Zealand sounds stationary at level and is statistically significant at all conventional levels for ADF and PP test for stationarity. Subsequently, the relative FOREX follows stationarity at first difference and is statistically significant at all conventional

levels for ADF and PP test for stationarity. The monthly GDP of Pakistan reports stationary at second difference and first difference for ADF and PP test for stationarity and is statistically significant at all conventional levels. However, the CPI and FOREX reports stationary at first difference and are statistically significant at all conventional levels for the ADF and PP test for stationarity.

The GDP of Philippines follows stationary process at levels and is statistically significant at 10 and 1 percent levels for ADF and PP test for stationarity. However, the CPI shows stationary at levels and is statistically significant at 1 and 10 percent level for the ADF and PP test for stationarity. The monthly FOREX of Philippines reports stationary at first difference and is significant at all conventional levels for the ADF and PP test for stationarity. The GDP of Singapore reports stationary at first difference for ADF and level for PP test for stationarity and is statistically significant at all conventional levels. The monthly CPI of Singapore owes stationarity at level and is statistically significant at all conventional levels for ADF and PP test for stationarity. However, relative FOREX of Singapore follows stationarity process at first difference and is statistically significant at all conventional levels for ADF and PP test for stationarity. The GDP of Sri Lanka opts to stationary at first difference for ADF and level for PP test for stationary and is statistically significant at 1 and 10 percent levels, respectively. The CPI and FOREX of Sri Lanka follows stationary at first difference and are statistically significant at all conventional levels for the ADF and PP test for stationarity. The GDP, CPI, and FOREX of Taiwan forms stationary at levels and first difference and are statistically significant at all conventional levels for the ADF and PP test for stationarity. Moreover, GDP, CPI and relative FOREX of Thailand accounts stationarity at levels and are statistically significant at all conventional levels for the ADF and PP test for stationarity as indicates in table 7 of panel C. Table 7 comprised of ‘Not Applicable’ (NA), which signifies data is not available for that specific periods respectively.

7 Empirical results

The parameters of GARCH and TGARCH models are estimated through employing the Maximum likelihood method. Table 8 of panel A, B, C and D recapitulates the estimates of the GARCH(1,1) model for pre-, during, post-liberalization and full period. However, table 9 offers the results of the TARCH(1,1) model for pre-, during, post-liberalization and full period, respectively.

7.1 Impact of financial stock market liberalization on returns volatility in Asian economies

The impact of past volatility on current volatility has been assessed through examining the GARCH coefficient (β_1) conditional variance. However, for degree of precision, the impact of stock market liberalization on returns volatility is estimated through the coefficients on the variables, namely, Duringlib (λ_2), Postlib (λ_3), and PreduringPostlib (λ_4) which represent the during-liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively.

The monthly GARCH coefficient of SHCOMP (China) stock returns for during-liberalization and full period is 0.930 and 0.543, respectively as shown in table 8 of panel B and are significant at 1 percent level, subsequently resembles coefficient in during-liberalization period surge highest. Hence, the evidences suggests that past volatility possess largest impact on current volatility in during-liberalization period. However, further for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent the during liberalization, post-liberalization and pre-plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 0.286 in the GARCH model (table 8), and 0.258 in the TARCH model (table 9), and both of which are significant at 1 percent level. These results signifies that volatility is significantly higher in during-liberalization period. The coefficient of post-liberalization period is meager and is statistically not significant in the GARCH and TARCH model. Therefore, there exist no concrete evidence to suggest that the volatility in post-liberalization period is very different from that in the pre-liberalization period. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is different relative to pre-liberalization period. The coefficient is 0.039 in GARCH model and 0.099 in the TARCH model, and both are statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization transformation compared with the period stock market was not unveiled to the foreign investors. However, the daily and weekly SHCOMP stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

The monthly GARCH coefficient of HSI (Hong Kong) stock returns for during-liberalization and full period is 1.043 and -0.018, respectively and the during-liberalization period coefficient is

significant at 1 percent level. However, the coefficient of full period is statistically insignificant and subsequently resembles coefficient in during-liberalization period surge highest and hence, suggests that past volatility possess largest impact on current volatility in during-liberalization period. However, further for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent the during liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. During-liberalization coefficient is -0.926 in the GARCH model (table 8), and 0.299 in the TARCH model (table 9), and both of which are significant at 1 percent level, respectively. These results signifies that volatility is significantly higher in during-liberalization period. The coefficient of post-liberalization period is 0.290 and 0.269 for the GARCH and TARCH model and are statistically significant at all conventional levels. Thus, there exist evidence that the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicates that past volatility possessed the greater impact on current volatility during the period of liberalization and hence, reveals that the stock market liberalization in Hong Kong prone to increase in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.198 in GARCH model and 0.188 in the TARCH model, and both are statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly HSI stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

The monthly GARCH coefficient of Nifty (India) stock returns for during-liberalization and full period is 0.920 and 0.393, respectively and the during-liberalization period coefficient is significant at 1 percent level. However, the coefficient for full period is statistically insignificant and subsequently resembles coefficient in during-liberalization period attain highest and hence, suggests that past volatility has had largest impact on current volatility in during-liberalization period. However, further for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and

PreduringPostlib which represent the during-liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 0.350 in the GARCH model (table 8), and 0.331 in the TARARCH model (table 9), and all of which are significant at 1 percent levels. These results signifies that volatility is significantly higher in during-liberalization period. The coefficient of post-liberalization period is 0.999 and 1.00 for the GARCH and TARARCH model respectively and are statistically significant at all conventional levels. Thus, there exist evidence that the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicates that past volatility possessed the greater impact on current volatility during the period of liberalization and hence, reveals the stock market liberalization in India leads to increase in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.285 in GARCH model and 1.00 in the TARARCH model, and both are statistically significant at all conventional levels. It specifies that there has been an increase in returns volatility following liberalization reform compared with the pre-liberalization period. However, the daily and weekly Nifty stock returns estimates of financial stock market liberalization on returns volatility further accumulates on the evidences to unlock and understand the magnitude of impact.

Moreover, the identical evidences experienced in Nifty stock returns has been reflected in SENSEX stock index (India) in relation with the impact of financial stock market liberalization on returns volatility.

The monthly GARCH coefficient of JCI (Indonesia) stock returns for during-liberalization and full period is 0.374 and 0.964, respectively and during-liberalization period coefficient is significant at 10 percent level. However, the coefficient for full period is statistically significant at 1 percent level and subsequently resembles coefficient in during-liberalization period declines and hence, suggests that past volatility has had largest impact on current volatility in full period. However, for precision the impact of stock market liberalization on returns volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 0.798 in the GARCH model (table 8), and 0.798 in the TARARCH model (table 9), and all of which are significant at 1 percent level. These results

signifies that volatility is significantly higher in during-liberalization period. The coefficient of post-liberalization period is 0.493 and 0.263 for the GARCH and TARCH model, respectively and are statistically significant at all conventional levels. Thus, there exist evidence that the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicate that past volatility has had the greater impact on current volatility during the period of liberalization and hence, reveals that the stock market liberalization in Indonesia leads to increase in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.306 in GARCH model and 0.181 in the TARCH model, and both are statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly JCI stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

The monthly GARCH coefficient of NKY (Japan) stock returns for during-liberalization and full period is 0.541 and 0.095, respectively and during-liberalization period coefficient is insignificant. However, the coefficient for full period is statistically significant at 1 percent level and subsequently indicates no evidence to infer the significant impact of past volatility on current volatility in during-liberalization period. However, for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during liberalization, post-liberalization and pre-plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 0.962 in the GARCH model (table 8), and 0.968 in the TARCH model (table 9), and both of which are significant at 1 percent level. These results signifies that volatility is significantly higher in during-liberalization period. The coefficient of post-liberalization period is 0.070 and 0.084 for the GARCH and TARCH model, respectively and are statistically significant at all conventional levels. Thus, there exist evidence for the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicates that past volatility has had the greater impact on current volatility during the period of liberalization and hence, reveals that the stock market liberalization in Indonesia leads to increment in returns volatility.

The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.027 in GARCH model and 0.411 in the TARCH model, and the GARCH and TARCH coefficients are statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly NKY stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

The monthly GARCH coefficient of KOSPI (Republic of Korea) stock returns for during-liberalization and full period is -0.060 and 0.616, respectively and during-liberalization and full period coefficient is insignificant. Subsequently indicates no evidence to infer the significant impact of past volatility on current volatility in during-liberalization period. However, for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 0.741 in the GARCH model (table 8), and 0.731 in the TARCH model (table 9), and both of which are significant at 1 percent level. These results signifies that volatility is significantly higher in during-liberalization period. The coefficient of post-liberalization period is 0.504 for both the GARCH and TARCH model, respectively and are statistically significant at all conventional levels. Thus, there exist evidence for the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicates that past volatility has had the greater impact on current volatility during the period of stock market liberalization and hence, reveals that the stock market liberalization in Republic of Korea leads to increment in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.383 in GARCH model and 0.344 in the TARCH model, and both the coefficient is statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly KOSPI stock returns

estimates of financial stock market liberalization on returns volatility further adds on the evidences to unearth and understand the magnitude of impact.

The monthly GARCH coefficient of FBMKLCI (Malaysia) stock returns for during-liberalization and full period is -0.009 and 0.243, respectively and during-liberalization period coefficient is insignificant. However, the coefficient for full period is statistically significant at 1 percent level and subsequently indicates no evidence to infer the significant impact of past volatility on current volatility in during-liberalization period. However, for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during-liberalization, post-liberalization and pre-plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 5.432 in the GARCH model (table 8), and 5.515 in the TARCH model (table 9), and both of which are significant at 1 percent level. These results signifies that volatility is significantly higher in during-liberalization period. The coefficient of post-liberalization period is 0.341 and 0.348 for the GARCH and TARCH model, respectively and are statistically significant at all conventional levels. Thus, there exist evidence for the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicates that past volatility has had the greater impact on current volatility during the period of liberalization and hence, reveals that the stock market liberalization in Malaysia leads to increment in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.977 in GARCH model and 0.979 in the TARCH model, and both the coefficient is statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly FBMKLCI stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

Table 8 Maximum likelihood estimates for the AR(s)-GARCH(1,1)

Panel A: ML estimates for pre-liberalization period															
Sl. No.	Asian economies		Model	Data frequency	ϕ_0	ϕ_1	α_0	α_1	β_1	λ_1	λ_2	λ_3	λ_4	LL	AIC
1	China	SHCOM P Index	AR(1)-ARCH(1)	D	0.83***	0.192***	0.019***	0.769***	-	0.868***	-	-	-	65.64	-0.48
				W	4.175***	0.4000***	1.998***	0.540***	-	0.744***	-	-	-	-100.24	4.08
				M	4.496	-0.361	24.927	0.300	-	0.884***	-	-	-	-31.85	7.17
2	Hong Kong	HSI Index	AR(1)-ARCH(1)	D	20.561***	0.988***	3.392***	2.112***	-	0.086***	-	-	-	-12625.43	5.26
				W	21.127***	0.946***	20.390***	2.053***	-	-0.022***	-	-	-	-3482.78	6.85
				M	-0.009	-0.043	0.119***	-0.003	-	0.998***	-	-	-	-143.55	1.26
3	India	NIFTY Index	AR(1)-ARCH(1)	D	0.384	0.118***	0.059***	1.984***	-	0.979***	-	-	-	-148.67	1.12
				W	153.102	0.982***	101.387***	0.463***	-	0.121*	-	-	-	-268.43	7.89
				M	6.493	-0.196	1.933	2.030	-	0.969***	-	-	-	-38.87	5.716
		SENSEX Index	AR(1)-ARCH(1)	D	20.973**	0.991***	2.375***	2.676***	-	-0.035	-	-	-	-6089.61	4.94
				W	0.212	-0.010	0.001***	1.108***	-	1.002***	-	-	-	953.76	-3.02
				M	22.965**	0.862***	37.719***	1.593***	-	0.468***	-	-	-	-564.85	7.687
4	Indonesia	JCI Index	AR(1)-ARCH(1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
5	Japan	NKY Index	AR(1)-ARCH(1)	D	7.110	0.991***	2.987	0.027	-	NA	-	-	-	-4731.21	3.96
				W	7.083	0.961***	5.725***	1.301***	-	0.027***	-	-	-	-1308.68	5.24
				M	7.783	0.850***	36.409***	-0.020	-	0.276***	-	-	-	-379.35	6.60
6	Republic of Korea	KOSPI Index	AR(1)-ARCH(1)	D	15.448	0.992***	0.832***	3.158***	-	-0.005***	-	-	-	-5361.34	4.07
				W	15.813	0.961***	8.157***	5.295***	-	0.084***	-	-	-	-1479.35	6.38
				M	17.606	0.850***	250.029***	-0.017	-	0.284***	-	-	-	-451.20	8.42
7	Malaysia	FBMKL CI Index	AR(1)-ARCH(1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
8	New Zealand	NZSE50 FG Index	AR(1)-ARCH(1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
9	Pakistan	KSE100 Index	AR(1)-ARCH(1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

10	Philippin es	PCOMP Index	AR(1)- ARCH(1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
11	Singapor e	FSSTI Index	AR(1)- ARCH(1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
12	Sri Lanka	CSEALL Index	AR(1)- ARCH(1)	D	-0.003	-0.474***	0.113***	2.621***	NA	0.939***	-	-	-	-757.33	1.69	
				W	-0.013	0.157**	0.066***	1.458***	NA	0.952***	-	-	-	-99.50	1.05	
				M	-0.248	0.176	2.175***	-0.030	NA	0.962***	-	-	-	-90.08	4.18	
13	Taiwan	TWSE Index	AR(1)- ARCH(1)	D	0.053	-0.018	0.0003***	0.452***	NA	1.003***	-	-	-	2050.35	-4.73	
				W	0.279	-0.006	0.0001***	0.665***	NA	1.001***	-	-	-	398.80	-5.12	
				M	0.462**	0.015	0.0001***	0.070	NA	0.999***	-	-	-	102.15	-5.60	
14	Thailand	SET Index	AR(1)- ARCH(1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Panel B: ML estimates for during-liberalization period																
Sl. No.	Asian economies		Model	Data frequency	ϕ_0	ϕ_1	α_0	α_1	β_1	λ_1	λ_2	λ_3	λ_4	LL	AIC	
1	China	SHCOM P Index	AR(1)- GARCH(1,1)	D	26.601**	0.992***	0.213***	0.844***	0.628***	-	-0.028***	-	-	-10404.34	5.32	
				W	26.132**	0.956***	10.829***	2.593***	0.163***	-	0.113***	-	-	-2812.70	7.01	
				M	26.984*	0.842***	14.038***	-0.008**	0.930***	-	0.286***	-	-	-815.71	8.54	
2	Hong Kong	HSI Index	AR(1)- GARCH(1,1)	D	4.631	0.988***	0.134***	0.464***	0.710***	-	-0.035***	-	-	-7147.36	4.31	
				W	0.038	0.002	9.170***	0.914***	0.0005	-	0.999***	-	-	2867.20	-8.189	
				M	5.015	0.787***	0.776***	-0.060***	1.043***	-	-0.926***	-	-	-536.88	6.73	
3	India	NIFTY Index	AR(1)- GARCH(1,1)	D	4.841	0.988***	0.160***	0.741***	0.614***	-	-0.021***	-	-	-8813.04	0453	
				W	5.621	0.946***	12.032***	-0.003	0.637***	-	0.094***	-	-	-2634.94	6.36	
				M	6.202	0.799***	11.570***	-0.036***	0.920***	-	0.350***	-	-	-709.91	7.44	
		SENSEX Index	AR(1)- GARCH(1,1)	D	5.377	0.989***	2.348***	-0.001***	0.700***	-	0.018***	-	-	-9472.16	4.90	
				W	0.192	0.987***	2.361***	-0.003	0.675***	-	0.014***	-	-	-2006.06	4.85	
				M	6.558	0.808***	38.168	-0.035**	0.707***	-	0.339***	-	-	-717.82	7.52	
4	Indonesia	JCI Index	AR(1)- GARCH(1,1)	D	13.710	0.989***	0.614***	2.253***	0.191***	-	-0.023***	-	-	-2896.24	4.27	
				W	13.900	0.950***	14.824***	1.175***	-0.003	-	0.022***	-	-	-878.23	6.11	
				M	13.485	0.789***	20.412	0.570*	0.374*	-	0.798***	-	-	-247.63	7.65	
5	Japan	NKY Index	AR(1)- GARCH(1,1)	D	13.982***	0.957***	0.073***	-0.004***	0.995***	-	0.054***	-	-	-772.21	4.44	
				W	14.292***	0.809***	-0.041	-0.043***	1.153***	-	0.245***	-	-	-183.76	5.10	

				M	-0.997	-0.201	2.230	-0.117	0.541	-	0.962***	-	-	-30.66	4.45	
6	Republic of Korea	KOSPI Index	AR(1)-GARCH(1,1)	D	5.917	0.988***	0.123***	0.554***	0.669***	-	-0.020***	-	-	-4518.99	4.20	
				W	5.040	0.926***	4.090***	0.602***	0.505***	-	0.231***	-	-	-1160.46	6.10	
				M	5.481	0.636***	23.322***	0.788***	-0.060	-	0.741***	-	-	-297.50	6.87	
7	Malaysia	FBMKL CI Index	AR(1)-GARCH(1,1)	D	9.770	0.992***	4.702***	1.770***	-0.0007***	-	0.034***	-	-	-11615.59	4.88	
				W	10.610*	0.960***	17.309***	0.024*	0.640***	-	0.073***	-	-	-3399.86	6.75	
				M	10.422	0.834***	33.684**	1.093***	-0.009	-	5.432***	-	-	-979.50	8.48	
8	New Zealand	NZSE50 FG Index	AR(1)-GARCH(1,1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
9	Pakistan	KSE100 Index	AR(1)-GARCH(1,1)	D	0.161	0.159***	0.030***	0.599***	0.098***	-	0.954***	-	-	2.70	0.01	
				W	0.541	0.021**	0.294*	-0.081***	0.521*	-	0.969***	-	-	-114.81	2.32	
				M	-0.252	0.146	0.077	-0.319	1.148***	-	0.968***	-	-	-18.73	2.06	
10	Philippin es	PCOMP Index	AR(1)-GARCH(1,1)	D	16.906	1.015***	3.259***	0.583***	0.210***	-	-0.005	-	-	-1212.73	4.91	
				W	1.667*	0.103	0.204***	0.626***	-0.035	-	0.979***	-	-	-85.84	1.74	
				M	16.272	1.316***	42.328	0.664	0.254	-	0.187*	-	-	-95.46	8.37	
11	Singapor e	FSSTI Index	AR(1)-GARCH(1,1)	D	NA	NA	NA	NA	NA	-	NA	NA	NA	NA	NA	
				W	NA	NA	NA	NA	NA	-	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	-	NA	NA	NA	NA	NA	NA
12	Sri Lanka	CSEALL Index	AR(1)-GARCH(1,1)	D	0.003	0.014	5.570***	1.603***	0.317***	-	0.997***	-	-	2735.52	-4.42	
				W	39.352*	0.959***	0.920***	-0.022***	1.006***	-	0.098***	-	-	-1004.52	7.39	
				M	-0.113	-0.075	4.000	-0.042	0.575	-	0.989***	-	-	-140.81	4.86	
13	Taiwan	TWSE Index	AR(1)-GARCH(1,1)	D	40.114	0.996***	1.554***	0.973***	0.515***	-	0.016***	-	-	-8755.33	5.46	
				W	40.599	0.974***	21.342***	1.293***	0.231***	-	0.055***	-	-	-2143.11	7.38	
				M	41.259*	0.867***	76.491***	0.834***	0.329***	-	0.216***	-	-	-602.43	8.99	
14	Thailand	SET Index	AR(1)-GARCH(1,1)	D	0.290**	-0.022***	0.0002***	0.687***	0.445***	-	0.995***	-	-	823.06	-3.34	
				W	1.303**	-0.014	0.136	0.199	0.718	-	0.948***	-	-	-171.86	3.38	
				M	5.545	0.280	7.900	-0.092	0.573	-	0.968***	-	-	-60.67	5.71	
Panel C: ML estimates for post-liberalization period																
Sl. No.	Asian economies		Model	Data frequency	ϕ_0	ϕ_1	α_0	α_1	β_1	λ_1	λ_2	λ_3	λ_4	LL	AIC	
1	China	SHCOM P Index	AR(1)-GARCH(1,1)	D	0.001	0.001	7.640***	0.693***	0.185***	-	-	0.998***	-	4650.59	-5.72	
				W	0.057	0.008	0.0001***	0.562***	0.175**	-	-	1.001***	-	857.00	-4.99	
				M	0.224	-0.036	0.0500***	1.346***	-0.004	-	-	1.008***	-	-33.73	0.96	
2				D	0.004	-0.032	0.001	-0.0008	0.790**	-	-	0.998***	-	1934.55	-2.34	

	Hong Kong	HSI Index	AR(1)-GARCH(1,1)	W	0.053	0.004	5.140***	0.554***	0.172*	-	-	1.000***	-	1087.54	-6.22
				M	4.233	0.842***	0.974***	-0.093***	1.071***	-	-	0.290***	-	-247.08	6.273
3	India	NIFTY Index	AR(1)-GARCH(1,1)	D	0.010	0.004	0.0001***	0.093***	-0.014***	-	-	0.999***	-	4381.33	-6.22
				W	0.135	-0.063	0.060	-0.009*	0.750***	-	-	0.995***	-	-186.19	1.21
				M	0.813	-0.022	0.066	-0.042***	0.574	-	-	0.999***	-	-21.02	0.68
		SENSEX Index	AR(1)-GARCH(1,1)	D	0.026	0.010	2.350***	0.666***	0.534***	-	-	0.998	-	4255.96	-5.35
				W	0.132	-0.064	0.124	-0.010***	0.589	-	-	0.995***	-	-209.12	1.29
				M	0.802	-0.018	0.043	-0.040***	0.577	-	-	0.999***	-	-5.69	0.28
4	Indonesia	JCI Index	AR(1)-GARCH(1,1)	D	8.947	0.992***	0.252***	0.690***	0.659***	-	-	-0.041***	-	-10779.22	5.17
				W	10.019	10.019***	154.81***	0.007***	-0.965***	-	-	0.080***	-	-3136.28	7.09
				M	10.611	0.842***	23.074***	1.038***	0.333***	-	-	0.493***	-	-837.23	8.17
5	Japan	NKY Index	AR(1)-GARCH(1,1)	D	21.881**	0.991***	4.396	-0.001***	0.720***	-	-	0.017***	-	-4262.37	5.53
				W	7.417*	0.963***	1.772***	-0.003***	0.942***	-	-	0.069***	-	-4777.93	6.19
				M	24.589**	0.960***	0.556***	-0.026***	1.009***	-	-	0.070***	-	-1156.88	6.52
6	Republic of Korea	KOSPI Index	AR(1)-GARCH(1,1)	D	8.304*	0.987***	0.111***	0.282***	0.805***	-	-	0.099***	-	-9737.27	4.93
				W	8.651**	0.938***	3.327***	0.747***	0.497***	-	-	0.234***	-	-2719.44	6.38
				M	9.194**	0.736***	19.575***	1.875***	0.169***	-	-	0.504***	-	-742.06	7.58
7	Malaysia	FBMKL CI Index	AR(1)-GARCH(1,1)	D	6.024**	0.989***	3.005***	0.380***	-0.006*	-	-	0.038***	-	-6206.3	3.77
				W	6.666***	0.953***	0.864***	0.689***	0.563***	-	-	0.174***	-	-1770.82	5.09
				M	6.623***	0.791***	8.441	-0.030***	0.755***	-	-	0.341***	-	-503.68	6.31
8	New Zealand	NZSE50 FG Index	AR(1)-GARCH(1,1)	D	7.406**	0.988***	0.011***	-0.003***	0.0006***	-	-	0.021***	-	-6450.42	4.04
				W	8.263**	0.946***	0.074***	-0.010***	1.013***	-	-	0.087***	-	-1811.00	5.48
				M	8.28**	0.799***	9.834**	-0.041***	0.0845***	-	-	0.324***	-	-501.97	6.62
9	Pakistan	KSE100 Index	AR(1)-GARCH(1,1)	D	11.497**	0.990***	0.279***	0.956***	0.536***	-	-	-0.040***	-	-12033.53	4.83
				W	10.528**	0.957***	0.062***	-0.007***	1.008***	-	-	0.078***	-	-3400.83	6.30
				M	11.199**	0.811***	101.118	-0.020***	0.490	-	-	0.350***	-	-970.69	7.83
10	Philippines	PCOMP Index	AR(1)-GARCH(1,1)	D	4.890	0.992***	-0.0005*	-0.001***	-0.001***	-	-	0.018***	-	-11749.77	4.58
				W	5.486	0.961***	30.145***	0.015**	0.015	-	-	0.076***	-	-3390.27	6.28
				M	5.638	0.837***	1.846***	-0.040***	1.023***	-	-	0.311***	-	-896.52	7.24
11	Singapore	FSSTI Index	AR(1)-GARCH(1,1)	D	3.393	0.991***	0.049***	-0.001***	0.988***	-	-	0.016***	-	-7777.93	4.22
				W	4.415	0.962***	11.939	-0.008***	0.590	-	-	0.075***	-	-2234.75	5.84
				M	4.371	0.836***	31.919	-0.023	0.552	-	-	0.256***	-	-609.91	6.94
12	Sri Lanka	CSEALL Index	AR(1)-GARCH(1,1)	D	7.786	0.993***	0.174***	0.711***	0.666***	-	-	-0.036***	-	-10325.46	4.38
				W	8.106	0.972***	7.678***	1.141***	0.247***	-	-	-0.030***	-	-3121.45	6.07
				M	8.468	0.877***	45.690***	1.650***	-0.002	-	-	0.230***	-	-882.16	7.48

13	Taiwan	TWSE Index	AR(1)-GARCH(1,1)	D	4.425	0.989***	0.144***	-0.0009***	0.974***	-	-	0.020***	-	-9328.57	4.51
				W	5.457	0.949***	0.004	-0.009***	1.009***	-	-	0.101***	-	-2405.35	5.68
				M	4.736	0.776***	16..97***	1.035***	0.142	-	-	0.595***	-	-687..50	6.98
14	Thailand	SET Index	AR(1)-GARCH(1,1)	D	7.915*	0.990***	0.177***	0.769***	0.610***	-	-	-0.041***	-	-14602.35	4.82
				W	8.576*	0.952***	1.149***	-0.004***	0.984***	-	-	0.087***	-	-4392.54	6.81
				M	8.336*	0.780***	26.568**	-0.018***	0.877***	-	-	0.358***	-	-1185.08	8.00
Panel D: ML estimates for full-period															
Sl. No.	Asian economies		Model	Data frequency	ϕ_0	ϕ_1	α_0	α_1	β_1	λ_1	λ_2	λ_3	λ_4	LL	AIC
1	China	SHCOMP Index	AR(1)-GARCH(1,1)	D	19.001*	0.991***	0.052***	0.573***	0.707***	-	-	-	-0.040***	-14652.57	5.06
				W	18.992*	0.955***	1.820***	1.396***	0.410***	-	-	-	-0.026***	-4062.30	6.79
				M	19,493*	0.828***	8.467**	0.744***	0.543***	-	-	-	0.039***	-1203.79	8.47
2	Hong Kong	HSI Index	AR(1)-GARCH(1,1)	D	11.790*	.990***	0.168***	0.624***	0.666***	-	-	-	-0.035***	-23112.65	4.72
				W	12.280***	0.953***	8.535***	1.505***	0.165***	-	-	-	-0.034***	-6666.48	6.44
				M	12.218*	0.810***	130.191***	0.421***	-0.018	-	-	-	0.198***	-1883.44	7.91
3	India	NIFTY Index	AR(1)-GARCH(1,1)	D	7.953*	0.991***	0.154***	0.767***	0.617***	-	-	-	-0.034***	-13017.59	4.67
				W	9.947**	0.962***	24.718	-0.007***	0.447	-	-	-	0.072***	-4044.40	6.58
				M	9.972**	0.830***	80.098	0.066	0.393	-	-	-	0.285***	-1114.11	7.84
		SENSEX Index	AR(1)-GARCH(1,1)	D	10.212**	0.991***	0.139***	0.646***	0.661***	-	-	-	-0.035***	-18396.24	4.64
				W	11.689***	0.962***	76.030***	-0.004***	-0.718***	-	-	-	0.062***	-5920.29	6.62
				M	11.833***	0.838***	10.199***	0.010**	0.919***	-	-	-	0.304***	-1620.37	7.79
4	Indonesia	JCI Index	AR(1)-GARCH(1,1)	D	10.094	0.991***	0.308***	0.857***	0.600***	-	-	-	-0.037***	-13762.47	4.98
				W	10.984*	0.959***	59.175***	0.525***	-0.008**	-	-	-	0.068***	-4145.43	7.07
				M	10.29	0.834***	12.558***	-0.017***	0.964***	-	-	-	0.306***	-1110.75	8.17
5	Japan	NKY Index	AR(1)-GARCH(1,1)	D	7.309**	0.991***	7.972***	0.253***	-0.003***	-	-	-	-0.002***	-23500.58	4.69
				W	7.472**	0.962***	1.965***	-0.0025***	0.924***	-	-	-	0.072***	-6419.25	6.06
				M	7.893**	0.861***	2.048***	0.017***	0.095***	-	-	-	0.027***	-1747.07	7.16
6	Republic of Korea	KOSPI Index	AR(1)-GARCH(1,1)	D	10.106***	0.990***	0.153***	0.934***	0.546***	-	-	-	-0.025***	-19759.83	4.52
				W	10.177***	0.949***	3.082***	0.871***	0.470***	-	-	-	0.209***	-5391.27	6.34
				M	10.755***	0.785***	69.206	-0.014	0.616	-	-	-	0.383***	-1568.66	8.00
7	Malaysia	FBMKL CI Index	AR(1)-GARCH(1,1)	D	10.164***	0.991***	0.048***	0.620***	0.705***	-	-	-	0.069***	-19937.10	4.42
				W	10.762***	0.958***	33.975***	0.023**	0.035	-	-	-	0.077***	-6136.95	6.42
				M	0.052	-0.090*	0.004***	0.485***	0.243***	-	-	-	0.977***	351.38	-1.57
8	New Zealand	NZSE50 FG Index	AR(1)-GARCH(1,1)	D	7.406**	0.988***	0.011***	-0.003***	1.000***	-	-	-	0.021***	-6450.42	4.04
				W	8.263**	0.946***	0.074***	-0.010***	0.013***	-	-	-	0.087***	-1811.00	5.48

				M	8.282**	0.799***	9.834**	-0.041***	0.845***	-	-	-	0.324***	-501.97	6.62
9	Pakistan	KSE100 Index	AR(1)-GARCH(1,1)	D	10.449**	0.989***	0.305***	1.001***	0.517***	-	-	-	-0.040***	-12883.38	4.81
				W	9.423**	0.957***	0.027***	-0.005***	1.007***	-	-	-	0.080***	-3756.00	6.34
				M	10.206**	0.805***	129.562	-0.020	0.336	-	-	-	0.362***	-1065.13	7.83
10	Philippines	PCOMP Index	AR(1)-GARCH(1,1)	D	7.658*	0.989***	0.317***	0.806***	0.578***	-	-	-	-0.036***	-13148.09	4.682
				W	7.964*	0.950***	24.2999***	0.872***	-0.004	-	-	-	0.0007	-3808.28	6.43
				M	8.237*	0.780***	56.454***	0.810***	-0.004	-	-	-	0.435***	-1012.59	7.45
11	Singapore	FSSTI Index	AR(1)-GARCH(1,1)	D	3.393	0.991***	0.049***	-0.001***	0.988***	-	-	-	0.016***	-7777.93	4.22
				W	4.415	0.962***	11.939	-0.008***	0.590	-	-	-	0.075***	-2234.75	5.84
				M	4.371	0.836***	31.919	-0.023	0.552	-	-	-	0.256***	-609.91	6.94
12	Sri Lanka	CSEALL Index	AR(1)-GARCH(1,1)	D	12.522**	.989***	18.373***	2.205***	-0.000***	-	-	-	0.022***	-19504.98	5.703
				W	13.182*	.970***	615.536***	0.381***	-0.057	-	-	-	0.951***	-6780.72	9.07
				M	13.299*	0.864***	26.902***	2.334***	0.159***	-	-	-	0.230***	-1348.67	7.84
13	Taiwan	TWSE Index	AR(1)-GARCH(1,1)	D	19.120	0.996***	0.779***	1.040***	0.530***	-	-	-	-0.0006***	-20109.42	4.89
				W	18.900*	0.975***	9.301***	1.465***	0.235***	-	-	-	-0.003***	-5237.99	6.61
				M	18.663*	0.876***	14.713***	0.194***	0.767***	-	-	-	0.258***	-1480.40	8.05
14	Thailand	SET Index	AR(1)-GARCH(1,1)	D	11.748**	0.990***	9.388	-0.001	0.584	-	-	-	0.019***	-18420.50	5.62
				W	12.386**	0.955***	0.074***	-0.005***	1.005***	-	-	-	0.081**	-4629.33	6.63
				M	12.121**	0.798***	136.225	-0.022***	0.567	-	-	-	0.374***	-1323.18	8.27

Note: ^a **LL** denotes value of log-likelihood function

^b **AIC** denotes Akaike Information Criteria

^c *, **, and *** denotes significance at 10, 5 and 1 % level, respectively

^d D, W, M denotes Daily, Weekly, and Monthly frequency data-set, respectively

^e NA denotes Not Applicable

The monthly GARCH coefficient of NZSE50FG (New Zealand) stock returns for full period is 0.845 and the coefficient for full period is statistically significant at 1 percent level. However, for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during-liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. The coefficient of post-liberalization period is 0.324 and 0.385 for the GARCH and TARARCH model, respectively and are statistically significant at all conventional levels. Thus, there exist evidence for the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicates that past volatility has had the greater impact on current volatility during the period of liberalization and hence, reveals that the stock market liberalization in New Zealand leads to increment in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.324 in GARCH model and 0.385 in the TARARCH model, and both the coefficient is statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly NZSE50FG stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

The monthly GARCH coefficient of KSE100 (Pakistan) stock returns for during-liberalization and full period is 1.148 and 0.336, respectively and during-liberalization period coefficient is significant at all conventional levels. However, the coefficient for full period is statistically insignificant and subsequently resembles coefficient in during-liberalization period surge highest. Hence, suggests that past volatility possess largest impact on current volatility in during-liberalization period. However, for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during-liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 0.968 in the GARCH model (table 8), and 0.952 in the TARARCH model (table 9), and both of which are significant at 1 percent level. These results signifies that volatility is significantly higher in during-liberalization period. The coefficient of post-liberalization period is 0.350 and 0.308 for the GARCH and TARARCH model, respectively and are statistically significant at all conventional levels. Thus, there exist evidence for the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicates that past volatility has had the greater impact on current volatility during the period of liberalization and hence, reveals that the stock market liberalization in Pakistan leads to increment in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.362 in GARCH model and 0.319 in the TARARCH model, and both the coefficient is statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the

daily and weekly KSE100 stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

The monthly GARCH coefficient of PCOMP (Philippines) stock returns for during-liberalization and full period is 0.254 and -0.004, respectively and both the coefficient are statistically insignificant. However, for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during-liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 0.187 in the GARCH model (table 8), and 0.171 in the TARCH model (table 9), and the GARCH coefficient is significant at 10 percent level and TARCH coefficient is insignificant. These evidence suggest that the volatility in the during-liberalization period is not different from that in pre-liberalization. The coefficient of post-liberalization period is 0.311 and 0.313 for the GARCH and TARCH model, respectively and are statistically significant at all conventional levels. Thus, there exist evidence for the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicates that past volatility has had the greater impact on current volatility during the period of liberalization and hence, reveals that the stock market liberalization in Philippines leads to increment in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.435 in GARCH model and 0.349 in the TARCH model, and both the coefficient is statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly PCOMP stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

The monthly GARCH coefficient of FSSTI (Singapore) stock returns for full period is 0.552 and the coefficient is insignificant. However, for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during-liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.256 in GARCH model and 0.302 in the TARCH model, and both the coefficient is statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly FSSTI stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

The monthly GARCH coefficient of CSEALL (Sri Lanka) stock returns for during-liberalization and full period is 0.575 and 0.159, and full-period coefficient is significant at all conventional levels. However, the coefficient for during-liberalization period is statistically insignificant and subsequently resembles coefficient in full-period surge highest. However, for precision the impact

of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during-liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 0.989 in the GARCH model (table 8), and 0.999 in the TARCH model (table 9), and the GARCH coefficient is significant at 1 percent level and TARCH coefficient is insignificant. These evidence suggest that the volatility in the during-liberalization period is significantly different from that in pre-liberalization. The coefficient of post-liberalization period is 0.230 and 0.233 for the GARCH and TARCH model, respectively and the GARCH coefficient is statistically significant at all conventional levels, in turn TGARCH coefficient is insignificant. Thus, there exist evidence for the volatility in the post-liberalization period is significantly and partially different from that of pre-liberalization period. Therefore, the results indicates that past volatility has had the high impact on current volatility during the period of liberalization and hence, reveals that the stock market liberalization in Sri Lanka leads to increment in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.230 in GARCH model and 0.235 in the TARCH model, and both the coefficient is statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly CSEALL stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

The monthly GARCH coefficient of TWSE (Taiwan) stock returns for during-liberalization and full period is 0.329 and 0.767, respectively and both of the coefficient is significant at 1 percent level. Subsequently resembles coefficient in during-liberalization period declines and hence, suggests that past volatility has had largest impact on current volatility in full period. However, for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during-liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 0.216 in the GARCH model (table 8), and 0.246 in the TARCH model (table 9), and both the coefficient is significant at 1 percent level. These evidence suggest that the volatility in the during-liberalization period is significantly different from that in pre-liberalization. The coefficient of post-liberalization period is 0.595 and 0.588 for the GARCH and TARCH model, respectively and both the coefficient is statistically significant at all conventional levels. Thus, there exist evidence for the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicates that past volatility has had the high impact on current volatility during the period of liberalization and hence, reveals that the stock market liberalization in Taiwan leads to increment in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.258 in GARCH model and 0.162 in the TARCH model, and GARCH coefficient is statistically significant at all conventional levels. However, the TARCH coefficient is insignificant. It indicates that there has

been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly TWSE stock returns estimates of financial stock market liberalization on returns volatility further adds on the evidences to unlock and understand the magnitude of impact.

The monthly GARCH coefficient of SET (Thailand) stock returns for during-liberalization and full period is 0.573 and 0.567, respectively and both of the coefficient are insignificant. However, for precision the impact of stock market liberalization on volatility is estimated through the coefficients on the variables namely, Duringlib, Postlib, and PreduringPostlib which represent during-liberalization, post-liberalization and pre- plus during plus post-liberalization periods, respectively. During-liberalization coefficient is 0.968 in the GARCH model (table 8), and 0.908 in the TARCH model (table 9), and both the coefficient is significant at 1 percent level. These evidence suggest that the volatility in the during-liberalization period is significantly different from that in pre-liberalization. The coefficient of post-liberalization period is 0.358 and 0.337 for the GARCH and TARCH model, respectively and the GARCH coefficient is statistically significant at all conventional levels, though TGARCH coefficient is insignificant. Thus, there exist evidence for the volatility in the post-liberalization period is significantly different from that of pre-liberalization period. Therefore, the results indicates that past volatility has had the high impact on current volatility during the period of liberalization and hence, reveals that the stock market liberalization in Thailand leads to increment in returns volatility. The PreduringPostlib coefficient is used to estimate if the volatility in the full period is significantly different relative to pre-liberalization period. The coefficient is 0.374 in GARCH model and 0.344 in the TARCH model, and both the coefficient are statistically significant at all conventional levels. It indicates that there has been an increase in returns volatility following liberalization reform compared with the period, where stock market was not open to the foreign investors. However, the daily and weekly SET stock returns estimates of financial stock market liberalization on returns volatility further accumulates on the evidences to unlock and understand the magnitude of impact.

7.2 Asymmetric shocks and returns volatility in Asian stock market

The result of impact of asymmetric shocks on volatility in the TGARCH model, are reported in table 9 of panel A, B, C, and D for the 14 Asian stock markets. The asymmetric coefficient (γ) captures the impact of negative and or positive shocks on returns volatility. If asymmetric coefficient significantly greater than zero will specify that negative shocks cause more volatility rather than positive shocks.

The asymmetric coefficient of monthly SHCOMP (China) stock index for full period is -0.719 and is significantly less than zero. Subsequently the asymmetric effect is significantly negative in the during-liberalization period (-0.704). Hence, the negative shocks are significantly negatively related with returns volatility in China for both the full and pre-liberalization periods. Moreover, the evidences suggests that SHCOMP stock index not experienced leverage effect. The asymmetric coefficient of monthly HSI (Hong Kong) stock index for full period is -0.331, which is insignificant, subsequently, the coefficient in during- and post-liberalization period is -0.144 and

0.034, and both are statistically insignificant. Hence, the evidences accounts that HSI stock index not experienced either positive and or negative asymmetric shocks in returns volatility.

The asymmetric coefficient of monthly Nifty and Sensex (India) stock indexes for full liberalization period is -0.112 and -0.437, which is statistically significant at all conventional level. Subsequently, the coefficient in during-liberalization is -0.091 and -0.080 and both are insignificant. The coefficient for post-liberalization period is -0.110 and -0.112, and subsequently both are statistically insignificant. Hence, the evidences accounts that Indian stock index experienced negative asymmetric shocks are significantly negatively related with the returns volatility in the full liberalization period and in turn owes stock market less volatile. However, no evidences found in both the liberalization periods for either positive and or negative asymmetric shocks in returns volatility. The asymmetric coefficient of weekly JCI (Indonesia) stock index for full period is -0.072, and is statistically significant at all conventional levels. Subsequently, the coefficient in during-, and post-liberalization periods are -1.305, -0.020 and both are significant at 1 percent levels. Hence, the evidences accounts that Indonesian stock index experienced negative asymmetric shocks are significantly negatively related with the returns volatility in the full period and successively for both the liberalization periods, and witnessed significant negative associationsip among asymmetric shocks and returns volatility.

The asymmetric coefficient of monthly NKY (Japan) stock index for full period is 0.542, and is statistically significant at all conventional levels. Subsequently, the asymmetric coefficient in post-liberalization period is 0.058 and is significant at 1 percent level. Thus, negative shocks are significantly positively correlated with returns volatility indicating that negative shocks made the NKY stock market volatile in both full and post-liberalization periods. Hence, the evidences suggests that NKY stock index experienced high volatility in post-liberalization period indicating the significant leverage effect. Thus, the result emphasis that the firms listed in NKY stock exchange simultaneously employs the leverage through external source for financing. The asymmetric coefficient of monthly KOSPI (Republic of Korea) stock index for full period is -0.097, and is statistically significant at all conventional levels. Subsequently, the coefficient in during-, and post-liberalization periods are -0.126, 0.060 and both are statistically insignificant. Hence, the evidences accounts that KOSPI stock index experienced negative asymmetric shocks are significantly negatively related with the returns volatility in the full period and successively both the liberalization periods has not possessed significant associationsip among asymmetric shocks and returns volatility.

The asymmetric coefficient of monthly FBMKLCI (Malaysia) stock index for full period is 0.548, and is statistically significant at all conventional levels. Subsequently, the coefficient in during-, and post-liberalization periods are -0.126, 0.004 and both are statistically insignificant. Hence, the evidences reveals that FBMKLCI stock index experienced negative asymmetric shocks are significantly positively related with the returns volatility in the full period and consecutively both the liberalization periods witnessed insignificant associationsip among asymmetric shocks and returns volatility. The asymmetric coefficient of monthly NZSE50FG (New Zealand) stock index for full period is 0.074, and is statistically insignificant. Subsequently, the asymmetric coefficient

in post-liberalization period is -0.074 and is statistically insignificant. Hence, the evidences reveals that NZSE50FG stock index not possessed any impact of the negative asymmetric shocks on the returns volatility in the light of stock market liberalization process. The asymmetric coefficient of monthly KSE100 (Pakistan) stock index for full period is -0.115, and is statistically insignificant. Subsequently, the asymmetric coefficient in during and post-liberalization periods are -0.128 and -0.117 and are statistically insignificant. Hence, the evidences reveals that KSE100 stock index not possessed any impact of the negative asymmetric shocks on the returns volatility in the light of stock market liberalization process. The asymmetric coefficient of monthly PCOMP (Philippines) stock index for full period is -0.213, and is statistically significant at all conventional levels. Subsequently, the asymmetric coefficient in during-, and post-liberalization periods are 0.489, -0.021 and both are statistically insignificant. Hence, the evidences reveals that PCOMP stock index witnessed negative asymmetric shocks are significantly negatively correlated with the returns volatility in the full period and successively both the liberalization periods witnessed insignificant negative associationsip among asymmetric shocks and returns volatility.

The asymmetric coefficient of monthly FSSTI (Singapore) stock index for full period is -0.060, and is statistically insignificant. Subsequently, the asymmetric coefficient in post-liberalization period is -0.060 and is statistically insignificant. Hence, the evidences reveals that FSSTI stock index not possessed any impact of the negative asymmetric shocks on the returns volatility in the light of stock market liberalization process. The asymmetric coefficient of monthly CSEALL (Sri Lanka) stock index for full period is -0.121, and is statistically insignificant. Subsequently, the asymmetric coefficient in during and post-liberalization periods are -0.127 and -0.606 and are statistically insignificant. Hence, the evidences reveals that CSEALL stock index not possessed any impact of the negative asymmetric shocks on the returns volatility for the instance of stock market liberalization process. The asymmetric coefficient of monthly TWSE (Taiwan) stock index for full period is -0.302, and is statistically significant at all conventional levels. Subsequently, the asymmetric coefficient in during-, and post-liberalization periods are -0.835, -0.409 and both are statistically insignificant. Hence, the evidences reveals that TWSE index witnessed negative asymmetric shocks are significantly negatively correlated with the returns volatility in the full period and successively both the liberalization periods witnessed insignificant negative associationsip among asymmetric shocks and returns volatility. The asymmetric coefficient of monthly SET (Thailand) stock index for full period is -0.114, and is statistically insignificant. Subsequently, the asymmetric coefficient in during and post-liberalization periods are -0.488 and -0.132 and the coefficient of during-liberalization period is statistically significant at 10 percent level. However, the coefficient in post-liberalization period owes insignificant. Thus, the evidences reveals that SET stock index not possessed any impact of the negative asymmetric shocks on the returns volatility in the light of stock market liberalization process with an exception of pre-liberalization period wherein the asymmetric coefficient is significant at 10 percent level. Hence, the negative asymmetric shock is significantly negatively correlated with returns volatility as present in table 9.

Table 9 Maximum likelihood estimates for the AR(s)-TGARCH(1,1)

Panel A: ML estimates for pre-liberalization period																	
Sl. No.	Asian economies		Model	Data frequency	ϕ_0	ϕ_1	α_0	α_1	β_1	λ_1	λ_2	λ_3	λ_4	γ	LL	AIC	
1	China	SHCOMP Index	AR(1)-TGARCH(1,1)	D	0.83***	0.192***	0.012***	0.550**	0.145**	0.868***	-	-	-	0.601*	68.91	-0.49	
				W	4.175***	0.400***	2.96**	0.806***	-0.486	0.763***	-	-	-	-0.57**	-94.93	3.95	
				M	4.496	-0.361	15.212	-0.260	0.433	0.888***	-	-	-	0.73	-30.13	7.22	
2	Hong Kong	HSI Index	AR(1)-TGARCH(1,1)	D	20.561***	0.988***	0.383***	1.299***	0.595***	-0.026***	-	-	-	-0.85***	-12101.16	5.04	
				W	21.127***	0.946***	51.426**	0.133*	0.588*	0.101***	-	-	-	-0.141**	-3710.38	7.308	
				M	-0.009	-0.043	0.077**	0.099***	0.546***	1.029***	-	-	-	-0.130***	-119.22	1.07	
3	India	NIFTY Index	AR(1)-TGARCH(1,1)	D	0.384	0.118***	0.030***	1.222***	0.145***	0.957***	-	-	-	0.274*	-128.80	0.99	
				W	153.102	0.982***	161.42	0.039	0.437	0.102	-	-	-	-0.110	-271.64	8.04	
				M	6.493	-0.196	2.073	-0.175	0.351	0.961***	-	-	-	2.427	-36.17	5.62	
		SENSEX Index	AR(1)-TGARCH(1,1)	D	20.973**	0.991***	0.170***	0.201***	0.723***	0.078***	-	-	-	0.354***	-5704.18	4.63	
				W	0.212	-0.010	0.0002***	1.163***	0.634***	1.001***	-	-	-	-1.185	1012.27	-3.20	
				M	22.965**	0.862***	35.840***	0.869***	0.0104***	0.460***	-	-	-	1.174	-562.62	7.68	
4	Indonesia	JCI Index	AR(1)-TGARCH(1,1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
5	Japan	NKY Index	AR(1)-TGARCH(1,1)	D	7.110	0.991***	0.570***	0.2448***	0.697***	-0.019***	-	-	-	-0.246***	-4499.44	3.77	
				W	7.083	0.961***	0.867***	0.534***	0.542***	0.150***	-	-	-	0.425***	-1322.67	5.30	
				M	7.783	0.850***	17.547	0.007	0.638	0.268***	-	-	-	-0.036	-3777.66	6.61	
6	Republic of Korea	KOSPI Index	AR(1)-TGARCH(1,1)	D	15.448	0.992***	0.238***	1.281***	0.641***	-0.007***	-	-	-	-1.282***	-5237.80	3.98	
				W	15.813	0.961***	65.901***	1.856**	-0.009	0.095***	-	-	-	-1.314	-1628.19	7.02	
				M	17.606	0.850***	0.691	0.805***	0.835***	0.142***	-	-	-	-0.870***	-393.07	7.39	
7	Malaysia	FBMKLCI Index	AR(1)-TGARCH(1,1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
8	New Zealand	NZSE50FG Index	AR(1)-TGARCH(1,1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
9	Pakistan	KSE100 Index	AR(1)-TGARCH(1,1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10	Philippines	PCOMP Index	AR(1)-TGARCH(1,1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

11	Singapore	FSSTI Index	AR(1)-TGARCH(1,1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
12	Sri Lanka	CSEALL Index	AR(1)-TGARCH(1,1)	D	-0.003	-0.474***	2.785***	0.430***	-0.0005	0.909***	-	-	-	0.255***	-1461.44	3.27
				W	-0.013	0.157**	0.113***	0.066	-0.013**	0.965***	-	-	-	1.295***	-108.61	1.16
				M	-0.248	0.176	2.175***	-0.030	-	0.962***	-	-	-	-	-90.08	4.181
13	Taiwan	TWSE Index	AR(1)-TGARCH(1,1)	D	0.053	-0.018	0.0003***	0.501***	0.001	1.003***	-	-	-	-0.116*	2050.80	-4.72
				W	0.279	-0.006	0.0001**	0.637***	0.413***	1.001***	-	-	-	-0.655***	399.22	-5.10
				M	0.462**	0.015	0.0002	0.0206	-0.539	0.998***	-	-	-	-0.264	103.25	-5.55
14	Thailand	SET Index	AR(1)-TGARCH(1,1)	D	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				W	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
				M	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Panel B: ML estimates for during-liberalization period																
Sl. No.	Asian economies		Model	Data frequency	ϕ_0	ϕ_1	α_0	α_1	β_1	λ_1	λ_2	λ_3	λ_4	γ	LL	AIC
1	China	SHCOMP Index	AR(1)-TGARCH(1,1)	D	26.601**	0.992***	0.438***	1.175***	0.599***	-	-0.026***	-	-	-0.634***	-10363.94	5.30
				W	26.132**	0.956***	11.850***	3.196***	0.220***	-	0.114***	-	-	-1.945***	-2802.88	6.99
				M	26.984*	0.842***	41.720***	0.665***	0.703***	-	0.258***	-	-	-0.704***	-797.43	8.36
2	Hong Kong	HSI Index	AR(1)-TGARCH(1,1)	D	4.631	0.988***	0.140***	0.081***	0.947***	-	-0.006***	-	-	-0.083***	-7358.27	4.44
				W	0.038	0.002	9.280***	0.617***	0.0002	-	0.999***	-	-	0.531***	2869.75	-8.19
				M	5.015	0.787***	0.190	0.114***	0.979***	-	0.299***	-	-	-0.144	-555.25	6.97
3	India	NIFTY Index	AR(1)-TGARCH(1,1)	D	4.841	0.988***	0.318***	0.796***	0.720	-	-0.008***	-	-	-0.713***	-8726.57	4.49
				W	5.621	0.946***	23.365***	0.221	0.352***	-	0.005	-	-	-0.229	-2636.22	6.37
				M	6.202	0.799***	43.97	0.052	0.667*	-	0.331***	-	-	-0.091	-717.14	7.53
		SENSEX Index	AR(1)-TGARCH(1,1)	D	5.377	0.989***	0.439***	0.499***	0.780***	-	-0.005***	-	-	-0.501***	-8846.50	4.58
				W	0.192	0.987***	5.970***	0.071	0.534***	-	0.014*	-	-	-0.07	-2070.97	5.01
				M	6.558	0.808***	50.916***	0.043	0.614***	-	0.315***	-	-	-0.080	-719.86	7.56
4	Indonesia	JCI Index	AR(1)-TGARCH(1,1)	D	13.710	0.989***	9.616	0.151	0.598	-	0.020*	-	-	-0.152	-3793.16	5.59
				W	13.900	0.950***	12.074***	1.304***	0.131**	-	0.032**	-	-	-1.305***	-859.49	5.98
				M	13.485	0.789***	20.412	0.570*	0.374*	-	0.798***	-	-	-	-247.63	7.65
5	Japan	NKY Index	AR(1)-TGARCH(1,1)	D	13.982***	0.957***	0.448	0.087***	-0.094***	-	0.938***	-	-	-0.094***	-797.64	4.59
				W	14.292***	0.809***	1.912***	-0.414***	0.956***	-	0.306***	-	-	0.386***	-191.81	5.34
				M	-0.997	-0.201	2.312	-0.0134	0.561	-	0.968***	-	-	-0.116	-30.61	4.57
6	Republic of Korea	KOSPI Index	AR(1)-TGARCH(1,1)	D	5.917	0.988***	0.197***	0.290***	0.649***	-	0.021***	-	-	0.795***	-4536.21	4.21
				W	5.040	0.926***	3.803***	0.651***	0.525***	-	0.231***	-	-	-0.139	-1160.12	6.10
				M	5.481	0.636***	22.006***	1.031***	-0.043	-	0.731***	-	-	-0.126	-296.35	6.87
7	Malaysia	FBMKLCI Index	AR(1)-TGARCH(1,1)	D	9.770	0.992***	3.316***	2.947***	0.022***	-	-0.022***	-	-	0.081	-11607.42	4.88
				W	10.610*	0.960***	17.756***	0.084***	0.723***	-	0.033***	-	-	-0.090***	-3404.56	6.76
				M	10.422	0.834***	34.014**	1.144***	-0.009	-	5.515***	-	-	-0.126	-979.38	8.49

8	New Zealand	NZSE50FG Index	AR(1)-TGARCH(1,1)	D	NA	NA	NA	NA	NA	-	NA	-	-	NA	NA	NA
				W	NA	NA	NA	NA	NA	-	NA	-	-	NA	NA	NA
				M	NA	NA	NA	NA	NA	-	NA	-	-	NA	NA	NA
9	Pakistan	KSE100 Index	AR(1)-TGARCH(1,1)	D	0.161	0.159***	0.028***	0.925***	0.129*	-	0.957***	-	-	-0.628***	6.16	-0.0009
				W	0.541	0.021**	0.296	0.051	0.528	-	0.962***	-	-	-0.132	-112.89	2.30
				M	-0.252	0.146	-0.032	-0.179	1.346***	-	0.952***	-	-	-0.128	-19.40	2.20
10	Philippines	PCOMP Index	AR(1)-TGARCH(1,1)	D	16.906	1.015***	2.845***	0.423***	0.265***	-	-0.005	-	-	0.305**	-1211.52	4.90
				W	1.667*	0.103	0.143***	-0.012	0.169	-	0.987***	-	-	1.450***	-79.74	1.64
				M	16.272	1.316***	49.672	0.410	0.203	-	0.171	-	-	0.489	-95.28	8.44
11	Singapore	FSSTI Index	AR(1)-TGARCH(1,1)	D	NA	NA	NA	NA	NA	-	NA	-	-	NA	NA	NA
				W	NA	NA	NA	NA	NA	-	NA	-	-	NA	NA	NA
				M	NA	NA	NA	NA	NA	-	NA	-	-	NA	NA	NA
12	Sri Lanka	CSEALL Index	AR(1)-TGARCH(1,1)	D	0.003	0.014	7.410***	-0.002***	0.834***	-	0.998***	-	-	0.210***	2284.35	-3.68
				W	39.352*	0.959***	2.675***	0.167***	0.913***	-	-0.021***	-	-	-0180***	-1025.33	7.55
				M	-0.113	-0.075	4.210	0.078	0.580	-	0.999	-	-	-0.127	-141.06	4.90
13	Taiwan	TWSE Index	AR(1)-TGARCH(1,1)	D	40.114	0.996***	1.547***	0.991***	0.517***	-	-0.016***	-	-	-0.044*	-8755.23	5.46
				W	40.599	0.974***	21.041***	1.613***	0.238***	-	0.062***	-	-	-0.61***	-2140.92	7.37
				M	41.259*	0.867***	59.218***	1.233***	0.418***	-	0.246***	-	-	-0.835	-599.62	8.97
14	Thailand	SET Index	AR(1)-TGARCH(1,1)	D	0.290**	-0.022***	0.0002***	1.126***	0.466***	-	1.003***	-	-	-0.759***	847.44	-3.44
				W	1.303**	-0.014	0.136	0.199	0.718	-	0.94***	-	-	-	-171.86	3.38
				M	5.545	0.280	8.888	-0.099	0.612	-	0.908***	-	-	-0.488*	-58.09	5.57
Panel C: ML estimates for post-liberalization period																
Sl. No.	Asian economies		Model	Data frequency	ϕ_0	ϕ_1	α_0	α_1	β_1	λ_1	λ_2	λ_3	λ_4	γ	LL	AIC
1	China	SHCOMP Index	AR(1)-TGARCH(1,1)	D	0.001	0.001	7.640***	0.690***	0.185***	-	-	0.998***	-	0.005	4650.60	-5.72
				W	0.057	0.008	0.0001***	0.353***	0.158*	-	-	1.001***	-	0.401**	859.33	-5.00
				M	0.224	-0.036	0.142	0.092	0.584	-	-	1.004***	-	-0.132	-59.04	1.62
2	Hong Kong	HSI Index	AR(1)-TGARCH(1,1)	D	0.004	-0.032	6.280	0.208***	0.944**	-	-	1.006***	-	-0.216***	2315.21	-2.80
				W	0.053	0.004	5.350***	0.338***	0.155***	-	-	1.000***	-	0.418	1089.69	-6.2
				M	4.233	0.842***	29.551	-0.065***	0.553	-	-	0.269***	-	0.034	-277.83	7.00
3	India	NIFTY Index	AR(1)-TGARCH(1,1)	D	0.010	0.004	8.170***	0.279***	-0.003***	-	-	0.999***	-	-0.145***	4611.47	-6.54
				W	0.135	-0.063	0.068***	0.275***	-0.015	-	-	1.008***	-	-0.113***	-91.57	0.58
				M	0.813	-0.022	0.066	0.065	0.578	-	-	1.000***	-	-0.110	-21.83	0.73
		SENSEX Index	AR(1)-TGARCH(1,1)	D	0.026	0.010	0.0004***	0.201***	-0.003***	-	-	0.998***	-	-0.076***	3728.19	-4.69
				W	0.132	-0.064	0.125	0.057	0.592	-	-	0.999***	-	-0.068	-212.54	1.32
				M	0.802	-0.018	0.044	0.069	0.579	-	-	1.000	-	-0.112	-5.97	0.31
4	Indonesia	JCI Index	AR(1)-TGARCH(1,1)	D	8.947	0.992***	1.044***	0.375***	0.827***	-	-	-0.008***	-	-0.377	-11151.37	5.35
				W	10.019	10.019***	3.129***	0.012***	0.960***	-	-	0.105***	-	-0.020***	-3134.95	7.09
				M	10.611	0.842***	117.801	0.068	0.564	-	-	0.263***	-	-0.101	-845.69	8.26

5	Japan	NKY Index	AR(1)-TGARCH(1,1)	D	21.881**	0.991***	2.871***	0.109*	0.864*	-	-	-0.001	-	-0.111	-4324.30	5.61
				W	7.417*	0.963***	2.032***	-0.021***	0.938***	-	-	0.081***	-	0.018***	-4773.03	6.19
				M	24.589**	0.960***	23.049	-0.065**	0.636**	-	-	0.084***	-	0.058**	-1225.74	6.91
6	Republic of Korea	KOSPI Index	AR(1)-TGARCH(1,1)	D	8.304*	0.987***	2.304***	0.619***	0.598***	-	-	-0.008***	-	-0.621***	-10017.32	5.07
				W	8.651**	0.938***	3.558***	0.595***	0.478***	-	-	0.231***	-	0.318***	-2716.27	6.37
				M	9.194**	0.736***	19.603***	1.857***	0.166***	-	-	0.504***	-	0.060	-742.06	7.59
7	Malaysia	FBMKLCI Index	AR(1)-TGARCH(1,1)	D	6.024**	0.989***	2.199***	0.581***	-0.003***	-	-	0.040***	-	-0.030	-6091.29	3.70
				W	6.666***	0.953***	8.959***	0.233***	0.205***	-	-	0.086***	-	-0.239***	-1825.73	5.25
				M	6.623***	0.791***	6.603**	-0.034***	0.815***	-	-	0.348***	-	0.004	-504.12	6.33
8	New Zealand	NZSE50FG Index	AR(1)-TGARCH(1,1)	D	7.406**	0.988***	0.009**	0.173***	0.964***	-	-	0.005***	-	-0.176***	-6510.20	4.08
				W	8.263**	0.946***	1.637***	0.019***	0.922***	-	-	0.133***	-	-0.030***	-1907.67	5.78
				M	8.28**	0.799***	22.918	0.029	0.639	-	-	0.385***	-	-0.074	-510.85	6.75
9	Pakistan	KSE100 Index	AR(1)-TGARCH(1,1)	D	11.497**	0.990***	3.410***	0.120***	0.778***	-	-	0.015***	-	-0.121***	-13175.46	5.28
				W	10.528**	0.957***	6.474***	0.394***	0.727***	-	-	0.019***	-	-0.404***	-3479.76	6.44
				M	11.199**	0.811***	89.407	0.093	0.588	-	-	0.308***	-	-0.117	-977.05	7.89
10	Philippines	PCOMP Index	AR(1)-TGARCH(1,1)	D	4.890	0.992***	6.788***	0.122***	0.363***	-	-	0.008***	-	-0.124	-12409.20	4.84
				W	5.486	0.961***	19.816	0.060	0.589**	-	-	0.079***	-	-0.074	-3429.63	6.35
				M	5.638	0.837***	40.890	-0.013	0.648	-	-	0.313***	-	-0.021	-921.98	7.45
11	Singapore	FSSTI Index	AR(1)-TGARCH(1,1)	D	3.393	0.991***	0.121***	0.104***	0.954***	-	-	-0.001***	-	-0.106***	-7759.01	4.21
				W	4.415	0.962***	11.999***	0.065	0.592***	-	-	0.080***	-	-0.073	-2242.59	5.87
				M	4.371	0.836***	20.588	0.036	0.712**	-	-	0.302***	-	-0.060	-609.40	6.95
12	Sri Lanka	CSEALL Index	AR(1)-TGARCH(1,1)	D	7.786	0.993***	2.984***	0.108***	0.718***	-	-	-0.005***	-	-0.109***	-11634.84	4.94
				W	8.106	0.972***	23.032**	0.092***	0.583***	-	-	0.038***	-	-0.097***	-3343.28	6.50
				M	8.468	0.877***	45.884***	1.994***	-0.002	-	-	0.233	-	-0.606	-881.84	7.49
13	Taiwan	TWSE Index	AR(1)-TGARCH(1,1)	D	4.425	0.989***	3.457***	0.134***	0.598***	-	-	0.019***	-	-0.136***	-9571.15	4.62
				W	5.457	0.949***	16.677*	-0.012***	0.493*	-	-	0.085***	-	-0.012***	-2569.12	6.07
				M	4.736	0.776***	15.889***	1.355***	0.155**	-	-	0.588***	-	-0.409	-687.10	6.99
14	Thailand	SET Index	AR(1)-TGARCH(1,1)	D	7.915*	0.990***	2.699***	0.947***	0.535***	-	-	-013***	-	-0.949***	-15171	5.01
				W	8.576*	0.952***	2.030***	0.177***	0.922***	-	-	0.027***	-	-0.188***	-4295.62	6.66
				M	8.336*	0.780***	114.356	0.107	0.578	-	-	0.337	-	-0.132	-1197.60	8.09
Panel D: ML estimates for full period																
Sl. No.	Asian economies		Model	Data frequency	ϕ_0	ϕ_1	α_0	α_1	β_1	λ_1	λ_2	λ_3	λ_4	γ	LL	AIC
1	China	SHCOMP Index	AR(1)-TGARCH(1,1)	D	19.001*	0.991***	2.849***	0.159***	0.792***	-	-	-	-0.038***	-0.160***	-16409.35	5.67
				W	18.992*	0.955***	97.478***	0.266***	0.547***	-	-	-	0.001	-0.273***	-4719.37	7.89
				M	19,493*	0.828***	5.966***	0.773***	0.773***	-	-	-	0.099***	-0.719***	-1182.75	8.335
2	Hong Kong	HSI Index	AR(1)-TGARCH(1,1)	D	11.790*	.990***	14.179***	-0.002	-0.002***	-	-	-	0.033***	0.278***	-25999.81	5.30
				W	12.280***	0.953***	16.514***	0.113***	0.685***	-	-	-	0.013***	-0.118***	-6947.11	6.71
				M	12.218*	0.810***	2.418***	0.277***	0.966***	-	-	-	0.188***	-0.331	-1791.42	7.53

3	India	NIFTY Index	AR(1)-TGARCH(1,1)	D	7.953*	0.991***	0.488***	0.150***	0.907***	-	-	-	-0.011***	-0.152***	-13495.32	4.84
				W	9.947**	0.962***	1.058***	0.138***	0.941***	-	-	-	0.020***	-0.149***	-3936.86	6.40
				M	9.972**	0.830***	84.982	0.073*	0.548	-	-	-	0.266***	-0.112***	-1112.37	7.84
		SENSEX Index	AR(1)-TGARCH(1,1)	D	10.212**	0.991***	0.126***	0.302***	0.742***	-	-	-	0.076***	0.211***	-8348.31	4.63
				W	11.689***	0.962***	0.633***	0.229***	0.939***	-	-	-	0.027***	-0.239***	-5.679.75	6.35
				M	11.833***	0.838***	7.808***	0.408***	0.836***	-	-	-	0.207***	-0.437***	-1579.20	7.60
4	Indonesia	JCI Index	AR(1)-TGARCH(1,1)	D	10.094	0.991***	9.466	0.160	-0.162	-	-	-	-0.0006	-0.162***	-15544.34	5.63
				W	10.984*	0.959***	22.952***	0.066***	-0.072***	-	-	-	0.054***	-0.072***	-4170.87	7.11
				M	10.29	0.834***	10.885***	0.590***	0.810***	-	-	-	0.181***	-0.629***	-1061.65	7.82
5	Japan	NKY Index	AR(1)-TGARCH(1,1)	D	7.309**	0.991***	0.457***	0.124***	0.879***	-	-	-	-0.007***	-0.125***	-22664.11	4.52
				W	7.472**	0.962***	2.520***	-0.016***	0.906***	-	-	-	0.080***	0.014***	-6417.24	6.06
				M	7.893**	0.861***	42.537***	0.090*	-0.003	-	-	-	0.411***	0.542***	-1727.25	7.08
6	Republic of Korea	KOSPI Index	AR(1)-TGARCH(1,1)	D	10.106***	0.990***	1.632***	1.527***	0.505***	-	-	-	-0.006***	-1.528***	-20568.44	4.70
				W	10.177***	0.949***	3.595***	0.661***	0.444***	-	-	-	0.207***	0.412***	-5383.77	6.33
				M	10.755***	0.785***	36.512***	0.080***	0.803***	-	-	-	0.344***	-0.097***	-1563.82	7.98
7	Malaysia	FBMKLCI Index	AR(1)-TGARCH(1,1)	D	10.164***	0.991***	0.062***	0.487***	0.701***	-	-	-	0.067***	0.202***	-19902.54	4.41
				W	10.762***	0.958***	1.079***	0.213***	0.920***	-	-	-	0.034***	-0.002***	-5924.19	6.19
				M	0.052	-0.090*	0.005***	0.137	0.230***	-	-	-	0.979***	0.548***	356.10	-1.59
8	New Zealand	NZSE50FG Index	AR(1)-TGARCH(1,1)	D	7.406**	0.988***	0.009**	0.173***	0.964***	-	-	-	0.005***	-0.176	-6510.20	4.08
				W	8.263**	0.946***	1.637***	0.019***	0.922***	-	-	-	0.133***	-0.030***	-1907.67	5.78
				M	8.282**	0.799***	22.918	0.029	0.639*	-	-	-	0.385***	-0.074	-510.85	6.75
9	Pakistan	KSE100 Index	AR(1)-TGARCH(1,1)	D	10.449**	0.989***	3.904***	0.208***	0.666***	-	-	-	-0.008***	-0.209***	-13848.37	5.18
				W	9.423**	0.957***	5.888***	0.324***	0.755***	-	-	-	0.028***	-0.332***	-3800.35	6.42
				M	10.206**	0.805***	89.73	0.090	0.588	-	-	-	0.319***	-0.115	-1071.94	7.97
10	Philippines	PCOMP Index	AR(1)-TGARCH(1,1)	D	7.658*	0.989***	6.807***	0.154***	0.595***	-	-	-	0.017***	-0.156***	-14705.54	5.23
				W	7.964*	0.950***	8.945***	0.350***	0.621***	-	-	-	0.099***	-0.350	-3796.40	6.41
				M	8.237*	0.780***	13.169***	0.194***	0.811***	-	-	-	0.349***	-0.213***	-1015.76	7.48
11	Singapore	FSSTI Index	AR(1)-TGARCH(1,1)	D	3.393	0.991***	0.121***	0.104***	0.954***	-	-	-	-0.001***	-0.106***	-7759.01	4.21
				W	4.415	0.962***	11.999***	0.065	0.592***	-	-	-	0.080***	-0.073	-2242.59	5.87
				M	4.371	0.836***	20.588	0.0368	0.712**	-	-	-	0.302***	-0.060	-609.40	6.95
12	Sri Lanka	CSEALL Index	AR(1)-TGARCH(1,1)	D	12.522**	.989***	17.006***	0.036***	0.559***	-	-	-	0.017***	-0.037***	-20563.02	6.01
				W	13.182*	.970***	3.561***	1.090***	0.001	-	-	-	0.553***	-0.091***	-5968.70	7.99
				M	13.299*	0.864***	24.573***	3.485***	-2.121	-	-	-	0.235***	-0.121	-1344.43	7.82
13	Taiwan	TWSE Index	AR(1)-TGARCH(1,1)	D	19.120	0.996***	1.075***	2.026***	0.400***	-	-	-	-0.018***	-1.482***	-19956.68	4.86
				W	18.900*	0.975***	9.680***	1.736***	0.235***	-	-	-	0.0007	-0.594***	-5235.11	6.61
				M	18.663*	0.876***	4.542***	0.262***	0.908***	-	-	-	0.162	-0.302***	-1456.24	7.92
14	Thailand	SET Index	AR(1)-TGARCH(1,1)	D	11.748**	0.990***	9.851***	0.145***	0.594***	-	-	-	0.014***	-0.147***	-18497.26	5.65
				W	12.386**	0.955***	43.347***	0.117***	0.595***	-	-	-	0.080***	-0.123***	-4983.90	7.14

				M	12.121**	0.798***	139.11	0.090	0.583	-	-	-	0.344***	-0.114	-1327.61	8.30
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Note: ^a **LL** denotes value of log-likelihood function

^b **AIC** denotes Akaike Information Criteria

^c *, **, and *** denotes significance at 10, 5 and 1 % level, respectively

^d D, W, M denotes Daily, Weekly, and Monthly frequency data-set, respectively

^e NA denotes Not Applicable

7.3 Impact of stock market liberalization on industrial growth in Indian economy

Table 10 presents the result indicating impact of financial stock market liberalization on industrial growth in Indian economy mimicking weekly and monthly frequency data set. Table 10 of panel A, B, and C offers the result of industrial growth regression for during-, post-liberalization period, and full period, respectively. The industries comprised of eleven sectorial industry engaging in non-financial and financial activities and the specifications about the sectorial industry has presented in table 2 of panel B.

The coefficient of Nifty stock returns is 0.016 for monthly and 0.014 for weekly indicates significant positive association with C&RE industry for during-liberalization period at 1 and 10 percent levels, respectively. Hence, the evidence supports that stock market liberalization spurs industrial growth as shown in table 10 of panel A. However, there exist no impact of stock market liberalization on the ELE and SERV industries owing to insignificant Nifty stock returns coefficient for liberalization impact. The coefficient of Nifty stock returns are 0.006 and 0.009 for weekly frequency and are prevails positively significant association with MANU and MIN industries at 5 percent levels, respectively. Thus, the results suggests the stock market liberalization leads to industrial growth. The weekly Nifty stock returns with coefficient (0.007) is positively associated with AFS industry and significant at 5 percent level as shown in table 10 of panel A. The weekly and monthly Nifty stock returns coefficients are 0.009 and 0.015 and is positively associated with BSERV industry at the 1 percent significant level. The FBFS industry is positively associated (0.045) with Nifty index returns at 1 percent significance level. Conversely, the financial stock market liberalization has not witnessed the industrial growth in INVS, OTFS and OFBFS industries.

Table 10 of panel B shows impact of financial stock market liberalization on industrial growth for the post-liberalization period. The Nifty stock returns is positively correlated with C&RE, MANU, MIN, SERV, BSERV, FBFS, INVS and OTFS, through the coefficients (0.016, 0.009, 0.014, 0.009, 0.009, 0.012, 0.004, 0.007) and are statistically significant at 1, 1, 1, 1, 5, 1, 5, 1 percent levels, respectively. Thus, the aforesaid industries witnessed growth after the liberalization of stock market that leads to openness of stock market for the foreign investors. In turn ELE, AFS and OFBFS industries not exposed to the impact of financial stock market liberalization. Table 10 of

panel C proposes impact of financial stock market liberalization on industrial growth for the full period. The Nifty stock returns is positively correlated with C&RE, MANU, MIN, SERV, AFS, BSERV, FBFS, and OTFS, through the coefficients (0.016, 0.007, 0.011, 0.011, 0.006, 0.009, 0.030, 0.007) and are statistically significant at 1, 1, 1, 1, 1, 1, 1, 5 percent levels, respectively. Thus, the aforesaid industries witnessed growth after the liberalization of stock market to the foreign investors. In turn ELE, INVS and OFBFS industries not exposed to the impact of financial stock market liberalization.

7.4 Impact of stock market liberalization on economic growth in Asian economies

Table 11 presents the result indicating impact of financial stock market liberalization on economic growth in Asian economy mimicking monthly frequency data. Table 11 of panel A, B, C and D offers the result of economic growth regression for pre-, during-, post-liberalization period, and full period, respectively after controlling the macro-economic factors, namely CPI and FOREX. The economic growth regression of respective 14 Asian economies is being tested whether there exist evidences claiming association among impact of financial stock market liberalization on economic growth in respective economies after controlling the macro-economic indicators.

Panel A of table 11 shows that prior to stock market liberalization in Asian economies no impact of economic growth been observed relative to stock market development for the pre-liberalization period. However, Panel B of table 11 indicates Hong Kong, Republic of Korea, Malaysia, and Taiwan owes and witnessed the economic growth in the during-liberalization period with statistically significant coefficients 0.014 (1 percent level), -0.015 (5 percent level), 0.047 (1 percent level) and 0.009 (1 percent level). Thus, the stock market liberalization proposes the significant positive association with the economic growth in Hong Kong, Malaysia and Taiwan. Conversely, Republic of Korea experienced negative impact of stock market liberalization on economic growth after controlling the macro-economic indicators. Panel C of table 11 specifies China possessed positive impact of SHCOMP stock index liberalization with significant coefficient of 0.017 (1 percent level) on economic growth. The CPI of China shows positive association with GDP coefficient 0.313 and is significant at 1 percent level. Hong Kong possessed negative impact of HSI stock index liberalization with significant coefficient of -0.053 (1 percent level) on economic growth in the post-liberalization period.

In line with China, Nifty and SENSEX stock index shows negative impact of stock market liberalization on economic growth in India with significant coefficient -0.021 (1 percent level) for both the indexes. However, the CPI of India reflects positive association with the GDP, coefficient of 0.164 (10 percent level) and 0.168 (10 percent level) for Nifty and SENSEX Indexes, respectively. Japan and Republic of Korea has witnessed negative impact of stock market liberalization on respective economic growth (GDP) with the significant coefficients 0.018 (10 percent level) and 0.022 (5 percent level), respectively. Hence, the evidences suggests that Japan and Republic of Korea has witnessed positive impact of stock market liberalization on the respective economic growth in the post-liberalization period, as represent in panel C of table 11. Panel D of table 11 marks the positive impact of SHCOMP, HSI, NKY, KOSPI, FBMKLCI, and FSSTI stock market liberalization on the economic development in the respective economies for the full period, namely, China, Hong Kong, Japan, Republic of Korea, Malaysia, and Singapore with significant coefficients 0.011 (1 percent level), 0.036 (1 percent level), 0.029 (1 percent level), 0.032 (1 percent level), 0.038 (1 percent level), 0.019 (10 percent level), respectively. However, the CPI of China, Hong Kong, Republic of Korea, and Singapore successively indicates positive impact except Singapore with negative impact on their respective economic growth (GDP). Subsequently, the statistically significant coefficient 0.551 (1 percent level), 0.848 (1 percent level), 1.193 (1 percent level), and -0.529 (10 percent level) evidenced and concretized the magnitude of impact of stock market liberalization on economic growth in China, Hong Kong, Republic of Korea, and Singapore, respectively in the full period. Conversely, Nifty and SENSEX stock market liberalization marks negative impact on the economic growth in India expressing significant coefficients -0.013 (5 percent level) respectively. Successively, the CPI and FOREX are negatively correlated with GDP of India owing coefficients -0.12 and -0.46, respectively for the full period. The PCOMP stock index liberalization possess negative impact on the economic growth (GDP) in Philippines with significant coefficient -0.017 at 5 percent level. However, the CPI shows negative association with GDP of Philippines with statistically significant coefficient -0.632 at 1 percent level.

Table 10 Regression modeling impact of financial stock market on industrial growth in India for the period Jan 1995-Sept 2014

Panel A: Industrial growth regression for during-liberalization period								
Serial No.	Industry	Industry specification	Data frequency	α_0	β_1	$SE(\beta_1)$	$\bar{R}^2$	Durbin Watson
1	Non-financial	C&RE	Weekly	-0.090	0.016***	0.003	0.022	1.931
			Monthly	0.063	0.014*	0.008	0.011	2.123
2	Non-financial	ELE	Weekly	0.114	0.001	0.004	-0.001	1.869
			Monthly	-0.035	0.009	0.008	0.001	2.034
3	Non-financial	MANU	Weekly	-0.052	0.006**	0.002	0.007	1.918
			Monthly	0.047	0.008	0.005	0.006	2.088
4	Non-financial	MIN	Weekly	-0.122	0.009**	0.004	0.004	1.956
			Monthly	-0.192	0.013	0.010	0.003	2.093
5	Non-financial	SERV	Weekly	-0.172	0.013	0.004	0.015	1.930
			Monthly	0.130	0.012	0.009	0.004	2.367
6	Financial	AFS	Weekly	-0.182***	0.007**	0.003	0.004	1.988
			Monthly	-0.155	0.011	0.007	0.008	2.080
7	Financial	BSERV	Weekly	-0.034	0.009***	0.003	0.008	1.953
			Monthly	-0.062	0.015***	0.008	0.015	2.287
8	Financial	FBFS	Weekly	-1.655***	0.045***	0.007	0.052	1.615
			Monthly	-1.188***	0.042	0.015	0.039	2.301
9	Financial	INVS	Weekly	2.139	-0.063	0.062	0.000	2.016
			Monthly	0.646***	5.710	0.011	-0.006	0.959
10	Financial	OTFS	Weekly	0.918***	-0.006	0.004	0.001	1.682
			Monthly	0.877***	0.004	0.008	-0.004	1.964
11	Financial	OFBFS	Weekly	0.918***	0.006	0.004	0.001	0.001
			Monthly	0.639**	-0.001	0.011	0.006	0.968
Panel B: Industrial growth regression for post-liberalization period								
Serial No.	Industry	Industry specification	Data frequency	α_0	β_1	$SE(\beta_1)$	$\bar{R}^2$	Durbin Watson
1	Non-financial	C&RE	Weekly	-0.061	0.016***	0.004	0.039	1.94
			Monthly	0.368***	0.002	0.008	-0.012	2.24
2	Non-financial	ELE	Weekly	0.118	0.004	0.003	0.001	1.83
			Monthly	0.323*	0.0008	0.006	-0.013	2.18
3		MANU	Weekly	-0.020	0.009***	0.003	0.020	1.899

	Non-financial		Monthly	0.502***	-0.001	0.005	-0.011	2.33
4	Non-financial	MIN	Weekly	-0.074	0.014***	0.003	0.043	1.97
			Monthly	0.613***	0.008	0.006	0.007	2.05
5	Non-financial	SERV	Weekly	-0.030	0.009***	0.003	0.013	1.86
			Monthly	0.550***	-0.005	0.005	0.0009	1.85
6	Financial	AFS	Weekly	0.013	0.006	0.004	0.003	2.206
			Monthly	0.742***	-0.006	0.008	-0.004	1.98
7	Financial	BSERV	Weekly	0.075	0.009**	0.004	0.011	2.05
			Monthly	0.707***	-0.0007	0.007	-0.013	2.02
8	Financial	FBFS	Weekly	-0.208	0.012***	0.004	0.021	2.01
			Monthly	0.125	0.012	0.008	0.015	1.95
9	Financial	INVS	Weekly	-0.051	0.004**	0.001	0.009	2.13
			Monthly	0.320**	-0.002	0.004	-0.010	2.07
10	Financial	OTFS	Weekly	-0.100	0.007***	0.002	0.024	1.84
			Monthly	0.132	0.006	0.005	0.002	2.03
11	Financial	OFBFS	Weekly	-0.070	0.002	0.001	0.003	2.10
			Monthly	0.216*	-0.002	0.004	-0.010	2.04
Panel C: Industrial growth regression for full period								
Serial No.	Industry	Industry specification	Data frequency	α_0	β_1	$SE(\beta_1)$	$\bar{R}^2$	Durbin Watson
1	Non-financial	C&RE	Weekly	-0.079	0.016***	0.002	0.028	1.93
			Monthly	-0.141	-0.0006	0.011	-0.013	2.30
2	Non-financial	ELE	Weekly	0.113	0.002	0.002	-0.0001	1.86
			Monthly	-0.188	-0.004	0.009	-0.010	2.14
3	Non-financial	MANU	Weekly	-0.043	0.007***	0.002	0.012	1.90
			Monthly	-0.199	0.007	0.007	0.001	2.25
4	Non-financial	MIN	Weekly	-0.108	0.011***	0.003	0.011	1.95
			Monthly	-0.649*	0.009	0.013	-0.007	2.12
5	Non-financial	SERV	Weekly	-0.120*	0.011***	0.002	0.014	1.91
			Monthly	-0.026	0.001	0.012	-0.012	2.39
6	Financial	AFS	Weekly	-0.117**	0.006***	0.002	0.005	2.06
			Monthly	-0.399	-9.260	0.009	-0.013	2.28
7	Financial	BSERV	Weekly	0.003	0.009***	0.002	0.010	1.98
			Monthly	-0.072	-0.006	0.011	-0.009	2.58

8	Financial	FBFS	Weekly	-1.157***	0.030***	0.004	0.036	1.59
			Monthly	-1.136*	0.016	0.021	-0.005	2.54
9	Financial	INVS	Weekly	1.379	-0.033	0.038	-0.0002	2.01
			Monthly	1.285***	1.285	0.015	-0.012	0.80
10	Financial	OTFS	Weekly	0.587***	0.007**	0.002	0.005	1.60
			Monthly	0.676**	0.005	0.011	-0.009	2.14
11	Financial	OFBFS	Weekly	1.364	-0.033	0.038	-0.0002	2.01
			Monthly	1.277***	0.002	0.015	-0.012	0.78

Note: ^a *, **, and *** denotes significance at 10, 5 and 1 % level, respectively

Table 11 Regression modeling impact of financial stock market liberalization on economic growth in Asian economies

Panel A: Economic growth regression for pre-liberalization period											
Sl. No	Countries	Data frequency	α_0	β_1	$SE(\beta_1)$	β_2	$SE(\beta_2)$	β_3	$SE(\beta_3)$	$\bar{R}^2$	Durbin Watson statistics
1	China	Monthly	9.20***	1.53	8.46	1.08	1.52	-2.26**	8.96	-	1.70
2	Hong Kong	Monthly	0.635	0.004	0.005	0.284	0.191	-0.095	0.329	-0.002	1.99
3	India	NIFTY	0.616	-0.0005	0.027	-0.030	0.608	0.057	0.602	-0.270	0.33
		SENSEX	4.066	0.023	0.017	-0.374	0.596	-0.081	0.470	-0.079	0.78
4	Indonesia	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
5	Japan	Monthly	4.010***	-0.013	0.090	-1.033	0.820	0.051	0.072	-0.009	0.81
6	Korea	Monthly	-0.360	0.003	0.005	0.741**	0.321	0.028	0.034	0.024	2.04
7	Malaysia	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
8	New Zealand	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
9	Pakistan	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
10	Philippines	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
11	Singapore	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
12	Sri Lanka	Monthly	-0.443	0.001	0.005	0.042	0.046	6.475**	2.933	0.054	1.91
13	Taiwan	Monthly	-0.037	0.0543**	0.026	0.214	0.347	0.207	1.021	0.048	2.03
14	Thailand	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
Panel B: Economic growth regression for during-liberalization period											
Sl. No	Countries	Data frequency	α_0	β_1	$SE(\beta_1)$	β_2	$SE(\beta_2)$	β_3	$SE(\beta_3)$	$\bar{R}^2$	Durbin Watson statistics
1	China	Monthly	-0.044	-5.920	0.0009	0.045	0.056	0.003	0.066	-0.012	2.01
2	Hong Kong	Monthly	-0.055	0.014***	0.006	-0.344**	0.168	-3.794	13.945	0.039	0.09
3	India	NIFTY	0.059	0.0004	0.002	-0.004	0.017	-0.043	0.113	-0.014	1.99
		SENSEX	0.057	0.0006	0.002	-0.004	0.017	-0.042	0.113	-0.014	1.99
4	Indonesia	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
5	Japan	Monthly	4.223***	-0.087	0.057	0.056	1.006	0.074	0.132	0.058	1.43
6	Korea	Monthly	0.023	-0.015**	0.007	-0.239	0.275	-0.004*	0.002	0.067	1.94
7	Malaysia	Monthly	3.344***	0.047***	0.008	-0.712	1.009	4.092	3.169	0.090	0.21
8	New Zealand	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
9	Pakistan	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
10	Philippines	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
11	Singapore	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA

12	Sri Lanka	Monthly	4.246***	0.004	0.003	-0.019	0.437	1.077	8.504	-0.026	0.44
13	Taiwan	Monthly	8.137***	0.009***	0.002	-0.064	0.121	-0.046	0.083	0.101	0.28
14	Thailand	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
Panel C: Economic growth regression for post-liberalization period											
Sl. No	Countries	Data frequency	α_0	β_1	$SE(\beta_1)$	β_2	$SE(\beta_2)$	β_3	$SE(\beta_3)$	$\bar{R}^2$	Durbin Watson statistics
1	China	Monthly	7.625***	0.017***	0.006	0.313***	0.085	-4.430	5.690	0.208	0.29
2	Hong Kong	Monthly	156.34**	-0.053***	0.016	-0.014	0.135	-20.139**	9.332	0.132	2.02
3	India	NIFTY	-0.018	-0.021***	0.007	0.164*	0.101	-0.090	0.087	0.120	1.91
			-0.020	-0.021***	0.006	0.168*	0.100	-0.088	0.087	0.125	1.89
4	Indonesia	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
5	Japan	Monthly	-3.128***	0.018*	0.010	0.157	0.481	0.039***	0.008	0.071	0.59
6	Korea	Monthly	13.983***	0.022**	0.010	1.221*	0.638	-0.008	0.002	0.107	0.16
7	Malaysia	Monthly	4.691***	0.002	0.020	1.220**	0.573	-7.699	6.032	0.021	0.21
8	New Zealand	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
9	Pakistan	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
10	Philippines	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
11	Singapore	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
12	Sri Lanka	Monthly	0.019	-0.0002	0.001	-0.022	0.056	-3.015	2.065	-0.003	1.97
13	Taiwan	Monthly	3.673***	-0.005	0.019	0.240	0.219	-1.506*	0.804	0.013	0.21
14	Thailand	Monthly	NA	NA	NA	NA	NA	NA	NA	NA	NA
Panel D: Economic growth regression for full-period											
Sl. No.	Countries	Data frequency	α_0	β_1	$SE(\beta_1)$	β_2	$SE(\beta_2)$	β_3	$SE(\beta_3)$	$\bar{R}^2$	Durbin Watson statistics
1	China	Monthly	9.310***	0.011***	0.002	0.551***	0.144	0.735	0.641	0.143	0.24
2	Hong Kong	Monthly	-2.454	0.036***	0.009	0.848***	0.275	0.876	0.721	0.064	0.21
3	India	NIFTY	7.213***	-0.013**	0.005	-0.129***	0.042	-0.468***	0.189	0.069	0.14
			7.193***	-0.013**	0.005	-0.126***	0.043	-0.467***	0.189	0.068	0.14
4	Indonesia	Monthly	5.430***	-0.0005	0.003	0.060	0.128	0.0003	0.0002	-0.005	0.21
5	Japan	Monthly	1.843***	0.029***	0.010	0.196	0.497	0.026	0.054	0.011	0.57
6	Korea	Monthly	5.790***	0.032***	0.007	1.193***	0.433	0.004	0.006	0.053	0.17
7	Malaysia	Monthly	3.752***	0.038***	0.007	0.594	0.570	1.514	2.745	0.053	0.18
8	New Zealand	Monthly	0.118	-0.0002	0.004	-0.233**	0.120	3.580	3.344	0.011	2.05
9	Pakistan	Monthly	-0.123	-0.003	0.006	0.358*	0.206	-0.084	0.146	0.002	2.91

10	Philippines	Monthly	7.829***	-0.017**	0.007	-0.632***	0.069	-0.095	0.150	0.299	0.21
11	Singapore	Monthly	-0.021	0.019*	0.010	-0.529*	0.292	-0.540	9.430	0.021	2.09
12	Sri Lanka	Monthly	0.019	-0.0002	0.001	-0.022	0.056	-3.015	2.065	-0.003	1.97
13	Taiwan	Monthly	7.192	0.004	0.018	0.249	0.238	-0.110	0.188	-0.003	0.17
14	Thailand	Monthly	6.406**	0.013	0.015	-0.531***	0.157	-0.050	0.078	0.053	0.22

Note: ^a *, **, and *** indicates significance at 10, 5 and 1 % level, respectively

8 Findings and summary

The study proceeds in three phases, firstly, examines the impact of financial stock market liberalization on returns volatility and subsequently impact of asymmetric shocks (leverage effect) on returns volatility in 14 Asian economies that comprised the instance of sub-prime crisis. Secondly, the impact of financial stock market liberalization on Industrial growth in India has been tested, and thirdly, the impact of financial stock market liberalization on economic growth in 14 Asian economies has been examined. To measure the intensity and magnitude of financial stock market liberalization on returns volatility, industrial growth and economic growth, the period of the study is further divided into four-phases, namely pre-, during, post-liberalization and full period, respectively. Furthermore, to entangle the dynamicity and variations in the financial stock market liberalization, daily, weekly, and monthly frequency data set has been employed.

The impact of stock market liberalization on returns volatility in 14 Asian economies, shows increasing trend (table 13) for China, India, Indonesia, Japan, and Republic of Korea in during and post-liberalization periods, respectively relative to full-period (table 12). However, Malaysia, Pakistan, Sri Lanka, and Thailand witnessed significant increase in returns volatility for during-liberalization period shown in table 13. Successively, Hong Kong, Philippines, and Taiwan experienced decrease in returns volatility in during-liberalization period relative to full-period as justifying through table 12 implying quantum of impact of stock market liberalization on returns volatility. Hong Kong, Philippines, and Taiwan subsequently signifies increase in returns volatility in post-liberalization period relative to full-period, mimicking in table 12. Malaysia, Pakistan, Sri Lanka, and Thailand resembles decrease in returns volatility in post-liberalization period relative to full-period as in table 12. Moreover, New Zealand and Singapore has witnessed moderate impact of stock market liberalization on returns volatility in post-liberalization period, respectively. Table 12 and 13 represent the magnitude and quantum of impact of stock market liberalization on returns volatility, evidenced the volatility persistence in stock returns after liberalization is statistically and significantly different from pre-liberalization period.

Table 12 Monthly estimates for AR(s)-GARCH(1,1)

Sl. No.	Asian economies		Phases of liberalization		
	Country	Indexes	During-lib	Post-lib	Full-lib
			λ_2	λ_3	λ_4
1	China	SHCOMP Index	0.286***	1.008***	0.039***
2	Hong Kong	HSI Index	-0.926***	0.290***	0.198***
3	India	NIFTY Index	0.350***	0.999***	0.285***
		SENSEX Index	0.339***	0.999***	0.304***
4	Indonesia	JCI Index	0.798***	0.493***	0.306***
5	Japan	NKY Index	0.962***	0.070***	0.027***
6	Republic of Korea	KOSPI Index	0.741***	0.504***	0.383***
7	Malaysia	FBMKLCI Index	5.432***	0.341***	0.977***
8	New Zealand	NZSE50FG Index	NA	0.324***	0.324***
9	Pakistan	KSE100 Index	0.968***	0.350***	0.362***
10	Philippines	PCOMP Index	0.187*	0.311***	0.435***
11	Singapore	FSSTI Index	NA	0.256***	0.256***
12	Sri Lanka	CSEALL Index	0.989***	0.230***	0.230***
13	Taiwan	TWSE Index	0.216***	0.595***	0.258***
14	Thailand	SET Index	0.968***	0.358***	0.374***

Note: *, **, and *** indicates significance at 10, 5 and 1 % level, respectively

Table 13 Impact of stock market liberalization on returns volatility

Sl. No.	Country(s)	Phases of liberalization	
		During-lib phase (λ_2)	Post-lib phase (λ_3)
1	China	Increase	Increase
2	Hong Kong	Decrease	Increase
3	India	Increase	Increase
4	Indonesia	Increase	Increase
5	Japan	Increase	Increase
6	Korea	Increase	Increase
7	Malaysia	Increase	Decrease
8	New Zealand	-	Moderate increase
9	Pakistan	Increase	Decrease
10	Philippines	Decrease	Increase
11	Singapore	-	Moderate increase
12	Sri Lanka	Increase	Decrease
13	Taiwan	Decrease	Increase
14	Thailand	Increase	Decrease

Table 14 Monthly estimates for AR(s)-TGARCH(1,1)

Sl. No.	Asian economies		Phases of liberalization					
	Country	Indexes	During-lib	Post-lib	Full-lib	During-lib	Post-lib	Full-lib
			λ_2	λ_3	λ_4	γ	γ	γ
1	China	SHCOMP Index	0.258***	1.004***	0.099***	-0.704***	-0.1324	-0.719***
2	Hong Kong	HSI Index	0.299***	0.269***	0.188***	-0.144	0.034	-0.331
3	India	NIFTY Index	0.331***	1.000***	0.266***	-0.091	-0.110	-0.112***
		SENSEX Index	0.315***	1.000	0.207***	-0.080	-0.112	-0.437***
4	Indonesia	JCI Index	0.798***	0.263***	0.181***	-	-0.101	-0.629***
5	Japan	NKY Index	0.968***	0.084***	0.411***	-0.116	0.058**	0.542***
6	Republic of Korea	KOSPI Index	0.731***	0.504***	0.344***	-0.126	0.060	-0.097***
7	Malaysia	FBMKLCI Index	5.515***	0.348***	0.979***	-0.126	0.004	0.548***
8	New Zealand	NZSE50FG Index	NA	0.385***	0.385***	NA	-0.074	-0.074
9	Pakistan	KSE100 Index	0.952***	0.308***	0.319***	-0.128	-0.117	-0.115
10	Philippines	PCOMP Index	0.171	0.313***	0.349***	0.489	-0.021	-0.213***
11	Singapore	FSSTI Index	NA	0.302***	0.302***	NA	-0.060	-0.060
12	Sri Lanka	CSEALL Index	0.999	0.233	0.235***	-0.127	-0.606	-0.121
13	Taiwan	TWSE Index	0.246***	0.588***	0.162	-0.835	-0.409	-0.302***
14	Thailand	SET Index	0.908***	0.337	0.344***	-0.488*	-0.132	-0.114

Note:

, and * indicates significance at 10, 5 and 1 % level, respectively

a *

Table 15 Impact of stock market liberalization on leverage effect and returns volatility

Sl. No.	Country(s)	Phases of liberalization		
		During-lib (γ)	Post-lib (γ)	Full-period (γ)
1	China	No leverage effect	No leverage effect	No leverage effect
2	Hong Kong	No leverage effect	No leverage effect	No leverage effect
3	India	No leverage effect	No leverage effect	No leverage effect
4	Indonesia	No leverage effect	No leverage effect	No leverage effect
5	Japan	No leverage effect	Leverage effect	Leverage effect
6	Korea	No leverage effect	No leverage effect	No leverage effect
7	Malaysia	No leverage effect	No leverage effect	Leverage effect
8	New Zealand	No leverage effect	No leverage effect	No leverage effect
9	Pakistan	No leverage effect	No leverage effect	No leverage effect
10	Philippines	No leverage effect	No leverage effect	No leverage effect
11	Singapore	No leverage effect	No leverage effect	No leverage effect
12	Sri Lanka	No leverage effect	No leverage effect	No leverage effect
13	Taiwan	No leverage effect	No leverage effect	No leverage effect
14	Thailand	No leverage effect	No leverage effect	No leverage effect

Table 14 presents the quantum of volatility persistence to diagnose if it is significantly different in during and post-liberalization period respectively relative to pre-liberalization period, and further asymmetric coefficient (γ) captures the impact of negative against positive shocks on returns volatility. An asymmetric coefficient, which is significantly greater than zero will indicate that negative shocks results more volatility than positive shocks. However, we discover reverse (table 14), for the 13 Asian economies in during and post-liberalization and full periods, respectively. Conversely, table 14 and 15 opines significant positive asymmetric coefficients for Japan in post-liberalization and full periods, respectively. Subsequently, implies that asymmetric effect results increase in returns volatility in post-liberalization period and full-period, respectively for Japan followed by increase in returns volatility for Malaysia for full-period showing leverage effect. Hence, ensuring the leverage employed by the Japanese firms in post-liberalization and so the Malaysia as present in table 14 and 15.

Table 16 presents the quantum and magnitude of impact of stock market liberalization on industrial growth in Indian economy. Among the 11 sectorial industries engaging in financial and non-financial activities, construction & real estate, manufacturing, mining, banking services, and fee based financial services industries has witnessed significant positive growth as an impact from stock market liberalization in during, post-liberalization, and full periods (table 17), respectively.

However, non-financial and other financial services industries has witnessed positive industrial growth in post-liberalization and full-periods, respectively.

Table 16 Regression modelling impact of stock market liberalization on industrial growth weekly frequency period Jan 1995-Sept 2014

Serial No.	Industry	Industry specification	Phases of liberalization		
			β_1	β_1	β_1
1	Non-financial	C&RE	0.016***	0.016***	0.016***
2	Non-financial	ELE	0.001	0.004	0.002
3	Non-financial	MANU	0.006**	0.009***	0.007***
4	Non-financial	MIN	0.009**	0.014***	0.011***
5	Non-financial	SERV	0.013	0.009***	0.011***
6	Financial	AFS	0.007**	0.006	0.006***
7	Financial	BSERV	0.009***	0.009**	0.009***
8	Financial	FBFS	0.045***	0.012***	0.030***
9	Financial	INVS	-0.063	0.004**	-0.033
10	Financial	OTFS	-0.006	0.004**	0.007**
11	Financial	OFBFS	0.006	0.002	-0.033

Note: ^a *, **, and *** indicates significance at 10, 5 and 1 % level, respectively

Table 17 Impact of stock market liberalization on industrial growth in India

Sl. No.	Industry(s)	Phases of liberalization		
		During-lib (β)	Post-lib (β)	Full-period (β)
1	Construction & real estate	Significant impact	Significant impact	Significant impact
2	Electricity	No impact	No impact	No impact
3	Manufacturing	Significant impact	Significant impact	Significant impact
4	Mining	Significant impact	Significant impact	Significant impact
5	Services (other than financial)	No impact	Significant impact	Significant impact
6	Asset financing services	Significant impact	No impact	Significant impact
7	Banking services	Significant impact	Significant impact	Significant impact
8	Fee based financial services	Significant impact	Significant impact	Significant impact
9	Investment services	No impact	Significant impact	No impact
10	Other financial services	No impact	Significant impact	Significant impact
11	Other fund based financial services	No impact	No impact	No impact

Asset financing services industry has experienced positive industrial growth in during-liberalization period and full-period respectively. Further, investment services industry felt positive growth in post-liberalization period. Among 11 industries, 8 industries felt significant positive impact of industrial growth in post-liberalization period. Moreover, electricity and other

fund based financial services industries has shown no impact of stock market liberalization on industrial growth (table 17) in Indian economy.

Table 18 Regression modelling monthly impact of stock market liberalization on economic growth in Asian Economies

Sl. No	Country(s)		Phases of liberalization			
			Pre-lib	During-lib	Post-lib	Full-lib
			β_1	β_1	β_1	β_1
1	China		1.53	-5.920	0.017***	0.011***
2	Hong Kong		0.004	0.014***	-0.053***	0.036***
3	India	NIFTY	0.0004	-0.021***	-0.013**	0.608
		SENSEX	0.0006	-0.021***	-0.013**	0.596
4	Indonesia		NA	NA	NA	-0.0005
5	Japan		-0.013	-0.087	0.018*	0.029***
6	Republic of Korea		0.003	-0.015**	0.022**	0.032***
7	Malaysia		NA	0.047***	0.002	0.038***
8	New Zealand		NA	NA	NA	-0.0002
9	Pakistan		NA	NA	NA	-0.003
10	Philippines		NA	NA	NA	-0.017**
11	Singapore		NA	NA	NA	0.019*
12	Sri Lanka		0.001	0.004	-0.0002	-0.0002
13	Taiwan		0.0543**	0.009***	-0.005	0.004
14	Thailand		NA	NA	NA	0.013

Note: ^a *, **, and *** indicates significance at 10, 5 and 1 % level, respectively

Table 19 Impact of stock market liberalization on economic growth in Asian economies

Sl. No.	Industry(s)	Phases of liberalization		
		During-lib period (β)	Post-lib period (β)	Full-period (β)
1	China	No impact	Significant impact	Significant impact
2	Hong Kong	Significant impact	Significant negative impact	Significant impact
3	India	No impact	Significant negative impact	Significant negative impact
4	Indonesia	-	-	No impact
5	Japan	No impact	Significant impact	Significant impact
6	Republic of Korea	Significant negative impact	Significant impact	Significant impact
7	Malaysia	Significant impact	No impact	Significant impact
8	New Zealand	-	-	No impact
9	Pakistan	-	-	No impact
10	Philippines	-	-	Significant negative impact
11	Singapore	-	-	Significant impact
12	Sri Lanka	No impact	No impact	No impact
13	Taiwan	Significant impact	No impact	No impact
14	Thailand	-	-	No impact

Table 18 presents the quantum and magnitude of impact of stock market liberalization on economic growth in 14 Asian economies. China, Japan, and Republic of Korea assured significant positive economic growth (table 19) in post-liberalization and full-periods, respectively after controlling for macro-economic factors. Hong Kong, Malaysia and Taiwan has witnessed positive economic growth in during-liberalization period as in table 19. However, Hong Kong, Malaysia, and Singapore has experienced significant positive economic growth in full-period. Conversely, Republic of Korea has witnessed significant negative economic growth in during-liberalization period, followed by Hong Kong in post-liberalization period. India witnessed significant negative economic growth in post-liberalization and full-periods, respectively after controlling macro-economic indicators as per table 19. Furthermore, Singapore witnessed significant negative economic growth in full-period as shown in table 19. Indonesia, New Zealand, Pakistan, Sri Lanka, and Thailand has witnessed no impact of stock market liberalization on economic growth as in table 18 and 19, respectively.

9 Conclusion

The extensive research has been propelled on the relationship between financial development and economic growth, both the finance and economic development literature limits and lacks the inclusive investigation of the impact of the stock market liberalization on returns volatility, asymmetric shocks, leverage effect, industrial growth, and economic growth, respectively.

The evidences of the present study exhibits that the financial stock market liberalization marks increment in returns volatility, industrial growth and economic growth in Asian economies. The impact of financial stock market liberalization on returns volatility embarks highest in during-liberalization and post-liberalization periods relative to full period for China, India, Indonesia, Japan, Republic of Korea, and Malaysia. Successively, conditional volatility measures are maximum in the liberalization period. Moreover, we find that positive return shocks caused more volatility than negative shocks indicating no leverage effects with an exception of Japan which exhibits asymmetric information effect in post-liberalization and full periods, respectively. The result is robust to a wide variety of investigations, comprising an alternative frequency data set for pre-, during, post-liberalization and full periods, respectively. Therefore, this increased volatility is probably the result of both new information and trading behaviour of the investors. Thus, stock

market liberalization provides a multitude of factors that may have contributed to higher market volatility.

Furthermore, the construction and real estate, manufacturing, mining, banking services, and fee based financial services industries experienced growth owing to positive impact of financial stock market liberalization on industrial growth in India for during, post-liberalization and full periods, respectively. China, Japan, and Republic of Korea witnessed positive economic growth followed by stock market liberalization in post-liberalization and full periods, respectively. Conversely, India confirms negative impact of stock market liberalization on economic growth in post-liberalization and full periods, respectively after controlling macro-economic factors.

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